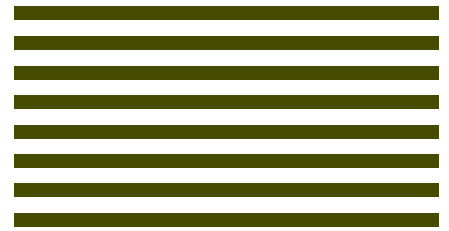


TRAIN PROTECTION SYSTEM TRACKSIDE NATIONAL ROLLOUT



Pre-Tender Issue

ETCS Project Scope of Works



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LIST OF ABBREVIATIONS

Abbreviation	Description
ADU	Aspect Display Unit
API	Application Programming Interface
APIS	Approval to Place in Service
APOM	Authorisation to Place on the Market
ASA	Application for Safety Approval
AsBo	Assessment Body
ATP	Automatic Train Protection
BTM	Balise Transmission Module
CAWS	Continuous Automatic Warning System
CCO	Control Command On-Board
CE	Compensation Event
CIÉ	Córas Iompair Éireann
CLD	Certification Level Documents
CME	Chief Mechanical Engineer
CR	Change Request
CRR	Commission for Railway Regulation
CSM-RA	Common Safety Method – Risk Assessment
D2G	Drogheda to Greystones project (existing project fitting ETCS in Greater Dublin area)
DART	Dublin Area Rapid Transit
DeBo	Designated Body
DfI	Department for Infrastructure – Northern Ireland
DMI	Driver Machine Interface
DMU	Diesel Multiple Unit
DR	Design Review
DVT	Driving Van Trailer
EBA	Euro Balise Antenna
EC	EC Certificate of Conformity
ECM	Entity in Charge of Maintenance
EMC	Electromagnetic Compatibility
EMD	Electrical Medium Distance
EMU	Electrical Multiple Unit
ERA	European Railway Agency

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ESC	ETCS System Compatibility
ETCS	European Train Control System
EVC	European Vital Computer
FEA	Finite Element Analysis
FRACAS	Failure Reporting, Analysis, and Corrective Action System
FRMCS	Future Railway Mobile Communication System
FTS	Fleet Technical Support
GASC	Generic Application Safety Case
HAZID	Hazard Identification
HFAP	Human Factors Assurance Plan
HFAR	Human Factors Assurance Report
HFIL	Human Factors Issues Log
HoS	Head of Series
HR	Hazard Record
HVP	Heuston Valeting Plant
IE	Iarnród Éireann
IM	Infrastructure Manager
IOD	Interoperability Directive
IRS	Irish Railway Standard
ISV	Intermediate Statement of Verification
JRU	Juridical Recorder Unit
LRU	Line Replaceable Unit
MSSC	Maintenance Support Service Contract
MTBF	Mean Time Between Failure
MTBSAF	Mean Time Between Service Affecting Failure
NL	Non-leading
NoBo	Notified Body
NRO	National Rollout
OSS	One Stop Shop
PMP	Project Management Procedure
PPE	Personal Protective Equipment
QA	Quality Assurance
RACI	Responsible, Accountable, Consulted, Informed (Matrix)
RFI	Request for Information

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RIU	Radio Infill Unit
RU	Railway Undertaking
SAP	Safety Approval Panel
SASC	Specific Application Safety Case
SCM	Safety & Compliance Matrix
SET	Signal, Electrical & Telecoms (Dept)
SMS	Safety Management System
SRAC	Safety Related Application Condition
STM	Specific Transmission Module
T&C	Testing & Commissioning
TF	Technical File
THR	Tolerable Hazard Rate
TPS	Train Protection System
TPWS/AWS	Train Protection Warning System/Automatic Warning System
TQ	Technical Query
TSI	Technical Specification for Interoperability
V&V	Verification & Validation

Table 1 - Abbreviations used in the present document

1 PROJECT OBJECTIVES & INFORMATION

This project shall modify the Irish railway infrastructure throughout the trackside rail network to enable European Train Control System (ETCS) Level 1 operation.

The objective of this procurement process is to appoint a Contractor to design, supply, install, integrate, test, and commission trackside equipment for an ETCS Level 1 system, referred to as Command Control Trackside (CCT) system, onto the Client's existing railway infrastructure. The project is called The TPS Trackside National Rollout which shall be referred to as the works. The combined onboard and trackside system will replace the near obsolete CAWS and ATP systems and at the same time extend the train protection system to the remaining 54% (1,166 track-km) of the Client's railway network, which is not currently equipped to operate with any form of driver warning or ATP system. This Scope refers to the Trackside National Rollout.

This Scope classifies all clauses as Mandatory or Functional, the definition of these terms is as follows:

Mandatory requirements are typically a Regulation, Statutory Requirement, Standard or a Requirement that IÉ believes cannot be deviated from in order to achieve the project objectives. It is necessary for the Contractor to provide a statement and acceptance of these requirements.

Functional requirements are project requirements, guidelines or good practice. IÉ can accept a deviation from these provided that the alternative proposed by the Contractor achieves the same outcome or function.

Information is provided in the form of a note in *Blue Italic text* to support the requirement with additional information, rationale, or guidance on the applicability of the requirement to the Client.

1.1 Existing Train Protection Systems

The current EMU fleet operating in the DART area is fitted with Automatic Train Protection (ATP). ATP monitors the speed of the train and alerts the driver if the speed of the train exceeds the permissible speed. However, should a driver not respond and continue attempting to drive the train, a full-service brake application is initiated and maintained until the driver responds. The ATP equipment senses the train's braking rate. If this is found to be inadequate, the ATP system initiates an emergency brake application.

Permissible speed is transmitted from the trackside to the train by means of coded information carried on the track circuits and picked up by coils mounted under the train. The codes are translated by the train equipment and presented to the driver on a display showing both the actual train speed and the permissible speed. ATP equipment will be removed from ATP fitted trains and the ETCS L1 onboard system will be installed in its place.

The current DMU, Loco and DVT fleets are fitted with a Continuous Automatic Warning System (CAWS). CAWS indicates to the driver each change in signal aspect, warns of a change to a more restrictive signal aspect, and initiates a brake application if the driver does not acknowledge the warning. In the CAWS system, the aspect of the next signal ahead is transmitted from the trackside to the train by means of codes carried on the track circuits and received by coils mounted under the train. The information is decoded and displayed to a driver in the form of a miniature signal display in the cab.

As part of the existing D2G project, ETCS trackside infrastructure is being overlayed on the existing ATP and CAWS trackside subsystems. This is a separate project to the TPS Trackside National Rollout (NRO). The existing ATP and CAWS trackside subsystems will remain in place until all trains operating on the IÉ network are fitted with ETCS.

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CAWS equipment will not be removed from CAWS fitted trains. As part of a separate project, the ETCS L1 onboard system will be installed, in addition to the CAWS system and both will operate simultaneously and independent of one another. This will remain in place until the ÍÉ network is fully fitted with ETCS trackside equipment.

1.2 ÍÉ Infrastructure Information

The Client intends to complete the overlay of ETCS Level 1 on the of the existing signalling systems which are composed of the following technologies.

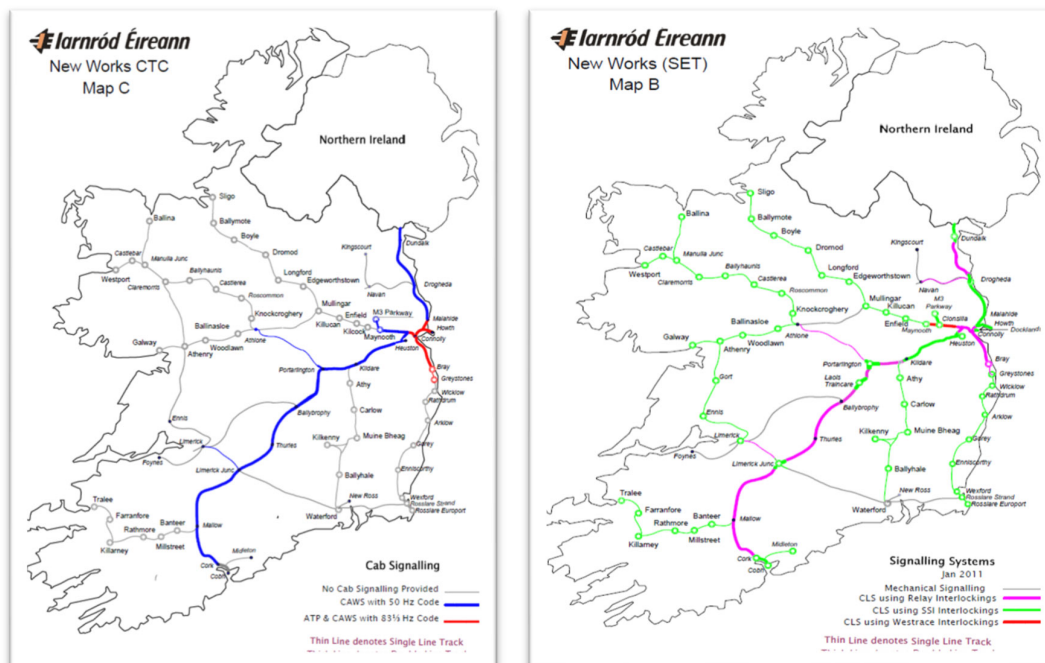


Figure 1 - ÍÉ Trackside existing signalling systems

1.3 Principles of the ETCS Level 1

The high-level requirements for ETCS Level 1 trackside are described in the Irish Railway Standard IRS-305, available on the [CRR website](#), and in the Irish Rail Trackside Common Application Requirement, referred to as CCT CAR and referenced as TPS-ETCS-P6-REQ-SL-IE-001. These requirements are derived from the following high-level principles.

1.3.1 Architecture Principles

The nominal architecture is to be distributed along the trackside with location cases and LEUs fitted adjacent to each signal requiring fitment. In specific circumstances (e.g., lack of power supply on the trackside) LEUs may be co-located in a centralised location. Innovative features are not precluded, e.g., to remove the needs for cutting in the circuits, or minimise overall Capex and Opex costs. These innovative features can only be envisaged post contract award via the Value Engineering process.

- All main, distant, and level crossing signals shall be fitted with a balise group.

- Shunt signals may be fitted with a balise group. The need for fitting shunt signals will be defined at specific application level.
- The balise installation rules shall be in accordance with the requirements of SUBSET-036 and SUBSET-040.
- Infill information shall be provided via balises. Euroloop, and RIU shall not be used.
- Signal balise group architecture: one fixed and one controlled balise for most locations. In some cases, two (2) controlled balises and one fixed balise may be required. Regardless of the preliminary design (Balise Layout Plans) produced by the Client prior to the Contract, it will be the responsibility of the Contractor to design and install the appropriate number of balises. The contractor will produce a revised balise plan showing the contractors proposed balise layout and submit this to IÉ for acceptance.
- Fixed balise groups shall be used for relocation, TSR warning messages, VBC or fixed signals and stop boards protection.

1.3.2 Functional Principles

The ETCS Level 1 trackside messages shall be configured to deliver the functionality of ETCS Level 1 overlaid on the existing signalling system to provide:

- Full Supervision Movement Authority derived from main signal aspects and block lengths, except mid-section level crossing main signals.
- Advanced Movement Authority information through infill balise groups.
- Linking principle making use of 'no linking reaction' as much as possible to improve system availability
- On Sight Movement Authority derived from shunt and call-on signals aspects and block lengths where they are required to be fitted.
- Train trip from any fitted signal displaying a 'Danger' (red) aspect.
- Train trip from any limit of shunt boards and stop and obtain instruction boards required to be fitted.
- Mid-section level crossing protection derived from mid-section level crossing signal aspects.
- Temporary speed restrictions are managed via the transmission of text messages from unlinked Balise Groups. No speed enforcement is required.
- Virtual balise cover orders through non-permanent fixed balises.

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2 THE WORKS SCOPE

2.1 Outline

IE has subdivided the network into 9 phases and has developed a migration strategy for the rollout of the Trackside NRO based on these 9 phases. The strategy is based on achieving incremental safety benefits as soon as possible and the sequence of rollout has been determined on a safety priority basis. This sequence can be seen in Figure 2 below.

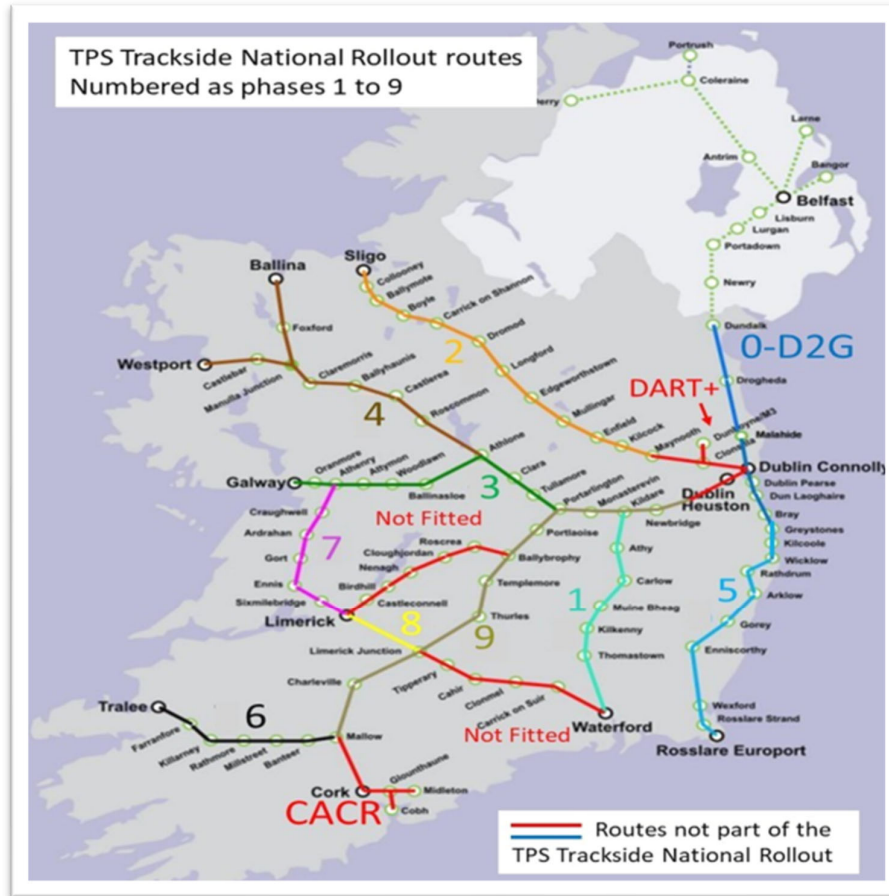


Figure 2 - Nine (9) TPS Trackside phases

This roll out of the nine (9) phases described in **Error! Reference source not found.** is estimated to correspond to the fitment of approximately 934 signals, 647 location cases, 2862 balises (including 519 infill balises), 523km of Balise Cable, and 328 km of new cable containment.

IE has developed a draft programme for delivery of the Trackside NRO; this programme shows completion of the first APIS 5 interim operation approval in 2029 (Phase 1) and the final APIS 5 interim operation approval in 2032 (Phase 9). Please refer to the draft project programme in Appendix D

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The Contractor shall:

- (i) complete a Generic Application Design that suits every portion of the Client's network
- (ii) for each Phase (up to 9 in total), complete the required Infrastructure Survey, take ownership of the Preliminary Design (Preliminary Balise Layout Plan) produced by Irish Rail, complete the Functional Design, complete Hardware Design (including new TPS Location cases and modifications to existing location cases), complete Data Design, manufacture and fitment of all ETCS Level 1 equipment.
- (iii) complete the manufacturing, supply and installation of location cases fitted with ETCS Level 1 trackside equipment and install ETCS Balises (with associated fixings, bars, tail cables etc.).
- (iv) complete all associated civils works in relation to the new TPS location case (as set out in section 7.2 of this document)
- (v) complete the manufacture of the required ETCS equipment including proprietary test tools.
- (vi) complete the modification (wiring mods) of the existing location cases to be prepared for integration with TPS location cases.
- (vii) complete all cable terminations as per design with respect to Trackside Testing Requirements TPS-NRO_TS-P1-REQ-TS-IE-001
- (viii) complete all the required Laboratory Testing, Static Testing and Dynamic Testing to demonstrate the achievement of performance and safety targets. Effort should be made to minimise onsite testing as much as possible.
- (ix) undertake all of the necessary works to achieve Iarnród Éireann, National and European safety approvals including.
 - IÉ Infrastructure Manager Safety Approval Panel approvals (SMS 14)
 - ERA Trackside Approval Process
 - CRR Authorisation to Place Into Service (APIS)
- (x) carry out various ancillary works and services in respect of the Scope (as more particularly set out in Section 16) ("**Ancillary Works**");
- (xi) provide training and maintenance services to the Client in respect of the supplied equipment (which will be administered under a separate contract which the Contractor shall be required to execute (the "**Maintenance Support Service Contract**" or "**MSSC**") (see Schedule B, Appendix 11 of the contract)).
- (xii) the tasks and deliverables set out above shall be delivered broadly in accordance with the draft project programme in Appendix D.

The Contractor shall achieve compliance with all of the Client's operational requirements when undertaking works on the railway in the most efficient and timely manner to ensure that no unplanned operational restrictions or delays may result from these works. For the avoidance of doubt, the Contractor is required to design the ETCS system so as to maximise train speed and performance.

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2.2 Contractor's Equipment Supply

The Contractor shall supply and install the equipment indicated as part of Contractor's Scope, as indicated in red in **Error! Reference source not found.** below in figure 3, i.e.:

- Balise
- Balise identification plate
- Fixing bars (or alternate where required to suit location)
- Tail cables
- Ducts between balise and disconnect boxes
- Ducts and cables between location cases and between the TPS location case and the Field Chamber. This includes all necessary cables, including signal, power and earth cables.
- TPS location cases inc. bases (and excavation of same), LEUs and all required wiring or components) within the LOC.
- Modification of the existing location cases and their integration with the TPS location cases
- Earth cable from earth pot to TPS and existing signal LOC and railing (note earth rod installed by others).
- LEU and Balises Programming and test tool

Please also refer to Section 7.2 in relation to division of responsibilities in relation to civils works to be undertaken by The Contractor and separate civils Implementation NRO contracts.

A civil engineering contractor (under a separate contract) will supply and install as indicted in blue **Error! Reference source not found.** in figure 3, i.e.:

- Cable containment (troughing etc.)
- Disconnect boxes
- LOC site materials & preparation, i.e. kingposts/modular block retaining structures, handrails etc.
- Field Cable between the Disconnect Box and the Field Cable Chamber. Please refer to civils drawings in Appendix for example of field cable chamber
- Earth rod.

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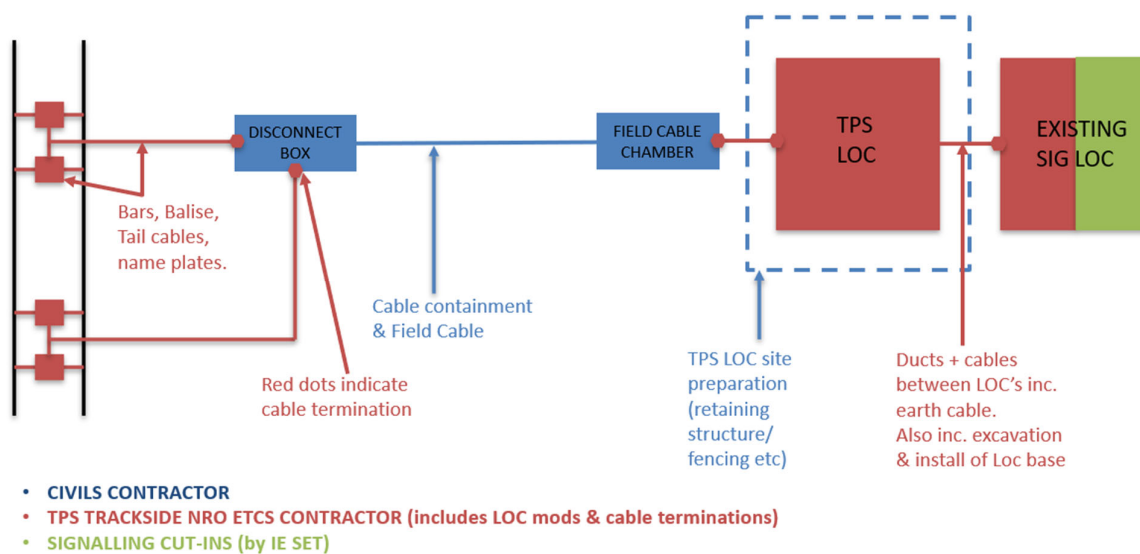


Figure 3 - Distribution of Responsibilities and Supply of Equipment

The Client will:

- Supply Field cable to the civil engineering contractor (separate to the ETCS Contract's civils works)
- Supply the Contractor with existing LOC case circuitry drawings and any other needed source record (e.g., signalling plans or control tables)
- Perform the Cut In to the signal circuit (as designed and prepared by the Contractor) or the avoidance of doubt, The Contractor shall design and implement the modification of the existing location case. The Client will undertake the connection activities and performs the no-regression testing of the live signalling system.

2.3 Breakdown of the Works

The works required are to be broken down as follows:

- Work Package A** works, which encompasses the Generic Application Design development and safety assurance documentation which shall be suitable for the subsequent design and implementation of each of the nine (9) Phases; and
- Work Package B** works, which comprises the Infrastructure Survey, Specific Design and associated system and safety assurance documentation of up to nine (9) Phases on the basis of the approved Generic Designs completed in **Work Package A**.
- Work Package C** works, which include the manufacture, integration, installation, test and commissioning of the system designed as part of **Work Package A** and **Work Package B** along with associated safety assurance documentation.
- Ancillary Works** and services required in relation to the **Work Package A**, **Work Package B**, and **Work Package C** works as more particularly described herein.

The Contractor may be authorised to commence **Work Package B** before **Work Package A** is completed so long as the Contractor demonstrates and implements risk mitigations that justify earlier

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achievement of the same results. Similarly, the Contractor may be authorised to commence **Work Package C** before **Work Package B** is completed.

2.4 Work Package A Works - Generic Application Design

The **Work Package A** works will include the following (without limitation):

Securing Product Acceptance for each component supplied by the Contractor, in respect of:

- Section 4 – Works Requirements
- Section 5.3.1 – Product Specifications
- Section 5 – Design Management
- Section 10 – Engineering Management

Securing the Generic Application Design approval which consist of any required standard rule, electrical and mechanical drawing, specification, template, process, or tools which enable **Work Package B** and **Work Package C** works to be undertaken for any of the nine (9) Phases in a repetitive production regime, and in respect of:

- Section 4 – Works Requirements
- Section 5.3 –Generic Application Designs
- Section 5 – Design Management
- Section 10 – Engineering Management

Testing of the Generic Application for each possible configuration present on the nine (9) Phases, in respect of which refer to:

- Section 4 – Works Requirements
- Section 8 – Testing & Commissioning
- Section 10 – Engineering Management

Safety Approval of the Generic Application, in respect of which refer to:

- Section 4 – Works Requirements
- Section 11- Systems Safety Approval

2.5 Work Package B Works - Infrastructure Survey and Specific Design

The **Work Package B** works will include, for each of the nine (9) Phases, the following (without limitation):

Any required Infrastructure Survey to collect the position of existing infrastructure, in respect of which refer to:

- Section 4 – Works Requirements

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- Section 5.4.1– Functional Designs.

Completing the Functional Design which includes the ETCS Scheme Plans, the signalling to ETCS interface table, the ETCS telegram specification, the aspect sequence charts, the geodetic survey and the infrastructure database, in respect of which refer to:

- Section 4 – Works Requirements
- Section 5.4.1– Functional Designs.

Completing the Hardware Installation Design which includes the existing location case modifications drawing, TPS location case modifications, and the cable plans, in respect of which refer to:

- Section 4 – Works Requirements
- Section 5.4.2 – Detailed Design

Completing the ETCS Level 1 Configuration Data of Lineside Electronic Units and Balises, the ETCS telegram specification, and the Infrastructure Database, in respect of which refer to:

- Section 4 – Works Requirements
- Section 5.5 – LEU and Balise Configuration Data

Safety Approval of the ETCS Level 1 Specific Application Design, in respect of which refer to:

- Section 4 – Works Requirements
- Section 11 - Systems Safety Approval

2.6 Work Package C Works - Manufacture, Integration, Installation, Test and Commissioning

The **Work Package C** works will include, for each of the nine (9) Phases, the following (without limitation):

Completing the manufacture and supply of the TPS products, in respect of which refer to:

- Section 4 – Works Requirements
- Section 6 – Manufacture and Supply.

Completing the manufacture and wiring of the TPS Location Case, in respect of which refer to:

- Section 4 – Works Requirements
- Section 6 – Manufacture and Supply.

Modification of Existing Location Case fitted to prepare interface with TPS Location cases, in respect of which refer to:

- Section 4 – Works Requirements
- Section 7 – Installation and Integration

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Installation of TPS location case fitted with Lineside Electronic Units and installation of Balises, in respect of which refer to:

- Section 4 – Works Requirements
- Section 7 – Installation and Integration

Integration of TPS location cases, cable terminations, Lineside Electronic Units, Field Cable, Disconnection Boxes, Tail Cables, and Balises, in respect of which refer to:

- Section 4 – Works Requirements
- Section 7 – Installation and Integration

Integration of the complete TPS trackside subsystem, in respect of which refer to:

- Section 4 – Works Requirements
- Section 7 – Installation and Integration

Static Testing of the TPS trackside subsystem, in respect of which refer to:

- Section 4 – Works Requirements
- Section 8 – Testing & Commissioning

Dynamic Testing of the TPS trackside subsystem, in respect of which refer to:

- Section 4 – Works Requirements
- Section 8 – Testing & Commissioning

APIS 5/6 Safety Approval of the TPS trackside subsystem, in respect of which refer to:

- Section 4 – Works Requirements
- Section 10 – Engineering Management.

Reliability Monitoring of the TPS trackside subsystem, in respect of which refer to:

- Section 12 – RAM Management

Hand Over of the TPS trackside subsystem, in respect of which refer to:

- Section 4 – Works Requirements
- Section 15 – Completion.

2.7 Ancillary Works

The Ancillary Works shall comprise:

- Value engineering and innovative features
- The delivery of system changes until all phases are completed

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- The supply of capability to perform system changes, data preparation, data validation, and testing including tools and process, and associated training for enabling IE staff to modify the ETCS trackside system.
 - The supply of training for maintenance and system knowledge.

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2.8 NEC4 Contract “Sections”

The Contract for the works (based on an NEC4 Engineering and Construction Contract, Option C), anticipates the works being undertaken in “Sections”. The “Sections” to be specified under the contract are shown in **Error! Reference source not found.**:

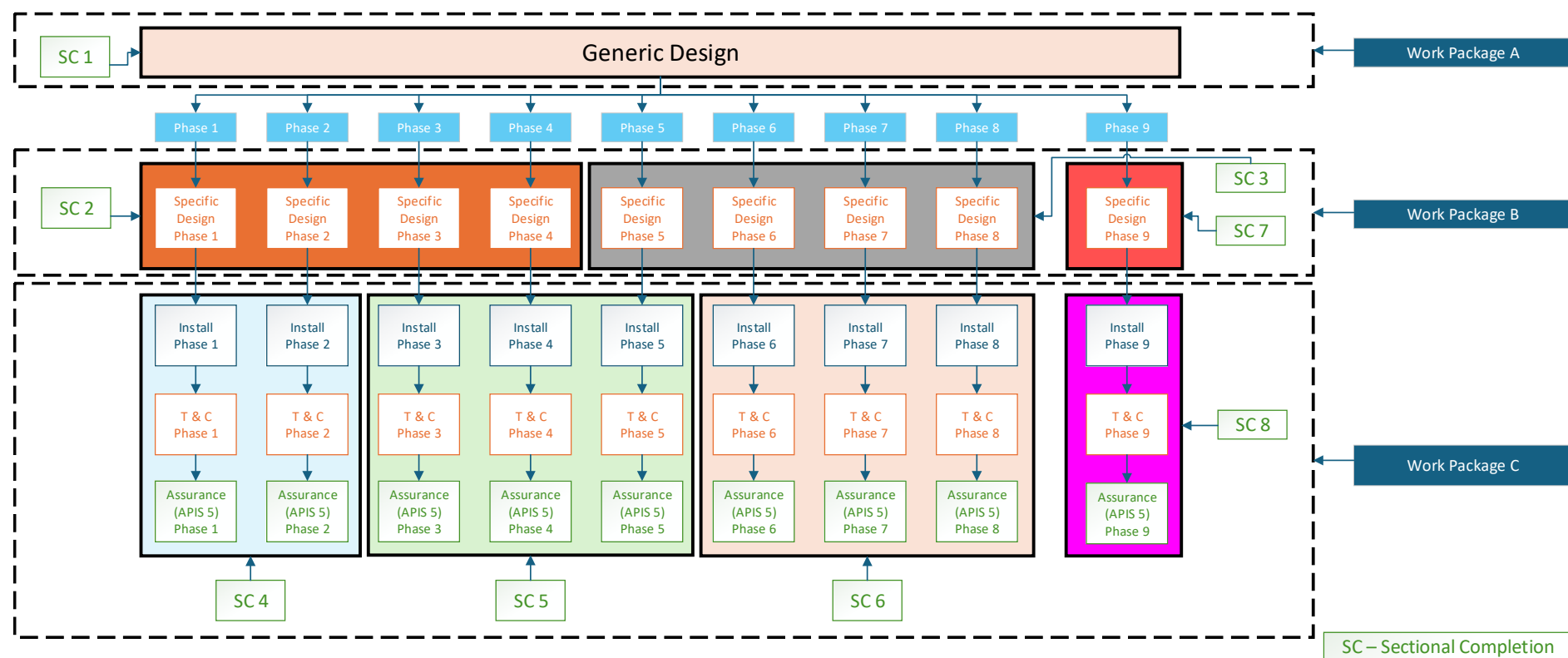


Figure 4 TPS Trackside NRO Sectional Completion: Sections

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- (i) The **Work Package A works** are common to all Phases and constitutes its own “Section”, with associated key date and sectional completion date.
- SC 1: Key Deliverable – Generic Design
- (ii) The **Work Package B works** for each Phase will be split into *up to 3 “Sections”*, with associated “Sectional” completion and key dates. They will be labelled as follows:
- SC 2: Specific Design – Phases 1-4
 - SC 3: Specific Design – Phases 5-8
 - SC 7: Specific Design – Phase 9
- (iii) The **Work Package C works** for each Phase will be split into up to 4 sections with associated “Sectional” completion and key dates. They will be labelled as follows.
- SC 4: Installation, T&C and APIS 5 approval - Phases 1-2
 - SC 5: Installation, T&C and APIS 5 approval - Phases 3-5
 - SC 6: Installation, T&C and APIS 5 approval - Phases 6-8
 - SC 8: Installation, T&C and APIS 5 approval – Phase 9
- (iv) The **Ancillary Works** will constitute their own “Section”.

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Phase	Section Deliverables	Sectional Completion	Key Task / Deliverables	Key Dates	Work Package	Completion Criteria
All	Generic Design	SC 1	Generic Design	KD 01	A	IE acceptance (Without site testing – included in SC - 4,5,6,8)
Phase 1 / Phase 2 / Phase 3 / Phase 4	Specific Design	SC 2	Specific Design Phase 1	KD 02	B	IE acceptance of Specific design
			Specific Design Phase 2	KD 03	B	
			Specific Design Phase 3	KD 04	B	
			Specific Design Phase 4	KD 05	B	
Phase 5 / Phase 6 / Phase 7 / Phase 8	Specific Design	SC 3	Specific Design Phase 5	KD 06	B	IE acceptance of Specific design
			Specific Design Phase 6	KD 07	B	
			Specific Design Phase 7	KD 08	B	
			Specific Design Phase 8	KD 09	B	
Phase 1 / Phase 2	Installation, T&C and APIS 5 approval	SC 4	Supply and Install Phase 1	KD 10	C	Handover certificate from Alstom to IE
			GA - T & C Phase 1	KD 11	C	Master test certificate issued to IE
			SA - T & C Phase 1	KD 12	C	Master test certificate issued to IE
			Assurance Phase 1	KD 13	C	APIS 5 Approval
			Supply and Install Phase 2	KD 14	C	Handover certificate from Alstom to IE
			GA - T & C Phase 2	KD15	C	Master test certificate issued to IE
			SA - T & C Phase 2	KD 16	C	Master test certificate issued to IE
			Assurance Phase 2	KD 17	C	APIS 5 Approval
Phase 3 / Phase 4 / Phase 5	Installation, T&C and APIS 5 approval	SC 5	Supply and Install Phase 3	KD 18	C	Handover certificate from Alstom to IE
			GA - T & C Phase 3	KD 19	C	Master test certificate issued to IE
			SA - T & C Phase 3	KD 20	C	Master test certificate issued to IE
			Assurance Phase 3	KD 21	C	APIS 5 Approval

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			Supply and Install Phase 4	KD 22	C	Handover certificate from Alstom to IE
			GA - T & C Phase 4	KD 23	C	Master test certificate issued to IE
			SA - T & C Phase 4	KD 24	C	Master test certificate issued to IE
			Assurance Phase 4	KD 25	C	APIS 5 Approval
			Supply and Install Phase 5	KD 26	C	Handover certificate from Alstom to IE
			GA - T & C Phase 5	KD 27	C	Master test certificate issued to IE
			SA - T & C Phase 5	KD 28	C	Master test certificate issued to IE
			Assurance Phase 5	KD 29	C	APIS 5 Approval
Phase 6 / Phase 7 / Phase 8	Installation, T&C and APIS 5 approval	SC 6	Supply and Install Phase 6	KD 30	C	Handover certificate from Alstom to IE
			GA - T & C Phase 6	KD 31	C	Master test certificate issued to IE
			SA - T & C Phase 6	KD 32	C	Master test certificate issued to IE
			Assurance Phase 6	KD 33	C	APIS 5 Approval
			Supply and Install Phase 7	KD 34	C	Handover certificate from Alstom to IE
			GA - T & C Phase 7	KD 35	C	Master test certificate issued to IE
			SA - T & C Phase 7	KD 36	C	Master test certificate issued to IE
			Assurance Phase 7	KD 37	C	APIS 5 Approval
			Supply and Install Phase 8	KD 38	C	Handover certificate from Alstom to IE

			GA - T & C Phase 8	KD 39	C	Master test certificate issued to IE
			SA - T & C Phase 8	KD 40	C	Master test certificate issued to IE
			Assurance Phase 8	KD 41	C	APIS 5 Approval
Phase 9	Specific Design	SC 7	Specific Design Phase 9	KD 42	C	IE acceptance of Specific design
Phase 9	Installation, T&C and APIS 5 approval	SC 8	Supply and Install Phase 9	KD 43	C	Handover certificate from Alstom to IE
			GA - T & C Phase 9	KD 44	C	Master test certificate issued to IE
			SA - T & C Phase 9	KD 45	C	Master test certificate issued to IE
			Assurance Phase 9	KD 46	C	APIS 5 Approval

Table 2 Sectional Completion Deliverables and Key Dates
Note:

- Sectional completion is achieved when all the task within the scope of that Section is complete.
- Each task within each Section is a key date item.

3 PHASE INFORMATION

Additional information about the Phases is provided in Appendix A. The exact boundaries and description of the existing infrastructure for each Phase is detailed in the source records and Specific Application Requirement (SAR) document for each Phase:

- Phase 1 Cherryville – Waterford
- Phase 2 Maynooth – Sligo
- Phase 3 Portarlinton – Galway
- Phase 4 Athlone – Westport / Ballina
- Phase 5 Wicklow – Rosslare Europort
- Phase 6 Mallow – Tralee
- Phase 7 Limerick – Athenry
- Phase 8 Limerick J. – Limerick
- Phase 9 Hazelhatch – Mallow

Within the timeframe of the TPS NRO there may be further extensions to the IÉ Network such as the Foynes line (currently freight only). Although these are not currently planned to be fitted it may be possible that IÉ will approach the supplier to provide further services for these lines.

Within these geographical areas, other projects will undertake infrastructure works that will affect the boundaries and scope of the TPS being delivered as part of this contract. The interface projects foreseen at the time of writing this Scope is illustrated in Appendix A, and includes:

- DART+ West
- DART+ Southwest
- Cork Area Commuter Rail
- Galway station alterations
- New passing loop in Oranmore
- Waterford new station

The Contractor shall regularly liaise with any interface project to ensure that inputs, outputs, boundaries and share of work are established and recorded in interface agreements.

Depending on the timing of these interface projects and the progress of the TPS trackside NRO works, the following impact can be foreseen:

- Change of input data to reflect modified track layout and signalling arrangement
- Rework of to remove a Level transition in case of an interface project being commissioned after the TPS trackside NRO

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- Non implementation of Level Transition in case of the TPS trackside NRO is commissioned after the interface project
 - Usage of Virtual Balise Cover orders.

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4 WORKS REQUIREMENTS

4.1 Regulatory Rules & Requirements

This section describes the regulatory rules and requirements that shall be fulfilled by the ETCS CCT Subsystems and by the Contractor in the execution of the works. The list is not exhaustive and the list in Appendix B may be updated during the works development.

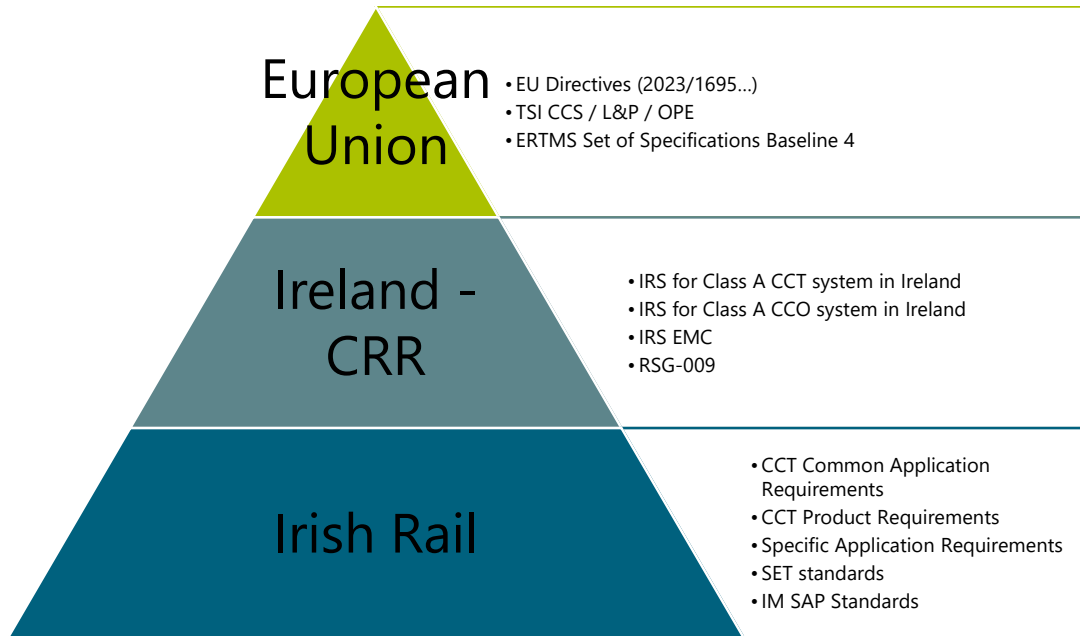


Figure 5 - Hierarchy of Requirements

Ref: 1 - Mandatory Requirement

The Contractor will otherwise comply with:

- i. all EU Directives, Regulations and Laws.
- ii. all CRR; standards and
- iii. all Irish Rail standards and requirements

as set out in **Error! Reference source not found.**, where there is any conflict between the above, the relevant requirements shall take precedence in the above order.

Ref: 2 - Mandatory Requirement

The ETCS CCT Subsystem shall be compliant with the following:

- i. Directive (EU) 2016/797 On the Interoperability of the Rail System within the European Union.
- ii. EU Commission Regulation (EU) 2023/1695 Technical Specification for Interoperability relating to the 'control-command and signalling' (CCS TSI) subsystems of the rail system in the EU.
- iii. Directive (EU) 2016/798 On Railway Safety: defines common principles for the management, regulation and supervision of railway safety. Any other EU Directives or Commission Regulations with applicability to the works.

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Ref: 3 - Mandatory Requirement

The ETCS CCT Subsystem shall be fully compliant with the EN Standards, and the Client's requirements and standards applicable to the works, a non-exhaustive list is provided in Appendix B:

Ref: 4 - Mandatory Requirement

In the event of Regulatory rules or standard changes during the course of the project, the contractor shall perform an impact analysis to identify if alignment to the latest releases is mandatory, preferable or not required.

4.2 Technical Requirements

Ref: 5 - Mandatory Requirement

The ETCS CCT Subsystem shall be fully compliant with the Irish Railway Standard (IRS) IRS-305 Requirements for Class A ETCS CCT Systems and for RU and IM Operating Rules in the Republic of Ireland, and compatible with IRS-304 Requirements for Class A ETCS CCO Systems and for RU and IM Operating Rules in the Republic of Ireland, and fully compliant to IRS-203-B EMC Co-ordination.

Note: No deviation or derogation to comply to any of these requirements is possible.

Ref: 6 - Mandatory Requirement

The ETCS CCT Subsystem shall be fully compatible with the existing signalling system, as installed in its working condition. The Contractor shall be responsible for identifying any sources of potential compatibility (e.g. current peaks, EMC) and for delivering a solution on either the existing infrastructure or the ETCS CCT..

Ref: 7 - Mandatory Requirement

The ETCS CCT Subsystem shall be fully compliant with the following Irish Rail requirements referenced in Appendix B:

- i. The Common Application Requirement (CAR). This set of requirements present two architecture options: Decentralised and Centralised. For the avoidance of doubt, the decentralised architecture is required to be designed and implemented by the Contractor
- ii. The Product Requirements (PR) for balises and LEUs
- iii. The ETCS Balise Layout Methodology
- iv. The Specific Application Requirements (SAR). The requirements contained in the SAR document are only applicable to the Phase they are referring to.

Ref: 8 - Mandatory Requirement

The Specific Application Requirement mentioned above are included in the Contract at a state of completion that is consistent with the level of information available at Contract signature. Before the commencement of the functional design for the Contractor shall seek an updated revision of the Specific Application Requirements and corresponding sources records which will form the first baseline requirement for the works.

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The Contractor shall request this updated revision from the Project Manager eight (8) weeks prior to commencing the Functional Design of the associated Phase.

The Contractor will not be entitled to any extra time or cost caused by the change between the two revisions of the SAR unless these changes can be deemed and demonstrated as having a substantial impact on the system delivered for this Phase.

4.3 General Requirements

Ref: 9 - Mandatory Requirement

The Contractor shall be responsible for the provision of all design, manufacture, installation, testing and commissioning staff required to undertake the works, with all staff having the relevant competence, tools, equipment, materials and training to undertake the tasks allocated to them. Their competence shall be demonstrated in accordance with SET standard '3rd Party Signalling Competency Requirements' SET-SMS-004-011.

Ref: 10 - Mandatory Requirement

The Contractor and the Project Manager shall develop a means to exchange and respond to Technical Queries (TQ) and Requests for Information (RFI) within two (2) weeks of the contract commencement date. Notwithstanding the contractual obligations, the Contractor and the Project Manager shall respond to RFIs within two (2) weeks of submission from either party.

Ref: 11 - Mandatory Requirement

The Contractor shall provide competent support throughout the installation and testing period to answer technical queries, efficiently manage site staff and provide installation / test logs.

4.4 Construction Regulation and Requirements

Ref: 12 - Mandatory Requirement

The Contractor shall carry out the role of Project Supervisor Design Process (PSDP) and Project Supervisor Construction Stage (PSCS) in accordance with the Safety, Health and Welfare at Work (Construction) Regulations 2013 (S.I. No. 291 of 2013) or as amended. Please also refer to section 16.

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5 DESIGN MANAGEMENT

5.1 Overall Design Management Process

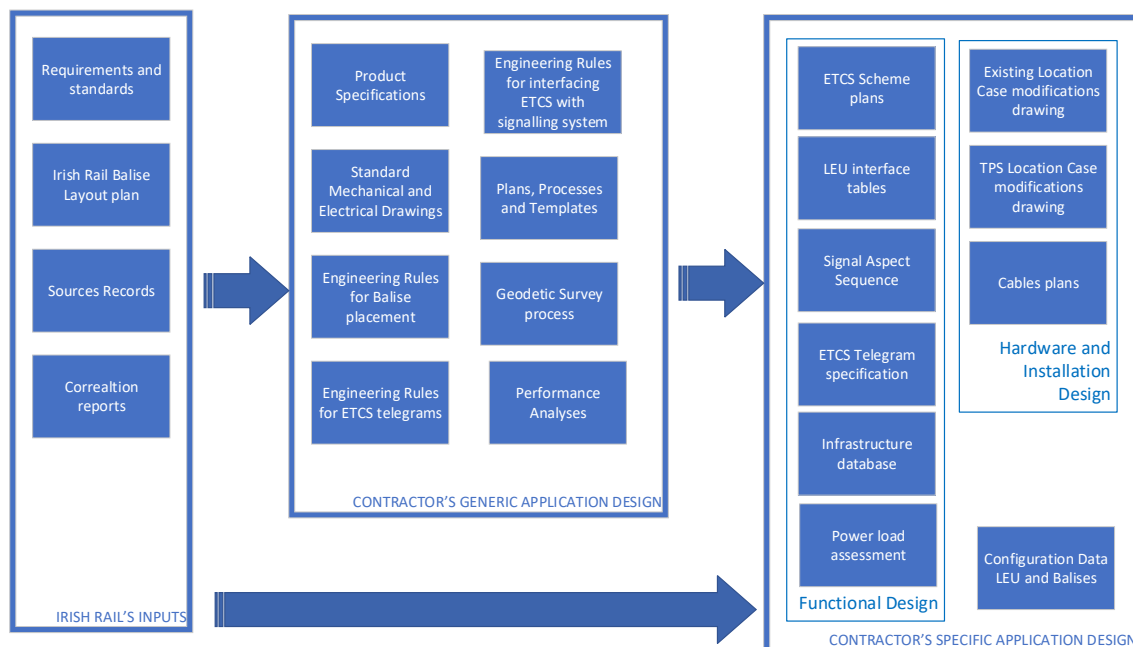


Figure 6 - Design Deliverables

Ref: 13 - Mandatory Requirement

The Contractor shall deliver the Generic Application Design and Specific Application Design artefacts, as listed in Figure 6 and further described in sections 5.3 and 5.4. Note that these packages will be delivered separately for each phase of the National Rollout.

Ref: 14 - Mandatory Requirement

The Contractor shall be the Design Authority for the works set out in this Scope; this is to include all designs artefacts listed in Figure 6.

Ref: 15 - Mandatory Requirement

Prior to commencing of the works the Contractor shall provide evidence that the engineering team in charge of producing, reviewing and verifying the designs are competent in the relevant field of expertise, have completed the minimum mandatory training requirements, and have been delegated engineering authority from the head of the design office.

Ref: 16 - Mandatory Requirement

Request for Information (RFI) or Technical Query (TQ) shall be sent to the client with sufficient anticipation by The Contractor such that the delivery dates requested in this Scope of Work are not delayed. The Contractor shall accommodate 10 days response period to the RFI or TQs.

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Ref: 17 - Mandatory Requirement

The Contractor shall submit all designs artefacts listed in Figure 6 to the Project Manager for a due diligence review and allow a four (4) weeks review period for each deliverable submitted. All Project Manager comments shall be responded and addressed by the Contractor within two (2) weeks. Any following iterations of the review form shall be returned one (1) week after being received. Comments shall be categorised as 'Blocking', 'Major' or 'Minor'. 'Blockings' comments do not allow the Contractor to carry on works related to the open comments. 'Major' comments allow the Contractor to carry on works related to the open comment at their own risk. Eventually all comments need to be addressed and closed out with acceptance by the Project Manager. The exact process and templates to be used for the due diligence review outlined above shall be agreed within four (4) weeks of the contract commencement date.

Ref: 18 - Mandatory Requirement

The Client will deliver the source records to the Contractor in their existing format:

- MicroStation and/or
- Pdf and/or
- MicroStation with Jpeg embedded
- Marked up correlation copies

Ref: 19 - Mandatory Requirement

All drawings shall be submitted by the Contractor in .pdf and MicroStation format.

5.2 Design Quality Assurance

Ref: 20 - Mandatory Requirement

The Contractor shall provide Design Management and Co-ordination - including Implementation of Design Change Control and issues log as per PMP-009-Design Management Procedure and PMP-005 – Change Control Procedure, and I-SET-9100 Engineering Management, and this Contract.

Ref: 21 - Mandatory Requirement

The Contractor shall carry out the role Designer for the trackside element, in accordance with the Guidelines on the Procurement, Design and Management Requirements of the Safety Health and Welfare at Work (Construction) Regulations 2013, for all Phases of this Contract.

Ref: 22 - Mandatory Requirement

The Contractor shall be fully responsible for the production, the review, the verification, the validation, and the approval of the designs. The Client will only perform due diligence checks of the Contractor's

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designs. The checks from the Client checks shall not be considered as a substitution to any work required to be undertaken by the Contractor.

Ref: 23 - Mandatory Requirement

All functional and detailed design shall be subject to Draft/Intermediate design review meetings, Interdisciplinary Design Checks and Acceptance reviews before they can be Approved for Construction (i.e. for data preparation).

Ref: 24 - Mandatory Requirement

To provide the Project Manager and Client with assurance that all the Generic Application and Specific Application deliverables are fit for purpose and met the requirement of the applicable standards, 'Quality Assurance' (QA) techniques shall be implemented by the Contractor. The Design Quality Assurance shall be described by the Contractor in a Design Management Plan. The various techniques shall include but not be limited to:

QA Design Validation & Checking		
Technique	Description	Frequency
Skilled staff	Skilled and competent staff to assure design quality by: - Ensuring that staff allocated to the <i>works</i> have experience in relevant <i>works</i> within this technical or business field. - Project Engineers dedicated to the <i>works</i> (who understand the requirements in detail).	Throughout the Generic and Specific Application Design and configuration data development.
Undertake Design Reviews	The Contractor shall organise quality design reviews by: - Assigning resources to perform regular 'Design Reviews' - Assigning the role Design Manager within the works with responsibilities for the quality of the Design produced by the Contractor.	At end of the Generic Application Design and at the end of functional Design and detail design stages of each sub-phase
Requirement verification	The Contractor shall perform verification activity to ensure that: Client requirements are correctly implemented in the Design	At end of the Generic Application Design and at the end of Functional Design and detail design stages of each sub-phase

Table 3 - QA Design Validation & Checking

Ref: 25 - Mandatory Requirement

The Contractor shall allow the Project Manager to perform quality audits of any design, survey or data production activity.

5.3 Generic Application Designs

Ref: 26 - Mandatory Requirement

The Contractor shall deliver a set of Generic Documents which are required to design and deliver the system, produce the Generic Application Safety Case and demonstrate compliance with the requirements of this Scope and any applicable requirements referred to in section 4. This set of Generic Documents shall include but not be limited to all the deliverables specified in this section 5.3.

Ref: 27 - Mandatory Requirement

The Generic Application Designs shall be delivered based on existing source records provided by the Client, i.e. the verification of control table and SSI data referred to in section 5.4.1.2 is not required to be done prior to producing the Generic Application Designs.

Ref: 28 - Mandatory Requirement

The Generic Application shall be documented such that any Generic Design Rule can always be deviated from or complemented at Specific Application level, so long as the deviation is supported by acceptable safety and performance analysis. Site specific changes required to accommodate isolated scenarios shall not always necessitate an update of the Generic Application.

Ref: 29 - Mandatory Requirement

The Contractor shall perform a critical analysis of the Client's requirements of this Scope and any applicable requirements referred to in section 4. This critical analysis shall identify possible incompatibilities between required, safety issues and performance issues. The result of this analysis shall be delivered to the Project Manager within four (4) weeks of the contract commencement date. The Project Manager may decide to update some of the requirements based on the feedback provided by the Contractor during this critical analysis.

Ref: 30 - Mandatory Requirement

The Contractor shall provide a matrix indicating the compliance status of the Generic Design against any applicable technical requirement indicated in this document (Scope of work) or in the Requirements Specifications referred to in section 4.

To do so the contractors shall propose a categorisation of each requirement with a type (Technical, Management, Information) and an allocation (Contractor or client).

For each Technical Requirement under the responsibility of the Contractor, the compliance status shall be indicated as 'Fully Compliant', 'Partially Compliant' or 'Not compliant'. In case of 'Partially Compliant' or 'Not compliant' the Contractor shall provide a justification and an alternative wording for the requirement. Each compliance statement shall be supported by references to documented evidence from the Generic Document listed in this section or equivalent. The alternative requirement shall be proposed to the client for Approval. The Client is entitled to reject the use of the Contractor ETCS Level 1 solution if it is 'Partially Compliant' or 'Not compliant' to one of the mandatory requirements. Note that compliance to mandatory requirements from any Irish Railway Standards IRS-305, IRS-304 and IRS-203 cannot be derogated.

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5.3.1 Product Specifications

Ref: 31 - Mandatory Requirement

The Contractor shall provide all necessary input to ensure that acceptance of their product can be made in accordance with the client standard I-SET-5051.

Ref: 32 - Mandatory Requirement

The Contractor shall provide, within four (4) weeks of the contract commencement date, for each product listed in section 2.2, the following set of specifications:

- Mechanical and electrical drawings of the product
- Interface specification
- Performance specification including RAM and response time.
- User Manual
- Configuration data tools user manual
- Installation rules
- Maintenance manual
- The list of Safety Related application conditions
- Any existing relevant certificates
- Safety Case, ISA and Interoperability Constituent conformity assessment certificates
- A list of known Change Requests that may have an impact the contracted products.

Ref: 33 - Mandatory Requirement

The Client may reject the use of a product if the Safety Related Applications are different from the ones supplied as part of the Contractor's tender return or subsequent proposed solution.

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Ref: 34 - Mandatory Requirement

To enable cross acceptance of the products, the Contractor shall provide for each product the following information for service in every application where this product is in service:

- Number of product in-service
- In-service demonstrated MTBF
- Number of failures reported to date,
- Severity of the failures reported to date
- The number of product replacements made to date
- The corrective actions that have been taken to address the defaults

The Client shall be authorised to contact other users of the products to investigate about their service performances. If required, the Contractor shall provide the contact details of the other users to the Project Manager.

Ref: 35 - Mandatory Requirement

The Contractor shall provide confirmation within four (4) weeks of a Project Manager request that the Field Cable be supplied directly by the Client is fully compatible with the Contractor's equipment referred to in section 2.2 of the Scope. Should the Contractor not confirm this compatibility within the above timescales, the Contractor shall be liable for any delay or cost incurred by a change of field cable.

5.3.2 Standard Mechanical and electrical Drawings

Ref: 36 - Mandatory Requirement

The Contractor shall provide standard Mechanical and Electrical Drawings required for manufacture, fitment, installation, and maintenance of their products integrated into an ETCS level 1 solution that is interfaced to the Client signalling system and compliant with the requirements of this Scope and the requirements referred to in section 4. These drawings must be in accordance with IÉ Standards and examples provided in Appendix B.

Ref: 37 - Mandatory Requirement

The mechanical and electrical drawings shall reflect any possible configuration of equipment supplied by the Contractor, included but not limited to:

- LEU housed in location case
- LEU on racks in Signalling equipment rooms
- Cable plans examples (containment length, cable routes, under track crossing, junction or termination boxes, types of cables and terminations)

These drawings must be in accordance with IÉ Standards.

Ref: 38 - Mandatory Requirement

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The standard electrical drawings shall directly reflect the twenty (20) most common signal configurations available on the 9 Phases and shall, with minimum adaptation, fit the need of any signal configuration present in the 9 Phases.

Ref: 39 - Mandatory Requirement

The mechanical and electrical drawings shall be made in accordance with SET standards I-SET-9308 and best practices used in SET design Office. Example of such drawings are provided in Appendix B.

Ref: 40 - Mandatory Requirement

First draft of representative mechanical and electrical drawings shall be issued to the Project Manager within eight (8) weeks of the contract commencement date for comment to ensure the quality and consistency of the production with the existing drawings in the Client's design offices.

Ref: 41 - Mandatory Requirement

Final drawings shall be submitted for acceptance by the Project Manager in accordance with Management and Review of Engineering Deliverables I-SET-9102. The Project Manager may perform due diligence review and/or sample checks of these submissions. Upon submission of the first set of comments by the Project Manager, the Contractor shall address them and resubmit within a maximum of ten (10) working days. Any subsequent comments shall be addressed within five (5) working days.

Ref: 42 - Mandatory Requirement

The final release of the drawings defining the modification required in an existing signal location case shall be submitted to the Project Manager twelve (12) weeks prior to the cut-in planned for that location case.

5.3.3 Engineering Rules

Ref: 43 - Mandatory Requirement

The Contractor shall provide a set of engineering rules that will enable the production and verification of the specific application design and the configuration data. This set of engineering rules shall be compliant with the requirement of this Scope and the requirement referred to in Section 4 and shall be applicable to any infrastructure configuration present in the Scope.

Ref: 44 - Mandatory Requirement

The Contractor shall provide a set of engineering rules that may be consistent with the engineering rules implemented in the TPS Drogheda to Greystones project:

- Alstom - ETCS Engineering rules. Ref Appendix B
- Alstom - ETCS Signalling and application Principles. Ref Appendix B
- Alstom - Encoder Standard Configuration Definition. Ref Appendix B
- Alstom - ETCS Braking Distance Table. Ref Appendix B
- Alstom - Gradient segmentation Methodology. Ref Appendix B

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It should be noted that the D2G set of engineering rules have been developed on previous versions of the Client requirements and may have been granted special derogations for full compliance. This does not mean that the Client will grant the same derogation for this Contract.

Ref: 45 - Mandatory Requirement

The engineering rules shall be composed of:

- ETCS Balise Positioning Rules which specify the rules for locating the Balise in relation to existing Infrastructure element.
- ETCS Telegram Rules which specifies the rules for configuring the ETCS data in the LEUs and Balises
- Signalling to ETCS Interface rules which specifies the rules for interfacing the ETCS system with the existing signalling system. The types and characteristics of lamps to which the ETCS system has to be interfaced is described in TPS Lamps Parameters (see Appendix B)
- Gradient simplification rules which explain the rules for converting the raw gradient from the infrastructure database into an optimised profile that can be used for the configuration of Balise and LEUs telegram. This optimised profile shall maximise the length of the braking distance calculated by the Onboard ETCS subsystem.
- Any other engineering Rules required to configure the Contractor's ETCS Level 1 subsystem such that it complies with the requirements of this Scope and the requirements referred to in section 4.

Ref: 46 - Mandatory Requirement

First draft or example of Engineering rules documents shall be submitted to the to the Project Manager within eight (8) weeks of the contract commencement date for due diligence review to ensure that their content matches Client expectations.

Ref: 47 - Mandatory Requirement

The final validated Engineering rules shall be submitted to the to the Project Manager within twenty-four (24) weeks of the contract commencement date for Project Manager acceptance. All internal and external document review cycles shall be completed at that time.

5.3.4 Infrastructure Survey Process

Ref: 48 - Mandatory Requirement

The Contractor shall provide the Infrastructure survey process which will describe how the Contractor will collect the position and vertical height of any infrastructure element required for the implementation of the Contractors' ETCS system, and how it verified, processed, recorded, reported, and updated and how they comply with the requirement of this Scope and the requirements referred to in section 4. An example of the geodetic survey specification used for D2G project is given in Appendix B: Generic Specification for Survey of Infrastructure. This example shall be considered by the Contractor as minimum requirements for the National Roll Out. The final specification and the process used by the Contractor shall be certified by the appropriate certification bodies as required for the Trackside NRO. The Contractors scope includes carrying out the Infrastructure Survey.

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Ref: 49 - Mandatory Requirement

The accuracy of the position and the vertical height collected shall +/-100mm

Ref: 50 - Mandatory Requirement

The Infrastructure Survey Process shall include the measurement that are required during design, installation and test and commissioning of the Contractor's ETCS Level 1 system.

Ref: 51 - Mandatory Requirement

The Infrastructure Survey Process shall be as automated and efficient as possible but shall also include the possibility for manual wheel measurements to be undertaken by the Contractor during the course of this project, or by the Client during revenue in service.

Ref: 52 - Mandatory Requirement

The output of the Infrastructure Survey Process completed by the Contractor shall be a database which level of quality and integrity is commensurate with the ETCS Level 1 SIL 4 application.

Ref: 53 - Mandatory Requirement

This database and any associated tool or software shall become the property of the Client at contract completion. The Client shall be trained by the Contractor to make updates to this database whilst maintaining its level of quality and integrity.

Ref: 54 - Mandatory Requirement

The output of the Infrastructure Survey Process shall be provided in Microsoft Excel format and may also be provided as a BIM database – see Section 31.

5.4 Specific Application Designs

Before the commencement of the Functional Design, the Contractor may agree with the Project Manager a split of phases into smaller manageable lots to fast track design deliveries and review. These sub-phases may be aligned to existing signalling scheme plans or possession maps as best suits the specific configuration of each phase and implementation strategy.

Ref: 55 - Mandatory Requirement

The Contractor shall provide for each phase a set of Specific Application Design artefacts which are required to implement the Generic Application described in section 5.3 and produce the Specific Application Safety Case for the phase.

Ref: 56 - Mandatory Requirement

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The Specific Application Design shall demonstrate compliance with the requirements of this Scope and any applicable requirements referred to in section 4. This set of deliverables shall include but not be limited to all the deliverables specified in this section 5.4.

Ref: 57 - Mandatory Requirement

After Installation and Testing and Commissioning, all Specific Application Design's shall be updated by the Contractor to reflect 'as-built' configuration. They shall be submitted by the Contractor in .pdf and MicroStation format.

5.4.1 Functional Designs

Ref: 58 - Mandatory Requirement

Before the commencement of each Phase, the Contractor shall ensure that he has all relevant input data to commence the works without delaying the delivery of the design and installation. The Contractor shall request the latest revision of source records and specific application requirements eight (8) weeks prior to the commencement of the functional design for each phase.

Ref: 59 - Mandatory Requirement

Before the commencement of each Phase, the Contractor shall ensure that he has an interface agreement in place with interface projects to ensure that all inputs, outputs and share of work are completely defined prior to the commencement of the functional design for each Phase

Ref: 60 - Mandatory Requirement

As part of functional design development, the Contractor shall facilitate and document any required consultation workshop and produce any required risk assessment to refine the fitment or functional requirement applicable to each phase (e.g. infill balise fitment, shunt signals fitment, lineside boards fitment, deficient overlap assessment).

Ref: 61 - Mandatory Requirement

For each phase, the Contractor shall first deliver the functional designs which shall include all deliverables required to produce a consistent set of installation design and configuration data. The functional design shall include but not be limited to the deliverable described in this section 5.4.1.

Ref: 62 - Mandatory Requirement

Eight (8) weeks after the start of the functional design development for each phase, the Contractor shall identify any engineering work that the Client may be responsible for (e.g. SSI or relay interlocking changes required to implement last wheel replacement feature, relocation of impedance bonds, increase of approach locking timers).

Ref: 63 - Mandatory Requirement

For each Phase, the Contractor shall perform a railway verification and validation analysis of the functional design for each Phase to ensure that railway performance is not affected by the introduction

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of ETCS level 1. This shall be based on a desktop verification exercise and rail system modelling tools (e.g.; Railsys), and shall include but not be limited to:

- Ensuring that no brake indication or intervention is triggered when operating on consecutive green aspects.
- Ensuring that timetable can be met with sufficient resilience to delays
- Ensuring that signalling ETCS section timers expiry are sufficient dimensioned, so they do not result in undesirable MA shortenings
- Ensuring that trains can stop at their usual stopping point. The usual stopping position of the trains has to be defined via consultation with the client

This Verification and Validation activity shall be delivered for each Phase within four (4) weeks of completing the ETCS Scheme Plans for that Phase and shall be updated within two (2) weeks of a change made to the Functional Design which may affect the analysis.

5.4.1.1 ETCS Scheme Plans

Ref: 64 - Mandatory Requirement

The contractor shall review and comment on the accuracy of the Clients balise layout plan within 8 weeks of contract award.

Ref: 65 - Mandatory Requirement

The Contractor shall deliver ETCS scheme plans which indicates the position and functions of the Balises Groups and TPS Location cases, in accordance with I-SIG-2103. The ETCS scheme plan shall be delivered as TPS layers added to the existing signalling master signalling plans (1:1 scale). The extra TPS layers shall be the following:

- Layer for TPS location cases
- Layer for speed changes configured in TPS
- Layer for all other ETCS information (balise, functional information, level transition, speed changes). An example of that main layer is provided in Appendix B.

Ref: 66 - Mandatory Requirement

The Contractor shall base ETCS scheme plans on the Balise Layout Plan provided by the Client (see Appendix B) and shall clearly identify any deviation from the Client Balise Layout Plan within the first release of the Contractor's ETCS scheme plan.

Ref: 67 - Mandatory Requirement

A Preliminary version of ETCS scheme plan may be submitted which does not consider the outcome of the infrastructure survey or the as built verification measurements. The Final revision of the ETCS scheme plan shall include the outputs of the infrastructure survey and the as built verification measurement.

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Ref: 68 - Mandatory Requirement

The ETCS scheme plan may require more equipment than envisaged by the Balise Layout plans provided by the Client (e.g. additional controlled balise at some signals). The Contractor shall be responsible for designing, supplying implementing all the necessary equipment to fulfil the Client requirements. The Balise Layout Plan delivered by the Client shall be considered for information only.

Ref: 69 - Mandatory Requirement

The Contractor shall deliver the ETCS scheme plan in MicroStation, and .pdf format.

5.4.1.2 ETCS Telegram Specification**Ref: 70 - Mandatory Requirement**

As a pre-requisite to develop ETCS telegram specification, the Contractor shall develop and verify Signal Aspect Sequence Charts based on existing control tables and/or existing relay interlocking circuit drawings. The aspect sequences charts shall be delivered to the Project Manager as part of the final design deliverables.

Ref: 71 - Mandatory Requirement

The Contractor will be provided with the SSI data in service and the Contractor shall verify that the Aspect Sequence implemented in the SSI data is compliant to the Aspect Sequence Chart derived from the Existing Interlocking Control Table.

Ref: 72 - Mandatory Requirement

The Contractor shall deliver an ETCS Telegram specification which describes for each Balise Group and LEU Telegram the applicable generic engineering rules as specified in section 5.3.3 and clearly identify any deviation or refinement from these engineering rules.

Ref: 73 - Mandatory Requirement

The ETCS Telegram specification shall reflect for each signal aspect and for each balise but not be limited to the following information:

- The ETCS packets that are submitted for each direction of travelling.
- The variable of interest of each packet (linked balise ID and reaction, applicable speed profiles, Movement authority length and EoA characteristics, Distances to level Transition, etc.)
- The generic rule applicable for each packet
- The deviation from the generic rules applicable for each packet and the rational for the deviation
- Any infrastructure database information relevant for data preparation and validation

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5.4.1.3 LEU Interface Tables

Ref: 74 - Mandatory Requirement

The Contractor shall deliver an LEU interface table with the applicable Generic engineering rules as specified in section 5.3.3 and that clearly identifies and justifies any deviation or refinement from these engineering rules.

Ref: 75 - Mandatory Requirement

The LEU interface table shall include for each LEU but not be limited to:

- The number of inputs and outputs and their allocation
- The telegrams sent for each combination of inputs
- The generic rule applicable for each LEU
- The deviation from the generic rules applicable for each LEU and the rationale for the deviation
- The characterisation of input (lamp type, current threshold) and outputs (balise type and identification).

5.4.1.4 Infrastructure Database

Ref: 76 - Mandatory Requirement

The Contractor shall continuously maintain the Infrastructure Database issued from the Infrastructure Survey Process described in section 5.3.4 such that it is always consistent with the functional and detailed design artefacts, up to their as-built configuration.

Ref: 77 - Mandatory Requirement

Using appropriate methods, the raw gradient data from the asset (geodetic) survey shall be simplified by the Contractor into a profile that can be used by data designers and verifiers.

Ref: 78 - Mandatory Requirement

The gradient data shall be smoothed such that minor changes, the omission of which has a negligible impact on braking performance, are considered on a lesser scale.

Ref: 79 - Mandatory Requirement

The Infrastructure Database shall be delivered to the client in Excel format as part of the functional design.

Ref: 80 - Mandatory Requirement

The Infrastructure Database shall be transferred to the Client together training required to be able to maintain this database for the remaining life of the system.

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5.4.2 Detailed Designs

5.4.2.1 Cable Plan

Ref: 81 - Mandatory Requirement

Within two (2) weeks of the ETCS scheme plan being finalised, the Contractor shall deliver the cable plan to the Project Manager to enable the completion of civil works. Cable plans (A2 template) examples are provided in Appendix B and shall include:

- Containment length
- Cable routes
- Under track crossing
- Junction or termination boxes
- Types of cables and terminations

5.4.2.2 Hardware and Installation Designs

Ref: 82 - Mandatory Requirement

The Contractor shall provide for each Phase a set of specific application design artefacts which are required to manufacture, install and integrate, test and commission each CCT item of equipment. The Electrical and Mechanical drawings shall be based on the standard drawings defined in section 5.3.2 made in accordance with SET standards and best practices used in SET design Office. Any deviation from the standard drawings shall be clearly visible and justified. Example drawings are included in Appendix B.

Ref: 83 - Mandatory Requirement

The Contractor shall deliver power load analyses for each impacted 650v supply to ensure that there is enough capacity to feed the additional ETCS equipment delivered by the Contractor or identify the need for power upgrades. These power load analyses shall consider all equipment fed by the 650v supply at the time of producing the assessment.

5.4.2.3 Existing Location Case Modifications Drawings

Ref: 84 - Mandatory Requirement

The Contractor shall provide for each Existing Signal Location Case or Signalling Equipment Room impacted by the introduction of the TPS trackside subsystem, the set of existing drawings updated with the modification required to be implemented.

The Contractor shall update existing location case drawings in line with the output from the correlation survey supplied by the Client as an input to this Contract.

Example drawings and correlation survey reports references are included in Appendix B.

Ref: 85 - Mandatory Requirement

The Contractor shall provide for each Existing Signal Location Case or Signalling Equipment Room impacted by the introduction of the TPS trackside subsystem, the set of existing drawings updated with the modification required to be implemented.

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5.4.2.4 TPS Housing Drawings

Ref: 86 - Mandatory Requirement

LEUs shall be housed in a location acceptable to the Client, including but not limited to a Location case, or racks in Signalling Equipment Rooms.

Ref: 87 - Mandatory Requirement

The Contractor shall provide the set of drawings required to manufacture install test and commission the ETCS Level 1 system for each new Location Case or Signalling Equipment Room housing TPS trackside equipment for the TPS trackside subsystem.

Ref: 88 - Mandatory Requirement

The Contractor shall adequately dimension the TPS location case and racks such that it enables the future use of at least one infill balise replicating the signal balise group information, even if that infill balise is not required by the Specific Application Requirement at Contract commencement date.

- Mandatory Requirement

The Contractor shall propose the client for acceptance a distribution of LEU per TPS housing to interface with existing location cases.

5.5 LEU and Balises Configuration Data

Ref: 89 - Mandatory Requirement

The Contractor shall produce for each configurable trackside equipment the ETCS Level 1 Configuration Data that complies with the ETCS telegram specification and that are consistent with the Existing and TPS location drawings, delivering the functionalities specified by the Generic application and the requirements of this Scope and the requirements referred to in section 4.

Ref: 90 - Mandatory Requirement

The Contractor shall deliver the configuration data to the Project Manager in a format that can be easily readable by the Client, so that due diligence spot checks can be done without the need of any Contractor proprietary tool.

Ref: 91 - Mandatory Requirement

The Contractor shall deliver the configuration data to the Client in a secured support together with data management procedure that ensures no obsolete data can be stored in IE network and or maintenance tools, in accordance with the requirements of I-SIG-2172.

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Ref: 92 - Mandatory Requirement

The Contractor shall provide the Client with set of configuration management tools to enable safe and secured ETCS data file management. For avoidance of doubt these tools shall also be able to manage to the D2G configuration files. These D2G configuration files will be provided by the client to the contractor for integration in their secured file management system. The client shall provide the necessary rights of use and licenses required.

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6 MANUFACTURE AND SUPPLY

Ref: 93 - Mandatory Requirement

The Contractor shall allow the Project Manager to perform quality audits of their product manufacturing and testing plans.

Ref: 94 - Mandatory Requirement

The Contractor shall manufacture the Location Case in accordance with I-SET-6200 standard.

Ref: 95 - Mandatory Requirement

The Contractor shall enable the Project Manager to access the Contractor's facilities to perform specific products tests required for the product acceptance.

Ref: 96 - Mandatory Requirement

The Contractor shall ensure that products sent to the Project Manager are effectively working and each product shall be delivered with the required quality, conformity and test certificates to demonstrate this working state. As a minimum the following certificate shall be included for all products, including assembled location case:

- EC declaration
- Quality certificates
- Environmental certificate
- Test certificate
- EMC/EMI certificate

Ref: 97 - Mandatory Requirement

The Contractor shall deliver a representative set of prototypes for possible ETCS trackside installation so that the level of quality, installation and testing process and maintainability can be audited and accepted by the Client. This prototype shall be delivered in following sequence:

- Products, cables and location case delivered independently to Irish Rail Premises
- Full installation and testing, from existing location case interface to balises at a defined locations representative of all foreseeable configuration.

Ref: 98 - Mandatory Requirement

The Contractor shall maintain an Asset Register which lists all equipment that are supplied to the Client, and includes:

- Equipment description.
- Manufacturing date.
- Expiry date (if applicable).

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- Manufacturer / Supplier details.
 - Part number.
 - Version details.
 - Serial Number.
 - Batch number where applicable.
 - Other traceability information as required; and
 - Software/firmware descriptions, versions, dates etc.
 - Equipment location: transit/storage place/ target installation location where it is installed

Ref: 99 - Mandatory Requirement

The Contractor shall be fully responsible for all equipment (and material) including storage and keeping them in working condition until they are installed, tested and commissioned in their designated locations. No storage of materials or equipment shall be provided by Irish Rail. Refer to 7.2.4 for responsibilities of Contractor for providing secure compound areas for their materials.

Ref

The contractor is responsible to provide to the Client all strategic maintenance spares and tools for those assets that have a long lead time or are vulnerable and/or susceptible to failure or damage. The spares quantity shall account for minimum of 10% of the installed elements for the project. The list of materials is included in activity schedule under MSH – Minimum stock holding.

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7 INSTALLATION AND INTEGRATION

7.1 General ETCS Installation

Ref: 100 - Mandatory Requirement

The Contractor shall be responsible for the installation, connection, and integration of the following equipment (please also refer to Section 7.2.2 and Section 2.2 for the division of responsibilities between the Contractor and the civils contractor): -

- Balises
- Balise ID plates (bolted) on balises
- Balise ID plates ('white plates' fixed to 'pink plates') on sleepers
- Relocation/addition of 'pink plates' should it be required
- Balise Mounting system (removal and reinstatement of IÉ rail fastenings by the Contractor's staff may be required,
- competency assessments will be carried out by the Client.
- Tail cables
- Mandatory plug connectors on one end of the tail cable to facilitate installation testing and maintenance
- Desirable connectors on both ends of the tail cable
- Cable pipes located buried to facilitate disconnection and reconnection of tail cable during maintenance
- Route from TPS location case base to disconnect box/cable containment
- Cables between location cases
- Orange duct from balise to end of sleeper (containing tail cable), fixed to sleeper with straps.
- Modification of existing location cases
 - ◆ Note the designs are specific to each existing location case. Space constraints must be considered in each design to ensure designed equipment fits as allocated See Appendix B.
- TPS location cases (Inc. bases and LEU's, components, wiring, etc.)
- Ducting between TPS location case and existing location cases
- Termination of all Field Cables including in the disconnection or junction box and TPS LOC.
- Cable Containment for Field and Tail cables to connect from chamber to Location Cases, between location cases and also from disconnect box to balise (including any roping).

Ref: 101 - Mandatory Requirement

Before the commencement of the site works, the Contractor may agree with the Project Manager a split of phases into smaller manageable lots to facilitate site access or make the delivery more efficient. These

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lots may be aligned to existing signalling scheme plans or possession maps as best suits the specific configuration of each phase and implementation strategy.

Ref: 102 - Mandatory Requirement

Balise ID plates bolted on the balises and the 'white plates' fixed to the 'pink plates' on sleepers shall be identical to the plates used in D2G project.

Ref: 103 - Mandatory Requirement

Balise mounting systems shall be of same type than the ones used in D2G project for all type of support (concrete sleeper, wooden sleeper, concrete slab). References are provided in Appendix B

Ref: 104 - Mandatory Requirement

All cables, cable containments, pipes, balise mounting products and associated installation rules shall be accepted by the Client before installation, in accordance with I-SET-5051. Samples of each of these products shall be provided to facilitate acceptance.

Ref: 105 - Mandatory Requirement

Tail cable connectors and cable pipes shall be dimensioned such that the cables and connectors can be pulled off the pipe during maintenance works.

Ref: 106 - Mandatory Requirement

TPS location cases shall be of full-width or half- width cabinet types. Pedestal types are not authorised for the TPS National Roll Out.

Ref: 107 - Mandatory Requirement

The Contractor shall be responsible for the modification of the location cases currently controlling the existing signals, in accordance with the design produced in 5.4.2.3 the objective of these modifications is to enable the interfacing with the LEUs.

Ref: 108 - Mandatory Requirement

Prior to the commencement of works on site, the Contractor shall provide evidence that the installation team is competent and have completed the minimum mandatory training requirements. Specialty evidence in relation to technical training certification, manual handling training and completion of the Client's personal track safety courses is required.

Ref: 109 - Mandatory Requirement

Four (4) weeks prior to the commencement of works on site, the Contractor shall provide, for each phase, a detailed works Method Statement for the activities to be completed by the Contractor or their representatives. Compliance with the Client and Contractor's check list will be completed. The Contractor

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shall provide evidence that insurance is in place (noting indemnity to principal in favour of IÉ & CIÉ) for the works and be in possession of a valid permit to work issued by the Client and comply fully with the requirements of IM-SMS-005 Management of Contractor's and permit-to-work system. As set out in Section 18.3, The Project Manager, in consultation and coordination with the Contractor, must complete and submit a Track Access Request (TAR) form, along with approved Risk Assessment Method Statement (RAMS) for the proposed works to the IM Track Access Coordinator a minimum of five (5) weeks in advance of the commencement of the eight-week block.

Ref: 110 - Mandatory Requirement

The Contractor shall provide for each phase a detailed installation plan to the Project Manager for acceptance twelve (12) weeks prior to commencement of works. This plan will describe the site installation activities proposed with all technical detail, planning, method, tools, logistical constraints and interfaces.

Ref: 111 - Mandatory Requirement

The Contractor shall be solely responsible for planning of installation works. This planning will require the Contractor to produce a "Box Plan" (please see appendix of template of Box Plan) for each phase that lists the items to be installed including any associated constraint, support or interdependency required to complete the successful installation. This Box Plan must be submitted to the Project Manager twelve (12) weeks prior to installation and will be used to book track access in a wide area. The box plan shall be defined on the basis of track access assumptions given in appendix Z and make use of standard possession maps available at the time of planning. For the avoidance of doubt, the Contractor shall be responsible for reviewing existing signalling drawings and documentation, analysing same and efficiently planning site testing works. The Contractor shall provide the Client with Wheels Free maps that clearly indicate the test areas that are required to be clear to all trains and machines. These maps shall be presented to the Project Manager as part of the possession planning. The extent of Wheels Free Maps shall take into consideration the Client's maintenance activities and shall only cover the area necessary for the testing on a night.

Ref: 112 - Mandatory Requirement

During Installation and for each phase, the Contractor is to maintain a register of works indicating the installation status of each component with the following minimum information:

For each controlled balise

- Junction box installed
- Balise Installed
- Cable pipe installed
- Field Cable Pulled
- Field Cable terminated
- Tail Cable Terminated
- Cable Labelled
- Cable Tested

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- Cable connected to balise

For each Balise:

- Balise ID
- Mounting system installed
- Balise Installed
- Balise Labelled
- Sleeper 'pink plate' labelled

For each existing location case

- Modification done
- Non-regression testing done

For each TPS location case/Signalling Equipment Room

- Concrete base done
- Cable duct to existing location case done
- TPS Location case/racks manufactured
- TPS Location case/racks installed
- Existing location case/racks modified
- TPS and existing location cases/racks connected (cut in)
- TPS location cases power-up

Ref: 113 - Mandatory Requirement

Following installation, the Contractor shall demonstrate conformity of the installation via appropriate forms and shall update all designs to an As-Built status that can later be used for maintenance purposes.

TPS location case requirements:

- Drawings to be available in TPS location case when installed on site (White paper, black print)
- As -Built drawings to be installed by ETCS Contractor

Existing location case (modifications):

- Check marks and modifications testing record (Yellow paper, red and green print). To be issued to IÉ TIC upon handover.
- Maintenance copy (White paper, red and green print) signed and dated when modification is complete and left in the existing location case for information. Clearly shows where connections are not terminated.

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7.2 Civils ETCS Works

7.2.1 General Requirements of the Civils works

The following section outlines a non-exhaustive list of the Contractor's general responsibilities in relation to the civils ETCS works: -

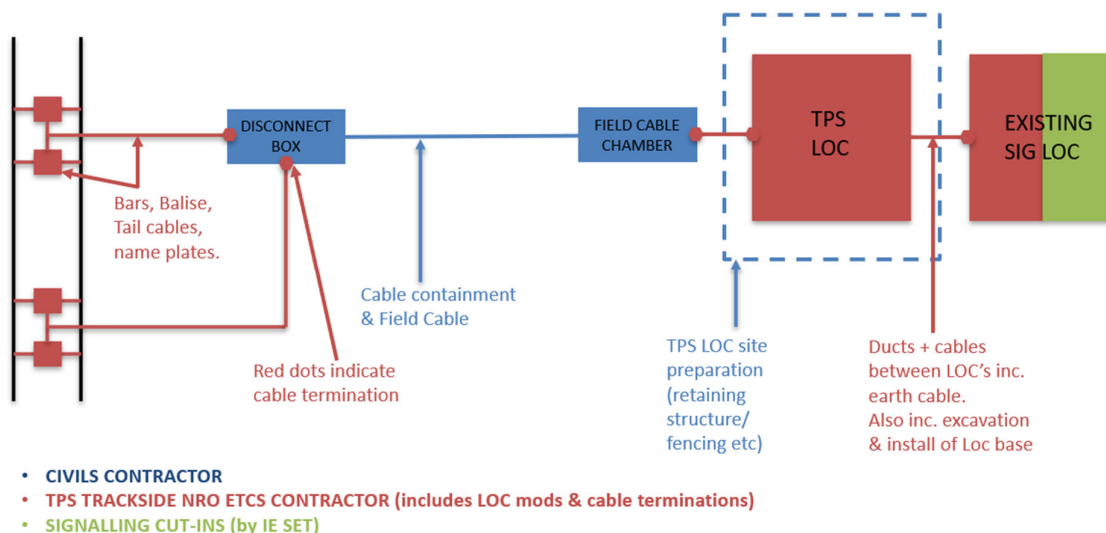
- a) The Contractor is responsible for the design and safe installation of new TPS Location Case and associated base, and all other associated works as set out on Section 7.2.2 below.
- b) Adhere to any guidance, requirements and procedures outlined in the project documents and specifications.
- c) The Contractor is obliged to familiarise himself with all details of the proposed works including the boundaries of those works.
- d) Fully plan the proposed Works at the multiple site(s) as and when required (refer to section 18.2 where No. of work sites is set out).
- e) Manage, Plan & Execute Track Possessions and Isolations of Overhead Power Lines (where applicable).
- f) Liaise and coordinate with the Project Manager and with the Client's CCE Track Access Staff to attend the works.
- g) Perform Cat & Genny surveys to determine the presence of existing cables and services, see document cable scanning in Appendix B.
- h) Produce as-installed documentation & handover sections to the Client (as described in Section 7.2.14 below).
- i) The Contractor will remove all construction materials, equipment, plant, spoil or waste from the site before the shift is completed.
- j) All construction waste to be disposed of off-site in accordance with the waste management act 1996-2011.
- k) Adhere to the general duties of a Contractor and others as outlined in the Safety, Health and Welfare at Work (Construction) Regulations 2013.
- l) Adhere to any additional safety guidance, requirements and procedures set out in the project documents and specifications.
- m) Carry out Contractor's safety tours and provide reports to the Client.
- n) Coordinate with the Client and Project Manager for joint safety tours.
- o) Community liaison where the works will generate nuisance to nearby residents.
- p) Earthing works.
- q) The excavation of TPS Location Case Base, and any associated ducting as specified in the Civils Drawing packs.
- r) Any cables to be tested will be as per I-SET-9603 - Iss.1.0 - Cable Route and Cable Installation. Results shall be recorded in Cable Testing Record Sheet.
- s) Contractor shall be responsible during the Contract for all protection of work on the Site.

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- t) Generally, works shall be required to be carried out under total possession of the railway at night; unless otherwise arranged and agreed with the relevant Client CCE Local Area Control Room.
- u) Access to the railway shall be from the access points shown on the project drawings and possession plans, as set out in Section 7.2.8 below.
- v) Access via track will require possession of the track infrastructure and will therefore be undertaken under possession (& isolation at Greystones) hours which are set out below in Section 18. The Contractor shall Manage, Plan & Execute Possessions required for the Works.
- w) Possessions are reserved via a process of work planning at control room meetings held weekly, the dates will be confirmed with the Contractor.
- x) Representation by the Contractor at possession planning meetings may be required and will be coordinated with the Project Manager.
- y) The Contractor is required to allow for the loss of a **minimum of 10% of the number & 25% of the duration of possessions booked due to emergency and/or other unforeseen occurrences**. Any costs associated with this loss of access is deemed to be included within the Contractor's Fee.
- z) The Client will provide the following protection personnel during possessions, PICOP, HS, ES and TSC. The Client will make available ES's to the Contractor in this contract for the undertaking of its works in possession, refer to Section 18.

7.2.2 The Works: Division of Responsibilities

As set out in Section 2.2 and indicated in red **Error! Reference source not found.**, a breakdown is provided in terms of the responsibilities of the Contractor (for civils works related to the ETCS installation scope), for ease of reference this diagram is provided again below.



In addition to this, a further breakdown of responsibilities is provided within the civils Implementation drawings. Please refer to Appendix B for these drawings, where the civils drawings for Phase 1 (Cherryville to Waterford) and Phase 2 (Maynooth to Sligo) have been included as part of this Contract for

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information purposes only (these works will be carried out by a separate contractor). Once the other 7 phases have been completed, these will be shared with the Contractor. A standard pack of 4 No. drawings have been produced that outline the following: -

- a) **Drawing 01:** Site Location Map with extent of works, including access points to the sites.
- b) **Drawing 02:** Ariel View of Proposed Location case site, showing local surroundings in close proximity (including associated photographs of same).
- c) **Drawing 03:** Site details, including but not limited to retaining structures, cable troughing, new location case base (and associated cases).
- d) **Drawing 04:** Extent of cable containment from new location case to associated balise.

In particular, The Contractor is directed to **Drawing No. 3**, where a “Stages and Division of Responsibilities of Works” has been provided. For completeness, the information contained in this section of the drawing is described below, where the Contractor, in particular, should pay close attention to items (i), (ii), (iv), (v), (vi), (vii), (viii) and (ix). Please also refer to Figures 6 and 7 on the follow page: -

- i) The civils contractor shall install concrete troughing, containment, (ducting where specified), retaining structures, key clamp and disconnection boxes, and backfill with class 1A backfill material up to the level of the existing ground / loc case. Sufficient space for a full width TPS location case has been shown and allowed for on this drawing for space proofing purposes, however, the specific size and the design of the TPS location cases for each site has yet to be fully determined but will be by the Contractor.
- ii) In the event that the installation of a retaining structure creates a void between the retaining structure and the track the civils contractor will be responsible for backfilling the void with class 1A backfill material up to existing ground level. The Contractor shall then excavate for the TPS location case base at a later date.
- iii) The civils contractor shall also pull field cables within new concrete troughing and leave with sufficient slack within concrete troughing or chamber in close proximity to the location of the new TPS Loc cases. The Contractor (ETCS supplier) will be responsible for ducting from the trough/chamber to the TPS LOC and between the two LOC's.
- iv) The Contractor shall be responsible for the design, procurement and installation of the Loc bases and Loc cases and will be responsible to deliver same to site.
- v) The Contractor shall be responsible to excavate to required level and install the Loc base. Loc base to be mounted on concrete blocks by the Contractor for ducting to run in under the base. In order to carry this out, the Contractor shall, as necessary, remove any key clamp fencing and reinstall same on completion of the works including earthing and bonding.
- vi) The Contractor shall be responsible to excavate and install a duct between the troughing and new loc case (as per the plan view on typical civils **Drawing 03**). The Contractor shall also pull the field cable from within the troughing to the new TPS location case. The Contractor shall be responsible for any final connections from new to existing Locs, including testing and commissioning of same.
- vii) The Contractor shall backfill around the new Loc base (as per the build-up noted in Section A-A of typical civils **Drawing 03**).

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- viii) The Contractor shall also be responsible to install the new Loc case onto the top of the Loc base.
- ix) The Contractor shall be responsible to excavate and install ducting between the existing loc case and the new loc case, including associated cabling between each. The Contractor shall be responsible for any final connections from new to existing Locs, including testing and commissioning of same.
- x) For location of the fixed balises, please refer to the finalised balise layout plans.
- xi) The Contractor will be responsible for installation of earthing cables between earth pot and the two location cases and the fencing.

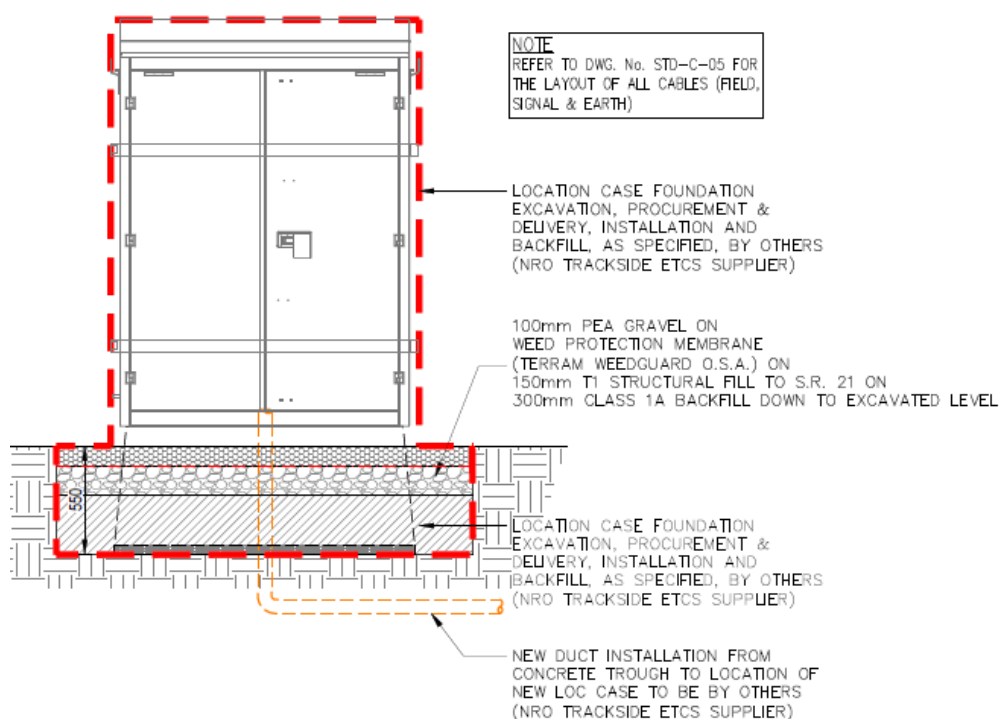


Figure 7 - Typical Ground Reinstatement, including new Location Case and associated Base

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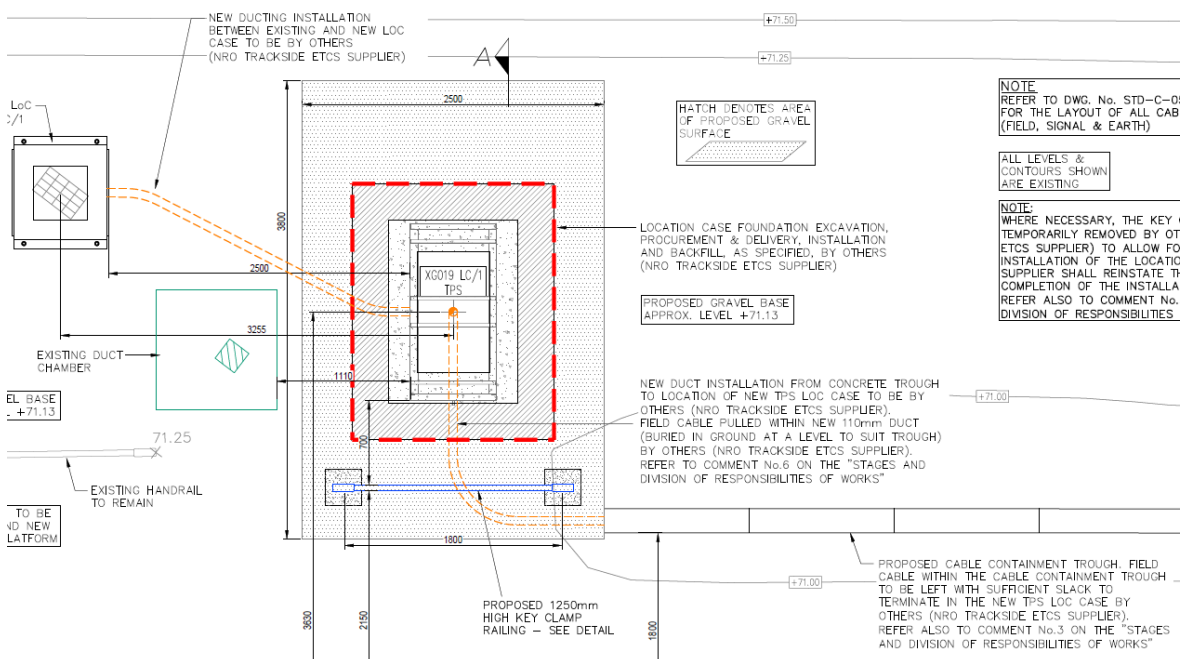


Figure 8 - Plan view of works associated with installation of new location Case

7.2.3 Excavation Works

Ref: 114 - Mandatory Requirement

Prior to any excavation works taking place, The Contractor shall perform Cat & Genny surveys & pot holing to determine the presence of existing cables and services.

Ref: 115 - Mandatory Requirement

All surplus excavated material is to be disposed of off site in accordance with relevant legislation.

7.2.4 Area for Site Establishment

Ref: 116 - Mandatory Requirement

The Contractor must note that there is limited space available within the site boundaries, and when undertaking works within the section of track during possession, the Contractor is required to use their own off-site compounds for its welfare and office set up as well as secure storage for materials, equipment and vehicles.

Ref: 117 - Mandatory Requirement

The Contractor is responsible for identifying and securing adequate areas and/or facilities outside the site boundaries. The Contractor must also arrange for all necessary permits, security measures, utility connections, and service connections required to support its work and facilities at these locations.

7.2.5 Traffic Management

Ref: 118 - Mandatory Requirement

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The Contractor is responsible for vehicle traffic and pedestrian management control for the duration of the works. This function is to be coordinated and arranged at the weekly Client's CCE Local Area Control Room.

Ref: 119 - Mandatory Requirement

The Contractor is responsible for all vehicle traffic and pedestrian/commuter management control for the duration of the works. The Contractor must submit for acceptance to the Project Manager all traffic/pedestrian management plans prior to commencement of works. The Contractor is responsible for coordination with the relevant local authority and must apply to the local authority for all road closures/road opening licenses as required.

Ref: 120 - Mandatory Requirement

The Contractor will not be in sole possession of the site during the project duration. Other Client IM Staff and Client's contractors could also utilise the site in the undertaking of ongoing maintenance requirements for the existing track and signalling systems etc. and the general public during operational use. These are in addition to specific works associated with the TPS.

Ref: 121 - Mandatory Requirement

The Contractor will be required to coordinate and accommodate his access in accordance with the possessions and isolation strategy agreed with the Client's CCE Local Area Control Room. The Contractor will be required to coordinate with other contractors well in advance of construction and ideally includes workshop meetings with all stakeholders including other contractors and relevant IE staff. In addition, access must be always maintained / given for emergency vehicles and situations.

Ref: 122 - Mandatory Requirement

The Contractor must ensure that passengers / commuters can access the platform areas and stations safely at all times. The works must be phased to accommodate this requirement. Persons with reduced mobility must be fully considered in all proposals developed. The Contractor is responsible for community liaison at instances where level crossing works may affect access to homes/farming areas.

7.2.6 Possession and Isolation Planning - Track Access Request

Please refer to section 18.2.

7.2.7 Eight week lookahead and Detailed Method Statement

Please refer to section 18.3.

7.2.8 Means of Access to the Site Works

Ref: 123 - Mandatory Requirement

As mentioned in Section 7.2.1 above, track access points have been included in the Contract for the civils contractor's Packages of Phase 1 and Phase 2. However, for all other Phases not yet complete, this information shall be shared with the Contractor.

Please see the individual TPS location case drawings for vehicular access gates closest to each of the proposed Location Case sites on the railway, Drawing No. 1 for the individual sites. Access to the linear worksite will be from these gates. Works access will be controlled by CCE track access staff and possession availability.

Ref: 124 - Mandatory Requirement

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The Contractor is required to plan works well in advance and to facilitate and coordinate communication with the relevant IÉ CCE Local Area Control Room to arrange and confirm track possession for these works. This is where nightly infrastructure works are planned and controlled between all IÉ's infrastructure departments.

7.2.9 Safety Health and Welfare at Work

The Contractor will be appointed as PSCS for the duration of their works.

Ref: 125 - Mandatory Requirement

The Contractor shall submit a contract-specific Construction Health and Safety Plan to the Project Manager for review within ten (10) working days of the contract commencement date.

This includes all works with the designated area as set out in the project drawings. These works will/may include works undertaken by Others including contractors who are employed by the Client on the TPS project.

Ref: 126 - Mandatory Requirement

The Contractor must always have a competent safety presence on site to discharge the function of the PSCS when works are underway including those works which may be undertaken by others employed by the Client.

Ref: 127 - Mandatory Requirement

All site personnel must have appropriate training and certification for the works to be undertaken, including manual handling

Ref: 128 - Mandatory Requirement

The Contractor shall comply with "IM-SMS-023 Implementation of the Construction Regulations" in relation to procedure for handover and hand back.

7.2.10 Clearance of Site

Ref: 129 - Mandatory Requirement

On completion of the works and as and when directed by the Project Manager, the Contractor shall clear away and remove from the site all construction plant, surplus materials, rubbish, debris, waste and temporary works of every kind and leave the whole of the site and works clean and in workmanlike condition to the satisfaction of Project Manager.

7.2.11 Supervisor on Site at all Times

Ref: 130 - Mandatory Requirement

The Contractor must have his supervisor on-site at all times when work is being carried out by either the Contractor's workers or his sub-contractor's workers. For the avoidance of doubt, the Contractor's supervisor must also be on-site when work is being carried out solely by their sub-contractor.

7.2.12 Control of Sub-Contractors

Ref: 131 - Mandatory Requirement

All proposed sub-contractors shall be approved by the Project Manager.

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Ref: 132 - Mandatory Requirement

The quality system shall include all works executed by sub-contractors, manufacturers and/or suppliers and the Contractor shall ensure that any sub-contractor adheres to and complies with the necessary procedures and controls required and outlined in the Contract documentation inclusive of drawings, specifications, guidelines and procedures.

7.2.13 Snagging Inspections

Ref: 133 - Mandatory Requirement

The works shall be complete in all respects at the dates identified in the Contractor's master programme for snagging of each section of the works and prior to Completion. The Contractor shall arrange for and complete any snagging in the presence of the Project Manager (or his designated person), along with any relevant Client Staff as deemed necessary.

Ref: 134 - Mandatory Requirement

A list of items requiring corrective action will be jointly prepared by the Project Manager and the Contractor immediately following the inspection. No area shall be declared ready for snagging when elements of work are incomplete. Following completion of rectification of all snags and in advance of Completion being certified by the Project Manager, the Contractor shall arrange for a final pre-inspection to be undertaken, and all areas cleaned and ready for the Client (and relevant Client staff deemed necessary) final inspection.

Ref: 135 - Mandatory Requirement

No later than the date designated for snagging as set out above, the Contractor shall provide all necessary test certificates for the relevant part or section of the Works and other documents required under the Contract for Completion to be achieved.

7.2.14 As-Built Documentation / Safety File

Ref: 136 - Mandatory Requirement

The Contractor shall provide a Safety File in accordance with the requirements set out in Section 33.

Ref: 137 - Mandatory Requirement

An electronic copy of draft As-Built drawings and Safety file information to be submitted to Project Manager (or his nominated person) for acceptance per location case site. This process is to continue as required taking the comments on board until a satisfactory quality of documentation is received. When the draft is accepted by the Project Manager, then the Contractor will be required to submit the final version to the Project Manager.

Ref: 138 - Mandatory Requirement

The as-built drawings shall be submitted to Project Manager in the same format type that they were provided (e.g. MicroStation) or marked-up pdf as agreed with the Project Manager. All constructed/installed works must be detailed and surveyed by the Contractor and represented accurately on the as-built drawings submitted to Project Manager (or his nominated person).

On completion of the works, the Contractor is required to undertake a survey of the following features:

- a) Start and end points of different containment types and different offsets from the running edge
- b) Ducting and chambers
- c) New TPS location cases

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- d) Existing location cases which form part of the works site.

Ref: 139 - Mandatory Requirement

This survey shall be AutoCAD dxf / DWG file compatible (or alternative format agreeable to the Client). All features shall be easily identifiable and named within the survey with descriptive layer/level naming. Surveyed heights (levels) shall be orthometric and quoted in metres above Malin Head (OGM15) and the survey coordinate reference system shall be to Irish Transverse Mercator (ITM).

Under exceptional circumstances, a clear hand marked version of an original paper copy of the drawing may be submitted by the Contractor in both paper and (high resolution scanned in PDF) electronic versions. However, this is by exception and pre-agreement with the Project Manager only.

Please also refer to Section 14 Completion.

7.2.15 Documentation in General

Ref: 140 - Mandatory Requirement

The Contractor shall maintain records to substantiate conformance to specified requirements, including work carried out by sub-contractors, Manufacturers, Suppliers and independent inspecting authorities.

Ref: 141 - Mandatory Requirement

The records shall include but not be limited to:

- a) Identification of the element, item, batch or lot.
- b) The nature and number of observation tests and monitoring.
- c) The number and type of deficiencies found.
- d) Details of any corrective action taken or proposed.
- e) Testing performed on elements or materials.

Ref: 142 - Mandatory Requirement

Any records which indicate that material or workmanship in any part of the works does not comply with the specified requirements or is defective or non-conforming shall be submitted to the Project Manager without delay together with the Contractor's proposals for rectification.

Ref: 143 - Mandatory Requirement

All records shall be retained on site and made available for examination. On completion of the works, a copy of all records shall be provided to the Project Manager in the Safety File unless directed otherwise.

Ref: 144 - Mandatory Requirement

The Contractor shall provide for the use of, and documentation compatibility with, cloud-based shared access systems, such as SharePoint, as hosted by the Project Manager and/or the Client.

Ref: 145 - Mandatory Requirement

The Contractor shall keep accurately dated records of the progress of the works. The detail and format to be defined by the Client. This will include but not be limited to quantities, meterage of units installed and their geographic location. The use of electronic devices for data collection and recording may be considered.

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7.2.16 PTS Training

Ref: 146 - Mandatory Requirement

The Contractor is responsible for protecting existing facilities against damage and ensuring that their works do not impact the safe operation of the railway.

Ref: 147 - Mandatory Requirement

All personnel on the project must have current IM PTS Training and Contractor must be fully CI-SMS-005 compliant. This applies to all the Contractor's staff, sub-contractors and sub-sub-contractors etc. The Contractor must allow a cost of €300.00 per person in its tender for this PTS training, payable to IÉ Training Centre if current employees/sub-contractors are untrained or previous PTS trained person's certification has expired.

7.2.17 Safe-Pass Training

Ref: 148 - Mandatory Requirement

All personnel on the project must have current SAFE PASS Training in accordance with the Safety Health and Welfare at Work (Construction) Regulations 2013 Safe Pass / Safety Awareness programme. It is the Contractor's responsibility to ensure that all planned site staff have undertaken the necessary training and received certification in advance of being required on site.

7.2.18 Manual Handling

Ref: 149 - Mandatory Requirement

All persons accessing Iarnród Éireann work sites are required to undertake and provide certification for manual handling. Evidence of same shall be passed to the Project Manager prior to works commencing.

7.2.19 Environmental Plan

Ref: 150 - Mandatory Requirement

The Contractor will minimise dust, smoke, odour, noise, vibration and impacts on Ecological Sensitive Areas. The Contractor's attention is brought to the proximity of the site to residential/ commercial areas, ecologically sensitive areas (treelines, hedgerows and/or other drainage conduits or watercourses) and the elevated nature of the site. The Contractor shall undertake all construction activities according to good working practice to minimise neighbour nuisance impacts from dust, smoke, odour, noise and vibration arising from construction activities and negative impacts on ecologically sensitive locations.

Ref: 151 - Mandatory Requirement

All contractors working on behalf of Iarnród Éireann are obliged to comply with the provisions of all applicable national and European environmental legislation.

Ref: 152 - Mandatory Requirement

All complaints received by the contractor related to the environment shall be forwarded to the client as soon as possible. Any corrective actions agreed upon with the complainant shall be actioned out on site by the contractor as instructed by the client. All records relating to environmental complaints will be retained by the contractor and the client.

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7.2.20 Noise and Vibration Mitigation Measures

Ref: 153 - Mandatory Requirement

In terms of Noise and Vibration, the Contractor shall implement mitigation measures, in accordance with BS 5228-1&2:2009 (Noise and Vibration Control on Construction and Open Sites) to minimise noise and vibration impacts. When working at night, the Contractor must ensure that its employees and plant operators make every effort to minimise noise emissions.

Ref: 154 - Mandatory Requirement

The Contractor will adhere to all guidelines and requirements outlined in BS 5228 and ensure that noise threshold limits are not exceeded during construction activities as provided in BS 5228.

Ref: 155 - Mandatory Requirement

The Contractor shall incorporate noise and vibration mitigation measures, including (inter alia) the following:

- a) Minimise the noise generated by construction activities through planning and scheduling of work activities.
- b) Provide instructions to site workers and vehicle haulage drivers on work schedule and precluded noise generating practices.
- c) Refrain from unnecessary noise-generating activities/actions, through the use of equipment, horns, sirens, etc. (unless safety critical).
- d) Refrain from utilising high noise generating saws, drills & hammers etc. unless necessary. In certain circumstances where this cannot be avoided, the Contractor is to utilise all means to ensure that only the minimum noise is generated, by the use of, where possible and safe, super silenced equipment, enclosure within proprietary acoustic enclosures, off-site assembly of materials, etc.
- e) Maintain plant and machinery in good working order.
- f) Position noisy plant and equipment as far as possible from neighbouring properties.
- g) Utilise, where possible, suitable acoustic enclosures/barriers/blankets for plant and equipment in continuous use.
- h) Shut down or throttle down plant and machinery when not in use.
- i) Ensure that loading and unloading of haulage vehicles is not conducted during night-time periods.
- j) Temporary barriers will be erected around noisy items such as generators or compressors where required.
- k) Construction plant will be selected for low potential for generating noise.

7.2.21 Dust Mitigation Measures

Ref: 156 - Mandatory Requirement

Dust prevention measures shall be included to reduce airborne particulate pollution during construction activities. Dust suppression shall be used where prevention is not feasible. The Contractor shall

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implement and adhere to best practice guidance from Ireland, the UK ((IAQM 2014), BRE (2003), The Scottish Office (1996), UK ODPM (2002)) and the USA (USEPA, 1997).

The Contractor shall incorporate air quality and dust emission mitigation measures, including (inter alia) the following:

- a) Burning of C&D material is illegal and prohibited at all site compounds and work locations.
- b) Maintain plant and machinery in good working order to ensure exhaust emissions are minimised.
- c) Employ, where possible, local screening when drilling/cutting concrete surfaces.
- d) Maintain a clean site compound/work location to minimise agitation of soiled surfaces.
- e) Store surplus spoil arising in appropriate areas and export when possible to minimise the potential for wind blow-off.
- f) Tarpaulin covers are required over stockpiles when high wind and dry weather are forecasted.
- g) Spray or dampen exposed surfaces during prolonged dry periods.
- h) Where applicable, tree and hedgerow protection measures will be provided for all trees and hedgerows to be retained in accordance with BS: 5837:2012. Trees in relation to design, demolition and construction. Work to remove trees or the demolition of existing structures on the site will take place outside of bird nesting season.
- i) Vehicles carrying fine materials will remain sheeted/covered for as long as practicable.

7.2.22 Ecological Mitigation Measures

Ref: 157 - Mandatory Requirement

The Contractor shall incorporate ecological mitigation measures, including (inter alia) the following:

- a) Sediment & Water Pollution Control: All works carried out as part of this contract will comply with all Statutory Legislation including the Local Government (Water Pollution) Act 1998 and 1990. Site drainage shall not be allowed to discharge to any watercourse.
- b) Provide temporary bunded areas or proprietary bunded containers for storage of all potentially polluting liquids, solvents, paints, etc.
- c) Maintain plant and machinery in good working order and inspect regularly for leakages of oils, lubricants or fuel.
- d) All refuelling should be conducted away from surface water gullies, drains or water bodies and refuelling / onsite maintenance shall incorporate the use of a drip tray.
- e) Spill kits and hydrocarbon absorbent packs shall be maintained and available for use within the site.
- f) A spill response plan shall be developed by the Contractor and all relevant site personnel shall be trained in preventative and corrective actions.
- g) Where made ground, soil and subsoil are excavated, the spoil shall be segregated into various waste fractions.

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- h) Run-off or leached water from spoil shall be managed to prevent discharges to water bodies or surface water drains.

Note: The Article 6 (3) Appropriate Assessment Screening Report, and Ecological Impact Assessment Report by MKO Consultants. These reports are included in Appendix B.

7.3 Hand Over

Ref: 158 - Mandatory Requirement

Once the installation of the TPS location case/racks and the modification of the existing location case/racks are certified by the Contractor, the Client will take responsibility for connecting the ETCS to the existing lamp circuits and undertake non-regression testing of the existing signalling system. During this 'cut-in' activity, the Contractor shall make competent staff available to assist the process on site.

Ref: 159 - Mandatory Requirement

Once the Cut-ins are finalised and the non-regression tests are completed, the Signal Location Case and TPS location will be handed over to the Contractor such that they can perform Test and Commissioning activities.

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8 TESTING & COMMISSIONING

8.1 General

Ref: 160 - Mandatory Requirement

The Contractor shall be responsible for providing competent staff throughout all Tests and Commissioning and any agreed post commissioning contingency cover to issue modifications required by testing activities. A record and certification of competence of staff shall be provided on request by the Project Manager.

Ref: 161 - Mandatory Requirement

The Contractor shall provide a Test and commissioning strategy to the Project Manager no later than twelve (12) weeks from contract commencement date. This strategy shall describe the different type of tests (laboratory, static, and dynamic) that are required to be undertaken by the Contractor to meet their internal requirements and their contractual obligations. This strategy shall describe the objectives, test environment and logistics, tests tools, and test resources.

Ref: 162 - Mandatory Requirement

The Test and commissioning strategy shall clearly indicate the tests that are required for the validation of the Generic Application and the test that are required for the validation of each Specific Application.

Ref: 163 - Mandatory Requirement

The Test and commissioning strategy shall demonstrate that all foreseeable operational scenarios are covered by test cases for all applicable trackside configurations.

Ref: 164 - Mandatory Requirement

The Test and commissioning strategy shall define a list of test cases that demonstrate both safety and performance of the system.

Ref: 165 - Mandatory Requirement

The Test and commissioning strategy shall reduce as much as possible the needs for accessing sites in a possession regime and the reliance on Client staff to perform the testing activities.

ETCS T&C shall rely as much as possible on laboratory testing compared to site testing.

Ref: 166 - Mandatory Requirement

The Test and commissioning strategy shall maximise the amount of laboratory testing compared to dynamic testing on site with a train. It shall demonstrate that each dynamic onsite test scenario cannot be achieved in a laboratory environment.

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Ref: 167 - Mandatory Requirement

The Contractor shall demonstrate compliance of the CCT system with the trackside testing requirements of CCS TSI, including item 6b of Table 6.3 related to the Operational Test with the onboard of two (2) different ETCS onboard suppliers. Note that this may be in real environment with two different Onboard suppliers, or partially done in a certified laboratory if the onboard subsystem of only one supplier can be supplied by The client. In this case the supply and configuration of the certified laboratory environment shall be under the responsibility of The Contractor.

Ref: 168 - Mandatory Requirement

The Contractor shall provide, for each Phase, a Test & Commissioning Plan (T&C Plan) to the Project Manager no later than twelve (12) weeks prior to the commencement of a specific testing activity for a dedicated area. The Test & Commissioning Plan shall include the Contractor's approach to the following:

- Laboratory Testing
- Static Testing
- Dynamic Testing

Twelve (12) weeks prior to the commencement of Static or Dynamic testing, the plan shall be updated to include details of the proposed tests, method statement, wheels free requirements and running test maps indicating the areas where the tests are proposed to take place. Where possessions are required to undertake activities, standard possession maps available at the time of planning shall be used by the Contractor. The Contractor shall provide all necessary information in order to allow the Client's Safety Staff to create and approve any necessary derogations that might be required for testing. Examples of the type of Derogations and method statement needed can be found in Appendix B. For the avoidance of doubt the Contractor shall be responsible for reviewing existing signalling drawings and documentation, analysing same and efficiently planning site testing works. The Contractor shall provide the Client with Wheels Free maps that clearly indicate the test areas that are required to be clear to all trains and machines. These maps shall be presented to the Project Manager and Client as part of the possession planning. The extent of Wheels Free Maps shall take into consideration the Client's maintenance activities and shall only cover the area necessary for the testing on a night.

Ref: 169 - Mandatory Requirement

The Contractor shall Define in the test plan the exact quantity and qualification of human resources (Contractor's, Client's or third party) and tools that will be mobilised for each test shift and demonstrate that the works can be achieved with this resources plan.

Ref: 170 - Mandatory Requirement

The Contractor shall provide Test & Commissioning test cases and forms (T&C Plan) to the Project Manager no later than four (4) weeks prior to the commencement of a specific testing activity. The Test & Commissioning Test cases and forms shall detail the tests context, starting condition, steps, observable, pass/fail criteria for each of the:

- Laboratory Testing
- Non-regression testing after cut ins
- Static Testing

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- Dynamic Testing

Ref: 171 - Mandatory Requirement

The Contractor shall provide an individual Test & Commissioning approved form for each test shift to the Project Manager no later than two (2) weeks after the end of the shift. This applies to:

- Laboratory Testing
- Static Testing
- Dynamic Testing

Ref: 172 - Mandatory Requirement

The Contractor shall provide, for each phase (or sub-phase), a final and validated Test & Commissioning report to the Project Manager no later than four (4) weeks after the end of the test. When a test session lasts more than four (4) weeks, the report shall be split into several controlled submissions to ensure it is never delivered to the Project Manager no more than eight (8) weeks after the commencement of the session. This applies to:

- Laboratory Testing
- Static Testing
- Dynamic Testing

8.2 Laboratory Tests

Ref: 173 - Mandatory Requirement

The Contractor shall provide a lab testing environment in Geographical Europe which makes use of a real or simulated EVC that is fully compliant, for the purpose of the laboratory testing, with the requirement of the IRS-304.

Ref: 174 - Mandatory Requirement

In the Test & Commissioning strategy, the Contractor shall clearly detail the limitation of the lab testing environment and what test shall occur with an ETCS fitted train to complement these limitations.

Ref: 175 - Mandatory Requirement

The Contractor shall perform laboratory testing to validate safety and performance of the system delivered.

Ref: 176 - Mandatory Requirement

The Contractor shall clearly indicate in the test strategy which test cases are safety related and required for the completion of the safety case.

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Ref: 177 - Mandatory Requirement

The Contractor shall perform laboratory testing that ETCS fitted trains can operate without degradation of performances in normal foreseeable condition of operations. In particular the test shall demonstrate:

- No entry in indication curves when consecutive green aspects are displayed to the driver
- No expiry of the section timer in operational condition as specified in the TPS Common Application Requirement document
- The correct positions of the infill balises ('green to green', "turnout" and "approach control" infill balises)

Ref: 178 - Mandatory Requirement

The Contractor shall perform laboratory testing and fix all errors detected during the laboratory testing before the dynamic testing can commence.

Ref: 179 - Mandatory Requirement

The Contractor shall use the Laboratory testing environment as much as possible to demonstrate compliance of the trackside with the requirement of CCS TSI, including item 6b of Table 6.3 related to the Operational Test with the ETCS onboard system on two different ETCS suppliers.

Ref: 180 - Mandatory Requirement

The Contractor shall always invite the Project Manager to participate to all laboratory testing sessions with a minimum of eight (8) weeks' notice.

Ref: 181 - Mandatory Requirement

The Contractor shall set up the laboratory testing environment such that it reflects as much as possible the real environment. In particular:

- The train acceleration and braking performance shall be adjustable.
- The Balise antenna position shall be adjustable
- The onboard odometry inaccuracy shall be adjustable from 0% to up to 10%
- The ETCS onboard parameters shall also be configurable to reflect any fleet potentially operating on the Client's network.
- The gradient used for simulating the trackside shall be as close as possible to the real gradient and not be the simplified gradient configured in the LEU and Balise.
- The balises not destined to the onboard (not in the linking chain) shall be included in the trackside description.
- The exact position of each balise of all balise group shall be included in the trackside simulation

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- Each telegram of each balise group shall be readable by the onboard unit independently from the other telegrams of the same balise group.

Ref: 182 - Mandatory Requirement

The Contractor shall perform laboratory testing to validate Generic Application standard functional and hardware configurations, including interface with the lamps circuits. .

8.3 Static Tests

Ref: 183 - Mandatory Requirement

The Contractor shall undertake all tests that are required to validate the correct installation, and integration of the ETCS trackside component with the infrastructure. That includes verification of the exact position of the equipment.

Ref: 184 - Mandatory Requirement

The Contractor shall undertake all tests that are required to validate the correct telegrams are sent for each possible signal aspect.

Ref: 185 - Mandatory Requirement

The Contractor shall minimise the dependence on the Signaller to change signal aspect during aspect to telegram correspondence testing, though the use of the 'white box' that enables aspect changes at local level. Please see example, specification and drawings of interface box in Appendix B. This 'white box' shall be used wherever and whenever it is possible to do so. In particular it is not possible to test early generation of SSI signal module with the 'white box'. Early generation SSI modules can be identified via photograph provided by the Client as part of a survey data.

Ref: 186 - Mandatory Requirement

In the event where the white box cannot be used, the Contractor shall ensure that each aspect of the signal tested is only requested once to the Signaller, i.e. if several control balises are connected to the same signal (e.g. infill balises), they shall be tested at the same time, ensuring sufficient Contractor's staff and tools in place to do so.

Ref: 187 - Mandatory Requirement

The Contractor shall be responsible for appointing a qualified person in charge of the static tests which can communicate with the Client's staff (PICOP, SET, signaller) during the execution of the shift. At the end of each shift this person in charge of the test shall deliver via email or web-based tool a summary and outcome of the shift.

Ref: 188 - Mandatory Requirement

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The Contractor shall be responsible for appointing a person that is available during the daytime and capable of giving detail reporting of the previous static test night shift and replanning the next night shift if necessary.

Ref: 189 - Mandatory Requirement

The Contractor shall provide the Project Manager a weekly progress update detailing what equipment (location case, LEU, Balise) has been tested, what equipment has outstanding issues (detailing the issue and action plan and tracking) and what equipment has not been tested. This tracker shall be visible to the Project Manager at all times.

Ref: 190 - Mandatory Requirement

The Contractor shall be solely responsible for planning of the static testing works. This planning will require the Contractor to produce a "Box Plan" that lists the items to be tested including any associated constraint, support and staff protection requirement or interdependency required to complete the successful test. This Box Plan must be submitted to the Project Manager twelve (12) weeks prior to testing and will be used to book track access in a wide area. Following on from the Box Plan the Contractor shall determine the exact extent of wheels free requirements. This will require the Contractor to study the Client's signalling system and produce detailed plans based on the Client's possession maps that allow track access to be planned efficiently, these "Wheels Free Maps" will be used by the Client to plan track access in detail for specific worksites. The Client's possession map and Box Plan examples can be found in Appendix B. The box plan and wheels free map shall be defined on the basis of track access assumptions given in Appendix B, including but not limited to, possession maps available at the time of writing the plan.

Ref: 191 - Mandatory Requirement

The Contractor shall present and explain the logic of the 'box plan' and associated test plan and method statement twelve (12) weeks prior to the commencement of any site activity. The Contractor shall be able to introduce reasonable changes to the box plan in a flexible manner as track access depends on railways conditions which are not always under the control of the Client.

Ref: 192 - Mandatory Requirement

The Contractor will resource the static testing activities so as to maximise progress during track access periods.

Ref: 193 - Mandatory Requirement

The Contractor shall ensure that the number of working test tools are available to perform the planned tests in the most efficient way and in accordance with the programme.

Ref: 194 - Mandatory Requirement

In addition to the Test report mentioned in Section 8.1, the Contractor shall produce the following individual test certificates as evidence that the following equipment have been successfully tested:

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- Cables
 - Balises
 - Location cases
 - LEU

8.4 Dynamic Tests

Ref: 195 - Mandatory Requirement

The Contractor shall complement laboratory testing and static testing with on-train dynamic tests, making use of any Client train fitted with ETCS. Note that the only accessible log will be the train event recorder and the ETCS Juridical Recorder. No change of ETCS onboard configuration will be possible.

Ref: 196 - Mandatory Requirement

The contractor is to minimise the number of dynamic tests using the Clients ETCS fitted train.

Ref: 197 - Mandatory Requirement

As a minimum the dynamic test cases shall include the validation of each generic function when they are first instantiated on one of the phases.

Ref: 198 - Mandatory Requirement

As a minimum the dynamic test cases shall also include for the first phase:

- All single route, i.e. all main or shunt routes of one block signalling ending on red signal, a limit of shunt board or a stop board.
- All Level transitions.
- All Green-to-green scenarios, i.e. proceeded on consecutive green aspect to ensure that no braking indication is displayed.
- All routes required to validate the correct enforcement of static speed profile.

As confidence grows in the accuracy of the platform testing, it may be possible to not undertake all these test cases for the subsequent phase. In the event where the dynamic test proves that the platform validation is not reliable enough to remove the needs for complete dynamic testing. The Project Manager is entitled to request all test cases to be performed in the real environment, with a test train, at the Contractor's own expense.

Ref: 199 - Mandatory Requirement

The Contractor shall be responsible for producing the dynamic test method statement, requesting, and explaining the associated need for derogation to the Client's procedure and safety management system, and requesting and explaining the associated needs for specific operational arrangement.

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Ref: 200 - Mandatory Requirement

The Contractor shall be responsible for allocating a qualified person in charge of the tests which can communicate with the driver and the other Client staff (PICOP, signaller) during the execution of the shift. At the end of each shift this person in charge of the test shall deliver via email or web-based tool a quick summary and outcome of the shift.

Ref: 201 - Mandatory Requirement

The Contractor shall be responsible for appointing a person that is available during the daytime and capable of giving detail reporting of the previous dynamic test night shift and replanning the next night shift if necessary.

Ref: 202 - Mandatory Requirement

The Contractor shall be solely responsible for planning of the Dynamic testing works. This planning will require the Contractor to produce a 'Dynamic test Sequence' that lists for each propose shift the possession map required to be booked, the sequence of transit movements and test cases for that shift - and for each test case , their start and end locations, their estimated durations, and any special logistic arrangement that are needed (e.g. Level Crossing controllers, shunters, signalling maintainers, wheels free tracks). The associated test plan shall include all items to be tested including any associated constraint, support or interdependency required to complete the successful test. The sequence and plan shall be submitted as a draft to the Project Manager twenty-four (24) weeks prior to testing and shall be finalised twelve (12) weeks prior to the start of testing so that track access can be booked in advance and planned efficiently. Possession map examples can be found in Appendix B. The 'test sequence s' shall be defined on based on track access assumptions given in Appendix B, including but not limited to, possession maps available at the time of writing the plans.

Ref: 203 - Mandatory Requirement

The Contractor shall be able to introduce reasonable changes to the 'dynamic test sequences' plan in a flexible manner as track access depends on railways conditions which are not always under the control of the Client.

Ref: 204 - Mandatory Requirement

The Client reserves the right to organise and undertake additional dynamic tests that may condition the final acceptance of the system. These additional 'acceptance tests' shall not drive the Contractor Safety Approval Process.

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9 MAINTENANCE

Ref: 205 - Mandatory Requirement

The Contractor shall deliver the updates to the existing Irish Rail Maintenance standards and create specific Irish Rail Maintenance Manual to incorporate the ETCS CCT system delivered under the Contract. Current Irish Rail maintenance standards, of manuals are provided as example in Appendix B

Ref: 206 - Mandatory Requirement

The Contractor shall deliver all tools necessary for the Client to maintain the system and performance at least the following tasks:

- Fault finding and diagnostic of any failure
- Replacement of any components
- Uploading/programming of LEU and Balise with configuration files
- Performing software updates for LEUs and other CCT components
- Exporting and archiving configuration and diagnostic data for documentation and traceability

Ref: 207 - Mandatory Requirement

The programming tool for LEU and Balises shall be able to store all or only portions of the whole NRO trackside configuration data. The Client shall be able to adjust the amount of configuration data stored in each programming tool as per site specific and maintenance organisation constraints.

Ref: 208 - Mandatory Requirement

The programming tool for LEU and Balises shall be able to report unambiguously to the user which version of configuration data is stored. This version shall correspond to the version indicated in the hand over certificate.

Ref: 209 - Mandatory Requirement

The programming tool for LEU shall be able download event log from LEU to monitor and investigate failures of the LEU.

Ref: 210 - Mandatory Requirement

The Contractor shall deliver maintenance training as specified in section 16.5.

Ref: 211 - Mandatory Requirement

The Contractor shall deliver maintenance training before completion and hand-over or the first phase to the Client. The training should be planned and completed in parallel with APIS 5 assurance approval for any given phase.

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Ref: 212 - Mandatory Requirement

The Contractor shall deliver a typical ETCS installation in the IÉ Thurles training facility prior to delivering the maintenance training. The layout for this installation is provided in Appendix B.

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10 ENGINEERING MANAGEMENT

10.1 General

Ref: 213 - Mandatory Requirement

The Contractor shall deliver a set of engineering management documents which demonstrates compliance of the Contractor with the engineering management requirements of TSI CCS, CENELEC 50126, CRR, IE PMP 001 to 010, and I-SET-9100. This set of engineering management plan shall cover but not be limited to the following streams of activities:

- Engineering Organisation chart and responsibilities
- Competence Management
- Engineering life cycle description
- Requirement Management
- Technical Issues Management
- Technical Queries and Request for Information
- Interface Management
- Configuration and Change Management
- Design Management
- Configuration Data Production
- Verification and Validation Management
- Installation Management
- Test & Commissioning Management
- Safety Management
- Quality Management
- RAM Management
- Human Factors Management
- EMC Management

Ref: 214 - Mandatory Requirement

The set of engineering management plans shall be delivered within eight (8) weeks of the contract commencement date and shall be submitted to the Project Manager for review and acceptance.

Note: The Contractor may commence the engineering works at their own risk, before acceptance by the Project Manager of the set of engineering management plans.

Ref: 215 - Mandatory Requirement

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The Contractor shall give full visibility to the Project Manager of existing/ potential issues, risks or change requests related to the project. In particular, the Project Manager shall be invited, with four (4) weeks' notice, to attend any of the Contractor technical gateway reviews related to the project.

Ref: 216 - Mandatory Requirement

The Contractor shall provide within eight (8) weeks of the contract commencement date any table, form, or templates applicable to the project, including but not limited to:

- Document Review Form
- Engineering Change Request Form
- Technical issues Register
- Generic Design Templates
- Functional Design templates
- Installation templates
- Test and commissioning test cases, test form, and test reports templates.
- Requirement matrixes templates
- Request for Information
- Asset Register
- Baseline Register

Ref: 217 - Mandatory Requirement

The Contractor may commence the engineering works at their own risk, before acceptance by the Project Manager of the set of templates.

Ref: 218 - Mandatory Requirement

The Contractor shall organise weekly technical meetings of 2 to 4 hours duration to follow-up the progress of technical deliverables, and follow-up the resolution of technical issues. The Contractor shall transmit the minutes of these meeting within two (2) working days after each occurrence of the meeting.

Ref: 219 - Mandatory Requirement

The weekly minutes of meeting shall reflect all technical aspect of the contractor's scope of work. In addition to accurate reporting of the discussion held and the actions decided during the meeting; the minutes shall include as a minimum:

- Status of inputs required from Irish Rail
- Status of each technical and safety deliverables for each phase, including generic application
- Progress on open issues
- Status and progress of change requests
- KPIs on requirement, issues, CRs

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- Status of risks and newly identified risks

10.2 Requirements Management

Ref: 220 - Mandatory Requirement

The Contractor shall provide to the Project Manager, within eight (8) weeks of the contract commencement date, a Requirements Management Plan that demonstrates how their project and company engineering management framework comply with EN50126 and best practices regarding Requirements Management.

Ref: 221 - Mandatory Requirement

The Contractor's Requirements management plan shall demonstrate how the Client requirements from this Scope and the Client requirements referred to in section 4 are analysed, refined into Contractor's own requirement or specification, and how compliance to these requirement is demonstrated throughout the life cycle of the project. The requirements management plan shall be delivered together with the Database structure and the templates and examples for the Requirement Matrices and Specifications that will be produced during the life of the project.

Ref: 222 - Mandatory Requirement

The Contractor shall deliver to the Project Manager, the Requirements database in Excel and IBM Doors formats.

Ref: 223 - Mandatory Requirement

The Requirements Specification produced by the Contractor shall clearly indicate the requirements numbers, attributes and text of the requirement within the body of the document. It is not acceptable to receive a specification delivered in tabular format only. When creating the requirements specification, the requirements of EN 50126-1 must be taken into account (e.g. must be complete, precise, unambiguous, verifiable, testable and maintainable)

224 - Mandatory Requirement

The Specification produced by the Contractor shall be delivered in .pdf and .doc formats. Each new release of the specifications shall also include the .pdf and .doc of the new release without track changes, and a .pdf and .doc of the new release with the track changes compared to the previously released version. The track changes shall show added text in blue and remove text in strikethrough red font. Intermediate revisions of the specifications submitted to the Project Manager for review shall also indicate the changes compared to the previous version using the same colour code mentioned above.

Ref: 225 - Mandatory Requirement

The Client requirements set includes optional requirements. If these requirements are agreed between the two parties to be implemented at contract signature or at the early development stage of the project, it becomes an obligation for the Contractor to implement these requirements.

Ref: 226 - Mandatory Requirement

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The Contractor shall provide and continuously manage a Client Requirement Matrix, which indicate for each Client requirement a compliance statement as follow:

- Full compliance
- Partial compliance
- Non-compliance

Ref: 227 - Mandatory Requirement

The Contractor shall propose an alternative requirement or set of requirements for any non-compliant or partially compliant statement. Such derogation shall be formally requested to the Client in the form of written and controlled justification which shall describe qualitatively and quantitatively how the alternative solution can be acceptable to the Client considering: Safety, performance, Cost, Operation and Maintainability. The Client has no obligation to accept the derogation request. Note that derogation requests to any IRS-305 mandatory requirement cannot be accepted.

Ref: 228 - Mandatory Requirement

The Contractor shall provide the Client's requirement compliance statement supported with precise references to V&V and deliver it at each key engineering milestone of the project:

- Within Four (4) weeks of the contract commencement date
- At completion of the Generic Application Design
- At completion of each Phase Specific Application – Functional Design
- At completion of each Phase Specific Application Test and commissioning

10.3 Technical Issue Management

Ref: 229 - Mandatory Requirement

The Contractor shall provide to the Project Manager, within eight (8) weeks of the contract commencement date, a Technical Issues Management Procedure that explains how their project and company engineering management framework identify, categorise, and address development, design and implementation issues.

Ref: 230 - Mandatory Requirement

The Contractor shall track in an Issues Register, all technical issues potentially impacting the design or implementation of the project. The Issues Register shall be submitted to the Project Manager at the end of each period and shall clearly identify what actions and decisions have been made since the previous period to progress the resolution of the issues.

10.4 Configuration Management

Ref: 231 - Mandatory Requirement

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The Contractor shall provide to the Project Manager, within eight (8) weeks of the contract commencement date, a Configuration Management Plan that demonstrate how their project and company engineering management framework comply with EN50126 and best practices regarding Configuration Management, in particular:

- Engineering Baseline Management, including template for Baseline Register and Baseline release report.
- Engineering Change Request (ECR) Management, including template for an Engineering Change Request form and Change request register.
- Configuration Items management, including template for asset register.

Ref: 232 - Mandatory Requirement

The Contractor shall continuously maintain a baseline register, based on engineering documentation evidence, that describes the scope, limitations, requirement compliance and non-compliance statement, and issues associated with the current Generic application and Specific Applications baselines. The Baseline Register shall be submitted to the Project Manager at the end of each period and shall clearly identify the change compared to the previous period.

Ref: 233 - Mandatory Requirement

The Contractor shall continuously maintain a baseline plan that describes the future baselines, their release dates, their scope and the engineering change requests that are planned to be implemented for each of them. The Baseline Plan shall be submitted to the Project Manager at the end of each period and shall clearly identify the change compared to the previous period.

Ref: 234 - Mandatory Requirement

The Contractor shall continuously maintain an engineering change request register plan that describes all change requests applicable the project delivery scope, include product, generic application and specific applications. The engineering change request register shall indicate for each ECR:

- The origin/source of the ECR
- The analysis of the ECR
- The Resolution of the ECR
- The target baseline in which the ECR is planned to be implemented.
- Status of the ECR

The engineering change request register shall be submitted to the Project Manager at the end of each period and shall clearly identify the change compared to the previous period.

Ref: 235 - Mandatory Requirement

The Contractor shall deliver a baseline release report at critical stages of the project so that system and safety approval statements can be made against a frozen set of consistent engineering evidence. Baseline release reports shall be delivered at least before each IMSAP and APIS submission and shall include but not be limited to:

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- The list of applicable engineering documents
 - The issues addressed/not addressed with that release.
 - The Engineering Change Request implemented/not implemented in that release.
 - The compliance statement for each Client requirement associated with that release.
 - Any limitations regarding safety or performance

Ref: 236 - Mandatory Requirement

The Contractor shall organise at the Client's premises a gateway review before the delivery of each baseline release report and shall invite the Project Manager to attend this gateway. Four (4) weeks' notice shall be given to the Project Manager.

Ref: 237 - Mandatory Requirement

The Contractor shall continuously maintain an asset register, which unambiguously identifies all equipment items that are planned to be installed or is installed in the Client premises. This register shall indicate for each piece of equipment:

- The status of the equipment: reference design/ detailed design/Transit/storage/ installed/tested/in service
- The configuration and function associated with the equipment (e.g. function of balise group)
- The equipment characteristics (cable type, cable length, type of location case, type of junction/connection box number of LEUs, number of balise in the group, type of interface 'relay' or SSI, type of balise mounting system,)
- Location (line, station, interlocking area, kilometric point, and GPS coordinate in Irish Transverse Mercator (ITM) Eastings and Northings format)
- Identifier of the equipment if applicable (e.g. NID_BG)
- Associated Location Case identification
- Equipment description.
- Manufacturing date.
- Expiry date (if applicable).
- Manufacturer / Supplier details.
- Part number.
- Version details.
- Serial Number.
- Batch number where applicable.
- Other traceability information as required; and
- Software/ firmware descriptions, versions, dates etc.

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Ref: 238 - Mandatory Requirement

The asset register shall be started by the Contractor at project inception based on the Balise Layout plans provided by the Client and shall be continuously updated to reflect the Engineering Change request and progress of the project stages (preliminary design, detail design, installation, T&C).

Ref: 239 - Mandatory Requirement

The asset register shall include a hyperlink to all relevant information associated to each of the equipment, at minimum:

- Functional Design Documents
- Detail Design Document (installation drawings)
- Configuration Data programmed in a readable format
- Installation form
- T&C forms
- Certificate of quality and conformity

Ref: 240 - Mandatory Requirement

The asset register and associated relevant information shall be accessible by the Project Manager for consultation at any time of the project in read-only mode and shall not require any specific tool or license to access it. The register shall be exported to MS Excel format.

Ref: 241 - Mandatory Requirement

The asset register shall be transferred to the Client together with any proprietary tools, right, and training required to be able to maintain this data base for the remaining life of the system.

10.5 Verification & Validation Management

Ref: 242 - Mandatory Requirement

The Contractor shall provide to the Project Manager, within eight (8) weeks of the contract commencement date a Verification and Validation Plan that demonstrates how their project and company engineering management framework comply with the requirements of EN 50126, EN 50716, EN 50129 and best practices regarding Verification and Validation Management

Ref: 243 - Mandatory Requirement

The V&V plan shall demonstrate how the Contractor will verify and validate the consistency between the various stages of the system design and implementation process, to assure that the requirements given to the ETCS Specifications have been correctly captured, cascaded, and implemented at each stage of the process. The V&V plan shall also demonstrate how compliance to all applicable standards is verified and validated along the engineering cycle of the project.

Ref: 244 - Mandatory Requirement

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The V&V plan shall demonstrate how the Contractor shall ensure that the implemented system, once integrated into the existing Client's railway environment, performs as required by EU Regulations, Irish Railway Standard, EN and the Client's Standards and requirements.

Ref: 245 - Mandatory Requirement

The V&V plan shall describe what Verification and/or validation activities are put in place at every stage of the life cycle, what input are required, what tools are used, what outputs are generated, and what level of competences are needed to perform these activities.

Ref 246 - Mandatory Requirement

The V&V plan shall ensure that verification and validation activities, including testing, of a change are limited to the scope of that change. A non-regression test analysis for each change shall be produced to justify the amount of Verification and Validation activity required and keep it to a minimum. The non-regression analysis will provide a comparison of the designs affected by the change and demonstrate that, all changes, and only changes are covered by the test cases and V&V activities. Aspects of the design that are not affected by the changes shall not be tested or be subject to V&V activities.

10.6 EMC Management

Ref: 247 - Mandatory Requirement

The Contractor shall provide to the Project Manager, within eight (8) weeks of the contract commencement date an EMC management Plan that demonstrate how their project and company engineering management framework comply with EN50126, EU EMC Directive 2014/30/EU and best practices regarding EMC Management.

Ref: 248 - Mandatory Requirement

The EMC Management plan shall describe all activities that the Contractor will undertake to identify and control EMC hazards throughout the life of the project and achieve all necessary technical and safety approvals (NoBo, AsBo, etc.).

Ref: 249 - Mandatory Requirement

The Contractor shall provide test data to support a statement of compliance with the EU EMC Directive 2014/30/EU for the location cases fitted with the ETCS equipment and wiring.

Ref: 250 - Mandatory Requirement

The Contractor shall identify any potential source of EMC threat from the existing infrastructure to their TPS trackside equipment.

Ref: 251 - Mandatory Requirement

The Contractor shall be responsible for the design, implementation and testing of all EMC mitigations required to ensure that:

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- The existing Client Infrastructure and Rolling stock is not affected by the CCT
 - The existing Client Infrastructure and Rolling stock is not affecting the CCT

This includes but is not limited to:

- EMC testing of representative fitted location cases for each type
- Any onboard or trackside EMC measurement required
- Any EMC desktop analysis required
- Any modification required to existing systems or ETCS equipment to fix an EMC issue.

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11 SYSTEMS SAFETY ASSURANCE AND APPROVAL

The TPS NRO project aims to enhance safety across the Client's rail network by implementing a unified, nationwide train protection system on all nine (9) passenger service routes outlined in the NRO scope – referred to as Phases. The implementation of the ETCS trackside subsystem impacts multiple departments inside the Client's (IÉ IM and RU) organisations and is classified as a significant change in accordance with the (EU No. 402/2013). Accordingly, it is subject to the internal safety management process and approval gates as specified by the IÉ IM-SMS-14 standard.

Additionally, the ETCS Subsystem is defined by the TSI CCS and requires an "Authorisation for Placing in Service (APIS) of Upgraded Fixed Installations" in accordance with (EU) 2016/797 Article 18. This modification also qualifies as a "new or material change to railway infrastructure of operational significance," as outlined in Section 42 of the Railway Safety Act 2005. Therefore, it requires acceptance from the CRR through a New Works Assessment in compliance with sections 42(1), 42(5), and 42(8) of the Railway Safety Act 2005.

Therefore, the project Assurance lifecycle process will follow the applicable gates as defined by the CNELEC Standards, the Client's IM-SMS-14 standard and CRR Guideline, as outlined in the following section.

The current safety assurance strategy is based on the submission of nine (9) different "Authorisation for Placing in Service (APIS) applications one for each of the NRO Phases. The legal applicant for the internal and external applications will be the IÉ IM, however the contractor remains responsible to satisfy all approval requirements (including but not limited to IM, RU, SAP, ERA and CRR).

The Contractor shall prepare the safety documentation required by all the applicable standards, and approval processes to be included or referenced by the IÉ IM in Safety approval Applications.

11.1 Systems Assurance Lifecycle

The Systems assurance lifecycle for the NRO is presented in the Table 4. The Client's IM-SMS-14 and RU-SMS-14 change management process, and the CRR guidance CRR-G-009-G are mapped to the EN 50126 Lifecycle. Table also shows the ERA Trackside Pre-authorisation process and the RSA 2005 gates for information.

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EN 50126			IM-SMS-014 & RU-SMS-014		CRR Stages	IOD	RSA 2005
1	Concept	E	Concept	1	Concept		
2	System Definition and operational context						
3	Risk Analysis and Evaluation	D	Preliminary Design	2	Preliminary Design	Trackside Pre-Authorisation	RSA-AFA-NW before commencement of works
4	Specification of System Requirements						
5	Architecture and Apportionment of System Requirements	C	Detailed Design	3	Detailed Design		
6	Design and Implementation						
7	Manufacture						
8	Integration	T	Testing	4	Testing		
9	System Validation						
10	System Acceptance	B	Interim Operation	5	Interim Operation	APIS	RSA-AFA-NW
			Close out	6	Operation		
		A	Completion				
11	Operation, Maintenance & Performance Monitoring						
12	De-Commissioning						

Table 4 - Assurance Lifecycle Stages

11.2 Safety Assurance Strategy

The Safety Approval strategy for the NRO project is described by the workflow in Appendix C and summarised hereafter.

As described in the lifecycle Table above, the NRO project will have to achieve internal and external approvals for the project elements. The Client's SMS-14 internal stages shall be completed before proceeding to the corresponding external CRR approval stage. The IM Safety approval Panels (IM SAP) reviews the safety deliverables and issue internal certificates for each stage authorising the activities of the next stage in the project.

The IM-SMS-14 **Stages E and D and the CRR Stages 1 and 2** are valid for all NRO routes. A single application is required for the whole NRO scope including all 9 phases. These phases of the NRO project have been already authorised by IM SAP and CRR, however they may be updated as required during the programme duration if the requirements or scope changes significantly during the delivery stages.

For the following stages in the approval workflow (i.e. IM Stages C to A, and APIS 3 to 6), a separate application is needed for each of the 9 phases of the NRO project. The timelines for the implementation of the enabling works modifications, and the ETCS specific application designs are different for each phase. The Specific Application workflow presented in Appendix C will be repeated for each phase to conclude the enabling works and acquire the IMSAP Cert C (detailed design) to Cert A (closure) and the CRR APIS stages 3 to 6.

The IM-SMS-14 **Cert C** typically approves the design and authorises the commencement of the installation activities on site including the installation quality inspections and tests. This submission is

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phase specific and is typically made to IM SAP in accordance with the IM-SMS-14 requirements. The Cert C allows the project to proceed to the CRR APIS Stage 3 approval from.

The IM-SMS-14 **Cert T** is critical for the commencement of dynamic testing activities on the respective phase and authorises tests to be carried out in possessions. If tests on the live railway are envisaged the project needs to acquire in addition to the Cert T a CRR APIS 4 approval for testing stage. However, this CRR approval is not required if the tests will only be conducted in possession.

The signalling and civil enabling works for the ETCS trackside project will typically include minor modification to the interlocking system, extension of the cable containment near the location cases and the installation of disconnect boxes.

The Client will seek an early authorisation of the civil enabling works installation activities including cabling, and cable containment activities through an early conditional “**Enabling Works Cert C and CRR Stage 3**” under the **Railway Safety Act 2005 (RSA)** process application. This application will be limited in scope and will not cover the full ETCS functional and Data designs. The Client will internally coordinate the submission of “**Enabling Works Cert C** (detailed design) under the IMSAP change application (Changes which require approval of IMSAP). The Client will also include certain signalling Enabling work activities such as SSI and relay room modifications as part of other ongoing interfacing projects, which will be delivered prior to the ETCS installation activities.

The Contractor shall start the hardware design and acquire new Enabling works “Cert C and CRR Stage 3 – approval of detailed design” under the IMSAP change application, and CRR RSA process, to allow the installation of the ETCS Location cases, after completing their design. This application will not allow full ETCS installation, however it may include the installation of passive ETCS elements that have no impact on the live railway signalling system, or the operation of the ETCS fitted trains, such as Eurobalises without functional information and the installation of the LEUs in the ETCS location cases.

The ETCS approval process plans for a **combined Cert C/T** (detailed design and testing) application for each NRO Phase, to be submitted to the Client’s IM SAP. These applications will be prepared following completion of ETCS designs and their corresponding design-stage V&V activities. At this stage, the Contractor must provide comprehensive V&V evidence encompassing all completed design and installation work. This evidence must demonstrate sufficient progress to verify, through design V&V, laboratory testing, and site inspections, that safety and compliance requirements have been met in accordance with TSI and CENELEC standards.

The **Cert C/T** application will seek approval for the detailed design and authorisation to proceed with the remaining ETCS installation and testing activities. These activities include balise programming, integration of the LEU into the signalling circuit, and conducting both static and dynamic testing. The required deliverables for this submission are detailed in Section 11.5 below.

Following the approval of the Cert C/T of the respective route, an **APIS Stage 3** application will be submitted to the CRR. The test activities will take place in possessions while the APIS 3 application is being reviewed by CRR since testing in possessions does not require authorisation from CRR.

Upon completing the test phase and all independent assessment activities, the project will move forward with the application for **Cert B Interim Operation** approval. This approval is a prerequisite for applying for **APIS Stage 5/6** certification from the CRR.

APIS Stage 5 certification is granted when interim operation is achieved with open conditions. If the project meets all technical and operational readiness requirements, and is supported by positive assessments from the AsBo, NoBo, and DeBo, the project may directly proceed with the application for **APIS Stage 6** approval.

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The Cert A is generally granted after twenty-four (24) weeks of (interim) operation when the system is proven to be functioning as planned and the project has closed all actions and completed the handover of all necessary resources and documentation to the Client's SET department.

11.3 Safety Assurance Responsibilities and Deliverables

The Contractor shall produce and submit the safety applications on behalf of the Client's Infrastructure Manager. The Client's Infrastructure Manager (IM) will support the implementation of safety assurance across all nine phases and will act as the applicant for safety approval applications, both internally and to the National Safety Authority, the Commission for Railway Regulation (CRR).

The Safety assurance responsibilities for this process are shared between the Asbo (Contractor) and relevant departments within the Client's IM, depending on the project's lifecycle stage.

The Client's IM will lead the safety management processes and requirements capture, up to the contract commencement date. Additionally, the Project Manager will manage the approval of civil detailed designs for enabling works to ensure the installation phase can begin. Throughout the project stages, the Project Manager will continue to supervise the Contractor's execution of safety assurance processes and will oversee the development of the safety approval deliverables. The Contractor's safety assurance responsibilities will encompass the following levels of the Safety Management process:

- ETCS Product-Level Safety Case and Certification deliverables (Interoperability Constituent Level).
- The Client's Generic Application Safety Case, along with supporting independent safety assessment records.
- Application-Specific Safety Case (ASPSC), which covers the integration of the ETCS trackside subsystem within the defined geographical boundaries and its specific interfaces with the existing Irish Rail system. The Application-Specific Safety Case will cover the closure of the O&M interfaces including all relevant project level SRACs to the IÉ IM and RU Stakeholders as required under section 16.4 Operational Readiness Support.
- The SASC shall be supported with positive NoBo, DeBo and AsBo statements and declarations or certificates.

The key deliverables for each of the safety management levels are outlined in the in the following sections for each stage of the approval process.

Ref: 252 - Mandatory Requirement

The Contractor shall be responsible for implementing all the required activities, as listed in the IM Phases and CRR stage description below.

Ref: 253 - Mandatory Requirement

The Contractor shall deliver incremental controlled revisions of each GASC and SASC to support every stage of IM SAP or CRR safety approval stages as described in the following sections. When required these incremental revisions shall be supported by associated NoBo and AsBo reports provided as part of the Contractor's works.

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Ref: 254 - Mandatory Requirement

In order to secure the safety approvals of the CCT system, the Contractor shall undertake the following activities:

- Drafting, Development and Submission of IM-SMS-014 Applications for Safety Approval Certificates C-A (detail design-completion).
- Drafting, Development and Submission of applications to the Commission for Rail Regulation (CRR) Approval for the stages 2 to 6.
- All liaison and management necessary to achieve the approvals including with the Client, the CRR (in conjunction with the Client as required) and with the ERA.
- Coordinate with the Client's stakeholders for the acceptance of the exported SRACs and follow up to capture the verification evidence provided by the Client.

11.4 IM Stage D and the CRR Stages 2 (Preliminary Design)

The first phase of the NRO project has already been initiated by the Client (IÉ IM). This phase included an assessment of the generic change description and an impact analysis on the IÉ Rail Network, along with preliminary risk assessments and requirements capture activities. These activities are recorded in the Client's Preliminary Hazard Analysis, NRO Safety and Compliance matrix and the traceability to the Client Requirements.

On Contract award, the existing safety baseline documents will be reviewed by the Contractor to evaluate any changes to requirements and identified hazards. Following the initial analysis of the safety baseline documents, the Contractor shall proceed with the design of the ETCS Level 1 trackside subsystem based on the suitable products and solutions readily available.

The ETCS solutions and products must comply with interoperability specifications (Interoperability Constituent, as defined in the TSI CCS) and shall be covered by a dedicated product-level generic safety case, independently assessed by a safety assessor (ISA/AsBo) in accordance with CENELEC Standards (EN 50126, EN 50129, EN 50159, EN 59716) and EU regulation (EU No. 402/2013).

Ref: 255 - Mandatory Requirement

The Contractor must have an established Safety Management System (SMS) that is fully implemented within the Contractor's organisation and compatible with the Client's SMS014.

Ref: 256 - Mandatory Requirement

The Contractor shall safety management processes shall comply with the latest version of CENELEC Standards 50126, EN 50716, and 50129.

Ref: 257 - Mandatory Requirement

The Contractor shall structure the safety assurance supporting deliverables according to the deliverable specified by the EN 50716:2023 standard for the Generic and Specific application levels.

Ref: 258 - Mandatory Requirement

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The Contractor shall prepare and submit a safety approval strategy document which defines how they propose to obtain IM SAP and APIS authorisation for their CTT system, including ERA trackside approval.

Approvals will be required from the relevant bodies in accordance with the IM-SMS-014 process, CRR-G-009-G, Railway Safety Act 2005 (and its amendments) and all applicable EU Directives and Regulations, international and national codes of practice and legislation.

Ref: 259 - Mandatory Requirement

The ETCS Trackside Safety Plan shall present the outline structure of the safety argument that will be developed in the ETCS Trackside Generic Application Safety Case (GASC) and subsequent Specific application Safety Cases. This Safety Plan must follow the structure described in section 6 of CENELEC EN50126. The ETCS Trackside Safety Plan shall be in place within eight (8) weeks of the contract commencement date.

Ref: 260 - Mandatory Requirement

The Contractor shall include within the ETCS Trackside Safety Plan all safety assurance deliverables defined by the CENELEC Standards and this Scope document.

Ref: 261 - Mandatory Requirement

The ETCS Trackside Safety Plan shall detail the number of submissions (updates) planned for each of the safety management deliverables.

Ref: 262 - Mandatory Requirement

The Contractor shall maintain the ETCS Trackside Safety Plan through to Completion of the whole of the works.

Ref: 263 - Mandatory Requirement

The ETCS Trackside Safety Plan shall clarify with a workflow diagram the dependency and ownership of all safety documentation and supporting evidence for the complete project lifecycle.

Ref: 264 - Mandatory Requirement

A preliminary list of hazards, not necessarily comprehensive, will be provided by the Project Manager in the early stage of the Contract, to ensure consistency in wording and level of definition.

For each hazard, the Contractor shall identify controls to eliminate the hazard, so far as reasonably practicable.

Ref: 265 - Mandatory Requirement

The Contractor hazard registers shall show the traceability of the transferred hazard from the Client's registers and the exported SRAC through the Hazard ID traceability.

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Ref: 266 - Mandatory Requirement

The Contractor Preliminary hazard analysis shall show traceability to the ETCS Hazard Log (Subset-113).

IM SAP Phase	CRR Stage	Authorises	Safety Assurance Tasks	Deliverables
Cert E – Concept Cert D - Preliminary Design	APIS Stage 1 APIS Stage 2	Further development of design	Safety Tasks: - Establishing the Contractor's project SMS and QMS - Defining the roles and responsibilities for safety and Personnel Independence in Tasks - Review of the Irish Rail Preliminary hazard analysis (Stage 2 Hazard Record) - Complete the System definition and Preliminary Hazard Identification in accordance with EU No. 402/2013 - Complete the SIL allocation for the subsystem functions. - Review and assign ownership of the IÉ Safety critical requirements. - Review the Safety and Compliance Matrix (TSI and National rule requirements) and develop the verification and validation and assessment plans.	Safety Deliverables - Contractor's safety approval strategy - Contractor's Safety Plan - Contractor's RAM Plan - Contractor's Preliminary Hazard Analysis with traceability to IÉ Prelim Hazard Analysis - Engineering Process Hazard Analysis Report (EPHA), including input data verification, engineering rules safety analysis, tools, configuration management, specific design and implementation activities, and skills). - Product Level Safety Cases (Interoperability Constituent level) - Interoperability Constituent EC certificates and NoBo Reports - AsBo Assessment Plan - NoBo And DeBo assessment Plans

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IM SAP Phase	CRR Stage	Authorises	Safety Assurance Tasks	Deliverables
			○ Review the RAM targets and develop the RAM delivery and demonstration strategy. ○ Engineering Process Hazard Analysis Report (EPHA), including input data verification, engineering rules safety analysis, tools, configuration management, specific design and implementation activities, and skills	

Table 5 – Safety Assurance Tasks and Deliverables for Cert E/D and CRR Stages 1/2

11.5 Enabling Works Cert C (Location Case Detailed Design)

An IM-SMS-14 internal application and a CRR Stage 3 Railway Safety Act 2005 application will be submitted for each route before installing new location cases and modifying existing location cases as necessary. These submissions will present the detailed ETCS location case design and authorize both the installation and connection of new ETCS location cases to existing interlocking location cases. The application will also permit LEU installation within the location case in preparation for signalling circuit integration (cutting-in). Cutting-in activities are excluded from this application scope unless the Client and IM Safety Panel provide specific agreement. Such agreement would require supporting evidence demonstrating that LEU cutting-in operations are non-intrusive, backed by appropriate risk assessments conducted in accordance with IM-SMS-14 procedures.

Ref: 267 - Mandatory Requirement

The Contractor shall submit a risk assessment report with an interface hazard analysis demonstrating the non-intrusiveness of the installation works carried out prior to the subsystem commissioning testing and commissioning stage, and their impact on the live Railway and operations.

IM SAP Phase	CRR Stage	Authorises	Safety Assurance Tasks	Deliverables
Cert C – Enabling works Detailed Design	Stage 3 RSA	ETCS non-intrusive Installation works	Safety Tasks - Enabling work HAZID - Interface Hazard analysis for the planned works - Safety Risk Assessment for the Installation, site inspection / early testing activities.	Safety Deliverables - IM Change Request form - Safety Plan Update - Installation Stage Safety Risk Assessment and Hazard Record - IM declaration and application for safety approval as required by SMS-014. - Safety and Compliance Matrix updated for enabling works - Application Specific Project Safety Case (ASPSC) - Stage 3 RSA CRR submission letter

Table 6 - Safety Assurance Tasks and Deliverables for Enabling works Cert C

11.6 ERTMS Trackside Approval

As part of the application for Authorisation for Placing In service (APIS) the Contractor shall engage with the ERA and complete the ERTMS Trackside approval process on behalf of the Client.

Ref: 268 - Mandatory Requirement

The Contractor shall Request and coordinate an Initial Engagement Meeting with the ERA, involving IE and the CRR.

Ref: 269 - Mandatory Requirement

The Contractor shall assess the interoperability risks and advise the Client regarding any issues that may affect ERA Trackside approval.

Ref: 270 - Mandatory Requirement

The Contractor shall compile all deliverables required for the Trackside approval (TA) application in accordance with the ERA guidelines, including the following requirements:

- ERA Positive Decision Letter
- Project planning
- List of ERTMS functions to implement
- IE Contract specification
- National rules related to ERTMS

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- Engineering rules
- Interoperability risk assessment
- Test strategy and test plan(s)
- Operational test scenarios

Ref: 271 - Mandatory Requirement

The Contractor shall propose and draft any clarifications or changes for the Client approval to address any issues raised by ERA through the One Stop Shop platform.

Ref: 272 - Mandatory Requirement

Upon ERA request, the Contractor must obtain an intermediate statement of verification (ISV) from a NoBo as evidence of review of the Engineering Rules and the Client's Contract specification.

Ref: 273 - Mandatory Requirement

The Contractor shall review and confirm the fees invoiced by ERA to the Client for the Trackside approval process. The Client shall pay the fees directly to the ERA.

Ref: 274 - Mandatory Requirement

The Contractor shall support the Client in general questions related to ERTMS technology (e.g. understanding of ERTMS Set of specification, responses ERA questionnaires, migration to Level 2).

IM SAP Phase	CRR Stage	Authorises	Safety Assurance Tasks	Deliverables
N/A	N/A			
ERA ERTMS Trackside Approval			Safety Tasks ○ ERA Initial Engagement ○ Submission of Trackside Approval application ○ Management of Issues log ○ Drafting required changes for IÉ review and approval ○ Coordination with CRR where required	Safety Deliverables ○ ERA ERTMS Trackside Approval Application Electronically submitted through the OSS, including: ○ NoBo ISV (If required by ERA)

11.7 ETCS Combined Cert C/T Application and APIS Stage 3 Application (ETCS Detailed Design Approval and Authorisation to Test)

The Contractor must follow their internal procedures to integrate the ETCS components into a system architecture that meets the Client's specific requirements. The safety assessment process will begin by addressing the hazards identified during the preliminary assessment and SRACs (Safety Requirements and Constraints) carried over from the product level.

Ref: 275 - Mandatory Requirement

During the design stage, the Contractor shall define the safety requirements specific to the Scope and Interface requirements related to the dependencies between the ETCS and the existing IÉ signalling system. in the respective route.

Ref: 276 - Mandatory Requirement

The safety tasks must demonstrate that all identified hazards have been mitigated through design, and the safety requirements have undergone an interim verification and validation process in line with the design development stage.

Ref: 277 - Mandatory Requirement

The Contractor shall identify controls to manage all hazards that cannot be eliminated associated with the Design, manufacture, supply, installation, Commissioning and Through Life Support of the ETCS Trackside Subsystem, so that the residual safety risk is tolerable in accordance with the risk matrix, and risk acceptance criteria provided in the Client's SMS-14.

Ref: 278 - Mandatory Requirement

The Contractor shall demonstrate that all risks have been reduced so far as is Reasonably Practicable (SFAIRP) in line with the Client's IM SMS-14 definitions and process.

Ref: 279 - Mandatory Requirement

The Contractor shall acquire the Project Manager consent whenever administrative or procedural means are proposed to minimise the risk.

Ref: 280 - Mandatory Requirement

The Contractor shall assess the influence of the ETCS subsystem failures on the safe operation of the Client's railway system as required by EN-50129.

Ref: 281 - Mandatory Requirement

The Contractor shall notify the Project Manager at an early stage of all exported Safety-Related Application Conditions (SRACs) which cannot be mitigated by the ETCS system (design to safety). SRACs shall follow the guidance of EN 50129. The Contractor shall provide all necessary information regarding

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physical, functional, environmental, operational and maintenance context to enable the Client to accept, assess and establish the necessary risk control measures by the Client. The Client has the right to reject any unreasonable SRAC not clearly and unambiguously communicated by the Contractor.

Ref: 282 - Mandatory Requirement

The Contractor shall ensure that where SRACs are notified to the Project Manager, for each SRAC the Contractor shall provide the following:

- The provision of a suggested solution to the SRAC identified.
- Documents referenced as supporting evidence are complete and have been submitted to the Project Manager.
- Only SRACs relevant to the works for the Client supply are notified or raised by the Contractor.

Ref: 283 - Mandatory Requirement

It is mandatory for the Contractor to clarify, at an early stage, whether the Client can comply with the SRACs or not and whether they are acceptable or not. Once these SRACs are accepted in writing by the Project Manager, it shall be responsible for ensuring that they are properly managed and responsible for the preparation of associated evidence.

The Client shall co-ordinate the submission of documents to the National Safety Authority (NSA) (if there are any required) and the European Union Agency for Railways (ERA). The Contractor shall be responsible for all remaining aspects of the process.

Ref: 284 - Mandatory Requirement

The Contractor hazard registers shall show the traceability of the transferred hazard from the Client's registers and the exported SRAC through the Hazard ID traceability.

Ref: 285 - Mandatory Requirement

For the installation stage, the quality inspections will be managed by the Contractor. The installation process shall be audited internally by the Contractor's own quality management system and externally by the AsBo, and the Client. The audit shall be against the requirements specifications, the applicable Client's IM Standards referenced by the specifications, and the IRS Standards (issued by the CRR).

The inspections may take place in parallel to the Installations, and the quality records will be provided to support the APIS application.

Ref: 286 - Mandatory Requirement

Appropriate control measures shall be specified by the Contractor in the application detailing the arrangements for managing the programmed balises installed on the live railway, and the use of Virtual Balises Codes to mask the non-commissioned programmed balises.

Ref: 287 - Mandatory Requirement

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The Contractor shall develop and deliver an ETCS Trackside Generic Application Safety Case (GASC) covering for all sections of track associated with the phases defined in this Scope.

Ref: 288 - Mandatory Requirement

The Contractor shall deliver controlled revisions of the GASC as necessary due to requirements or design changes during the delivery of the section of track associated with the phases defined in this Scope.

Ref: 289 - Mandatory Requirement

The ETCS Trackside GASC shall provide the associated, explicit evidence in support of the safety argument and as otherwise necessary to demonstrate the safety of the generic application of the ETCS Trackside Subsystem. The supporting deliverables shall be structured as specified in CENELEC standards.

Ref: 290 - Mandatory Requirement

The Contractor shall develop a specific application safety case (SASC) for each section of track associated with the phases defined in this Scope in accordance with the structure and requirements of the CRR Guideline RSC-G-009-E-Annex3 (SC) Checklist for evaluation of a Project Safety Case.

Ref: 291 - Mandatory Requirement

To demonstrate safety, the Contractor shall identify all reasonably foreseeable hazards that are introduced by the ETCS Trackside Subsystem and its operation in the Client Environment.

IM SAP Phase	CRR Stage	Authorises	Safety Assurance Tasks	Deliverables
Cert C/T – Combin ed Detailed Design approval and Testing	APIS Stage 3	Dynamic Testing in possessions under operation control measures	Safety Tasks The Safety Assurance activities are as specified by the EU No. 402/2013, and the CENELEC Standards. As a minimum the tasks will cover: ○ HAZID workshops, Risk Review Workshops engaging Client's Stakeholders ○ ETCS Safety Hazard Analysis and Safety Risk management ○ Failure Analysis (FTA, FMEA), Impact on Subsystem and system level operation	Safety and Approval Deliverables ○ Updated Safety Plan and Assessment Plans (if required) ○ NRO Generic Application Safety Case covering the NRO scope. ○ Specific Application Safety Cases (for each route) with positive conclusion allowing dynamic tests in possessions ○ Hazard Registers

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IM SAP Phase	CRR Stage	Authorises	Safety Assurance Tasks	Deliverables
			- environmental conditions impact, and Electromagnetic Compatibility and Interferences (EMC/EMI). - Hazard Analysis - Common Cause Failure Analysis - Interface Hazard Analysis - Testing Procedures Hazard Analysis and Safety Risk assessments - Hazard and Operability Analysis (HAZOP) - Human Factor analysis covering Contractor's O&M Procedures) - SRAC Transfer - Quality and Safety Audits - Quality Management records (Site Inspection and Installation release notes) - Interim Safety Verification and Validation Report for the design stage. - AsBo CSM safety Assessment - AsBo Assessment of the requirements capture. - AsBo Quality and Safety inspections on site	- Interim Safety Verification and Validation Reports (Preliminary V&V outcome based on design review and document/drawings checks) - Interim Human Error Analysis Report (HEA), including operation, configuration, programming installation, maintenance or fitment of ETCS trackside equipment. - Interim Quality Records supporting the safety cases (Quality Report) - Interim Safety and Compliance Matrix (design stage evidence) - Interim AsBo Assessment Report - NoBo Intermediate Statement of Verification (ISV) covering design stage - IM declaration and application for safety approval as required by SMS-014.

Table 7 - Safety Assurance Tasks and Deliverables for ETCS Cert C/T and CRR Stage 3

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11.8 ETCS APIS Stage 4 Application (testing on live railway)

The Client's safety assurance strategy stipulates that all tests be conducted within possessions, adhering to operational safety measures. As a result, Stage 4 approval from the CRR is not required. However, if the Contractor's delivery and testing strategy includes testing on the live railway, the Contractor must clearly specify this requirement in their safety approval strategy and obtain the Client's consent for the proposed testing in the early stages of the Project.

An APIS 4 application may be submitted by the Contractor, with the Client's support, subject to IM SAP approval of Cert C/T.

IM SAP Phase	CRR Stage	Authorises	Safety Assurance Tasks	Deliverables
N/A	APIS Stage 4	Dynamic Testing on live railway	Safety Tasks - Lab test and on-site Static Tests activities demonstrating compliance to the safety critical requirements. - NoBo Interim assessment - EMC measurements and tests as required.	Safety and approval Deliverables - Updated Specific Application Safety Case for the respective route, with positive conclusion allowing tests on live railway - EMC Reports - AsBo Assessment Report with positive statement supporting test readiness on live railway - NoBo Intermediate Statement of Verification (ISV) covering installation and Static testing

Table 8 - Safety Assurance Tasks and Deliverables for CRR Stage 4

11.9 ETCS APIS Stage 5/6 application (Interim Operation or Operation)

The Contractor shall complete all V&V activities and acquire evidence for the acceptance and closure of Safety Related application Conditions (SRACs) from the relevant Client departments and provide the safety case and Safety and compliance matrix for the AsBo and NoBo assessments. The Cert B Interim Operation, and CRR (APIS) stages 5 and 6 submissions shall be completed to authorise revenue service. This will conclude the works for the route and allows a Cert (A) Completion to be issued, following a 6-month interim operation period.

Ref: 292 - Mandatory Requirement

The Static and dynamic test activities will be witnessed by the Contractor's Asbo and NoBo and shall be covered by the safety case submitted for the subsystem. The Dynamic testing will take place after the approval of the design and installation under the APIS process (main IM and CRR application for the ETCS trackside subsystem).

Ref: 293 - Mandatory Requirement

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The Contractor shall complete all Operation Readiness support activities required under section 16.4, prior to the Cert B submission for the respective route.

An APIS 5/6 application may be submitted by the Contractor, with the Client's support, subject to IM SAP approval of Cert B.

IM SAP Phase	CRR Stage	Authorises	Safety Assurance Tasks	Deliverables
Cert B	APIS Stage 5/6	Entry into Operation (or Interim Operations)	Safety Tasks ○ Lab test and on-site static tests and V&V activities for safety, TSI and NR requirements ○ Dynamic testing activities ○ Assessment by Independent Verification and Validation roles ○ EMC measurements and tests as required. ○ AsBo and NoBo Test Witnessing ○ Maintenance Training delivery as required for operation readiness. ○ Safety Risk Assessment for the ETCS Commissioning procedures ○ Coordination with the Client and support in O&M SRACs clarification.	Safety and approval Deliverables ○ NRO Generic Application Safety Case (updated as required) ○ Specific Application Safety Case for the respective route, with positive conclusion allowing the entry into Operations ○ AsBo Assessment Report and Certificates ○ NoBo Report, Subsystem EC Certificate and accompanying Technical File ○ EMC Reports ○ Human Error Analysis Report (HEA), ○ Hazard Registers ○ Final Safety Verification and Validation Reports ○ Final Human Error Analysis Report (HEA), including operation, configuration, programming installation, maintenance or fitment of ETCS trackside equipment. ○ Training certificates

Table 9 - Safety Assurance Tasks and Deliverables for Cert B and CRR Stages 5/6

11.10 Appointment of the Assessment Bodies

The Contractor must appoint and manage the bodies below in order to fulfil the TSI, NR, Railway Safety Act 2005 and CSM assessment requirements. The NRO Contractor will engage the following independent conformity and safety assessment bodies as described below:

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- Notified Body (NoBo)
 - Irish Designated Body (DeBo)
 - Safety Assessment Body (AsBo)

Ref: 294 - Mandatory Requirement

The Contractor shall be responsible for the direct appointment of above listed assessment bodies (i.e. Notified Body, Designated Body and Safety Assessment Body) as required by the relevant legislation for its scope. These appointments are part of the Contractor's works. In addition to these appointments, the Client may engage an over-arching system CSM-AsBo if required. The assessment bodies will be required to review and provide reports for each safety submission and for all phases IM safety approval process. The Contractor shall be responsible for managing the Assessment Bodies (appointed by the Contractor) and ensuring that the output of reports etc. is fit for the intended purpose.

Ref: 295 - Mandatory Requirement

The AsBo appointed by the Contractor shall carry out 100% assessment on the suitability of the applied safety management process (CSM-RA (EU) 402/2013, Annex I and CENELEC) and additional assessment on a sample basis (at least 10%). They also inspect the installation and witness testing on a sample basis.

Ref: 296 - Mandatory Requirement

The scope of the AsBo appointed by the Contractor shall include the assessment activities related to the TSI-CSM Safety Requirements and the Requirements capture for all other non TSI Requirements.

Ref: 297 - Mandatory Requirement

The AsBo (appointed by the Contractor) shall create the Interim and final AsBo Assessment Reports, as required for the IM and APIS approval stages

Ref: 298 - Mandatory Requirement

The Contractor shall ensure that there are no gaps between the successive verification scopes of the AsBo, NoBo and DeBo.

Ref: 299 - Mandatory Requirement

The Contractor shall agree assessment modules with NoBo/DeBo based on full quality management systems approval and surveillance (module SH1) to minimize site inspection time and interruptions to the live railway.

Ref: 300 - Mandatory Requirement

The Contractor shall ensure the NoBo and DeBo assessment modules are selected to avoid overlapping assessments and reduce fixed costs across the NRO Phases.

Ref: 301 - Mandatory Requirement

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The Contractor shall Establish and maintain technical documentation in accordance with TSI requirements and NoBo/DeBo specifications

Ref: 302 - Mandatory Requirement

The Contractor shall Coordinate timing of certificate issuance with programme milestones

Ref: 303 - Mandatory Requirement

The Contractor shall provide the AsBo, NoBo and DeBo assessment plans to the client for review prior to the start of the independent assessment activities.

Ref: 304 - Mandatory Requirement

The AsBo, NoBo and DeBo, appointed by the Contractor shall provide regular progress reports on conformity assessment activities.

Ref: 305 - Mandatory Requirement

The Contractor shall establish clear communication protocols between NoBo/DeBo and Client's project team.

Ref: 306 - Mandatory Requirement

The Contractor shall Register all declarations with ERA as required by the TSI, in the relevant database systems (e.g. ERADIS).

ETCS Safety and operational Records

Ref: 307 - Mandatory Requirement

Recorded Data in the LEUs shall be readily made available in a readable format to the Infrastructure Manager, National Safety Authorities, and National Investigation Bodies.

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12 RAM MANAGEMENT

Ref: 308 - Mandatory Requirement

The Contractor shall provide to the Project Manager, within eight (8) weeks of the contract commencement date, a RAM Management Plan that demonstrates how their project and company engineering management framework comply with EN50126, the CCT Common Application Requirement referenced in Appendix B, and best practices regarding RAM Management.

Ref: 309 - Mandatory Requirement

The RAM Management Plan shall describe all activities that the Contractor will undertake to demonstrate that RAM target and specifications are met by the TPS trackside system.

Ref: 310 - Mandatory Requirement

The ETCS CCT Subsystem shall comply with all RAMS requirements of the following Subsets: ERTMS/ETCS RAMS Requirements Specification Chapter 2 RAM, issue 6, [R16]; ERA ERTMS/ETCS Subset 036, [R4]; ERA ERTMS/ETCS Subset 091.

Ref: 311 - Mandatory Requirement

As a minimum the RAM activities undertaken by the Contractor shall include:

- Allocation of RAM target to sub-systems and components
- A predictive RAM analysis to demonstrate the theoretical achievement of the target though the use of standard RAM techniques such as Failure Modes Effects and Criticality Analysis (FMECA) and Reliability Bloc Diagrams (RBD).
- Implementation of a monitoring system and associated corrective actions (FRACAS) to improve reliability and ensure that RAM targets and specifications are met in service and the system can enter in warranty period.

Ref: 312 - Mandatory Requirement

The hardware design, manufacturing and installation shall be made in accordance with best practice to ensure that the trackside system is resistant against all foreseeable mechanical and electromagnetic constraints. This shall be demonstrated through design review and prototype testing and review.

Ref: 313 - Mandatory Requirement

In service reliability monitoring of the system will have to be fully delivered by the Contractor, after APIS 5 and up until Cert A approval. This in service monitoring reliability shall be made in accordance with the following rules:

- The reliability monitoring period shall start after the testing for a phase has been completed
- The reliability Monitoring shall ensure that each equipment is initially screened for failure at least every month
- The Client will provide the required maintenance records and events recorder logs.

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- The Client will provide support for track access and protection should the contractor require detailed site inspection for failure investigation.
 - That frequency may be increased or decreased depending on the number of failures identified during the monitoring
 - The screening of the failure shall not only cover the failures that are present at the time of the check but shall also include failures that can have occurred since the previous check, via the download of the LEUs' event recorders.
 - The Reliability monitoring process shall be recorded in a FRACAS (Failure Reporting, Analysis, and Corrective Action System) developed by the Contractor and shared with the Client.
 - The warranty period shall only start once Cert A is granted (including demonstrating that the products Targets and Specification are met in service on the basis of the data gathered during the monitoring period. This correlation shall be made following a Reliability growth model proposed by the Contractor and accepted by the Project Manager

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13 HUMAN FACTORS MANAGEMENT

Ref: 314 - Mandatory Requirement

The Contractor shall provide to the Project Manager, within 8 (eight) weeks of the contract commencement date, a Human Factors Assurance Plan (HFAP) as per IM-SMS-014-OP1. The HFAP shall demonstrate how the Contractor shall comply with IM-SMS-014-OP1 and EN50126 in applying best practices for Human Factors Assurance/Integration. It shall also demonstrate the competence of human factors professionals allocated to the project team.

Ref: 315 - Mandatory Requirement

The Human Factors Integration Plan shall describe all activities that the Contractor will undertake to demonstrate that the TPS trackside design is fit for purpose and meets the needs of the following end-users: Drivers, Signallers, Level Crossing Controllers, CCE and the Client's SET maintenance staff.

Ref: 316 - Mandatory Requirement

As a minimum the Human Factors (HF) activities undertaken by the Contractor shall include:

- ◆ User Group Workshops
- ◆ Development and maintenance of a Human Factors Issues Log (HFIL)
- ◆ Delivery of HF Assurance Reports at key project stages
- ◆ Task and error analysis of high-risk tasks either introduced or modified by the Contractor
- ◆ HF analysis of operational or maintenance procedures introduced or modified by the Contractor
- ◆ Physical ergonomics screening, and full assessment where required, of any new equipment introduced by the Contractor
- ◆ Usability assessment of any new human-machine-interfaces (HMIs) introduced by the Contractor
- ◆ Alarm analysis, as per IE HF Guidance on Alarm System Management, for any new alarm systems introduced by the Contractor
- ◆ Identification and documentation of training needs.

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14 CYBER SECURITY MANAGEMENT

Ref: 317 - Mandatory Requirement

The Contractor shall provide to the Project Manager, within eight (8) weeks of the contract commencement date a Cyber Security management Plan that demonstrates how their project and company engineering management framework comply with EN50126 and best practices regarding Cyber Security Management.

Ref: 318 - Mandatory Requirement

The Cyber Security Management plan shall describe all activities that the Contractor will undertake to identify and control Cyber Security hazards throughout the life of the project.

Ref: 319 - Mandatory Requirement

The Contractor shall comply with IÉ I-SET-1101 cyber security standards, technical specifications including SET OT security or existing security measures as appropriate to the lifecycle stage of the TPS trackside subsystem.

Ref: 320 - Mandatory Requirement

The Contractor shall be responsible for the design, implementation and testing of all Cyber Security mitigations

This includes but is not limited to:

- Securing of equipment
- Securing of configuration data during and after project, including tools and systems transmitted to the Client to ensure security after the system is handed over.

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15 COMPLETION

This part of the document sets out:

- the requirements for Completion in respect of each Phase listed in Appendix A; and
- the procedures to be implemented to facilitate the Project Manager's determination as to when Completion in respect of any Phase has occurred.
- the process to administer the orderly and safe handover of all TPS trackside equipment to the Client as the works in respect of the train achieve Completion.
- outline requirements for training of Client's staff in respect of the maintenance and operation of the CCT system.

15.1 Requirements for Completion

The works in a Phase will have reached Completion when:

1. they are in a state which satisfies the definition of "Completion" as set out in Core Clause 11.2(2) of the Contract;

and that, without limitation:

2. in respect of any section of the works, the Contractor has done, supplied, submitted or procured all of the following:
 - a. completion of scope of works as stated in chapter 2 of this document for each of the relevant Sectional Completion as stated in chapter 2.8 of this document.
 - b. the completion, to the Project Manager's reasonable satisfaction, of:
 - i. all testing and commissioning in respect of the works comprised in each section as referred to in section 8 – Testing & Commissioning of the scope of works;
 - ii. all training and familiarisation defined as Operational Readiness of the Scope;
 - c. after the Contractor has, in respect of the design, manufacture and installation of the works in the relevant section, obtained all preliminary and intermediate certificates, approvals to proceed and/or necessary acceptances from ERA and/or CRR and/or the Client (as applicable) pursuant to the procedures set out in:
 - ♦ ERA trackside approval (TCS CCS 2023)
 - ♦ IM-SMS-014 Certificate B
 - ♦ CRR APIS Stage 5

the issue of:

- i. in respect of all sections of the works APIS 5 Certificate under CRR-G-009; and
- ii. Certificate B under IM-SMS-014; in each case, free of conditions;
- d. in respect of the works comprised in the relevant section, submitted to the Project Manager of:
 - i. a Baseline Release Note with all supporting engineering document in electronic format which is to include (without limitation):
 - ♦ As built functional design,

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- ♦ As built design drawings,
 - ♦ As built LEU and Balise configuration files for maintenance purpose and their equivalent readable format files,
 - ♦ calculations, designs, and request for information files,
 - ♦ As built Asset register,
 - ♦ installation manual,
 - ♦ certificates of compliance for equipment installed,
 - ♦ other certification as required e.g. certificates,
 - ♦ test & commissioning records for static and dynamic testing,
 - ♦ Client modified design drawings,
 - ♦ register of safety critical components,
 - ♦ any electronic systems data files contained within the works that the Client may be reasonably expected to retain,
 - ♦ maintenance manuals and tools, descriptive manuals, and system training materials,
 - ♦ systems sign off documentation.

together with such written instructions and materials as are reasonably required to enable the Client and/or Others to safely operate and/or maintain the works;

- ii. all other documentation reasonably necessary to enable the Client and/or Others to operate and/or maintain those works.

such documents and materials to be to a standard reasonably acceptable to the Project Manager;

- e. provided to the Client all strategic maintenance spares and tools for those assets that have a long lead time or are vulnerable and/or susceptible to failure or damage;
- f. provided to the Client all collateral warranties the Contractor is required to provide pursuant to Secondary Option Clause X8;
- g. provided to the Client any keys or other things required to access the works comprised in each section;
- h. done or provided anything else required to be done or provided as a pre-condition or requirement for Completion of each section, as otherwise specified in the contract.

15.2 Procedures for Assessment of Completion and Handover

The following procedures shall apply to each section of the works for the purposes of the Project Manager assessing Completion of each section. References to the "works" and "Completion" in this Clause will be references to the works of the relevant section and the Completion of that section.

The Project Manager instructs the Contractor to attend a pre-completion meeting with the Client to review the Contractor's progress and planning four (4) weeks prior to planned Completion of all Sections of the works as shown on the Accepted Programme or, if there is no updated Accepted Programme, when the Project Manager deems it to be the appropriate time ("Planned Completion Date"). This enables the Client to consider what action may be required or assistance given in order to achieve Completion.

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1. On or before the date which is two (2) weeks prior to the Planned Completion Date of all sections of the works, the Contractor prepares and submits to the Project Manager for acceptance a draft "Handover Plan" detailing all requirements and procedures to facilitate the assessment of Completion and to enable an orderly and safe transfer of the works to the Client. The Project Manager provides any comments he or she has on the plan submitted. The Contractor revises the Handover Plan accordingly until the Handover Plan is accepted by the Project Manager.
2. The Contractor:
 - a. notifies the Client and the Project Manager in writing not less than two (2) weeks in advance of the date on which the works are likely to reach Completion and
 - b. arranges for a joint inspection with the Project Manager of the works not later than the date which is five (5) days before the anticipated date of Completion of the works, the Client may invite the attendance of additional appropriately qualified third parties to advise the Client with respect to Completion and to assist with the inspection).
3. Within five (5) days of the joint inspection contemplated under the above for the works, the Project Manager shall notify the Contractor in writing either that:
 - a. he or she is of the opinion that the works have not achieved Completion, and in such case shall identify the items of work or any outstanding documents or other requirements which, in the opinion of the Project Manager, are necessary to achieve Completion (provided always that the list of such items shall not include items which are of a minor nature the existence or correction of which will not prevent the Client using the works for their intended purpose ("Minor Defects")); or
 - b. he or she has identified no further works or things that are required to be carried out or done to achieve Completion (but he or she may include with such a notice a list of Minor Defects (a "Minor Defect List")) and that the works have achieved Completion.
4. If the Project Manager issues a notice under the above, the Contractor shall:
 - a. proceed to correct or make good the items identified in the Project Manager's notice;and
 - b. issue a notice in writing to the Client and the Project Manager when it considers it has completed the identified work.

Following items 4a and 4b, the parties and the Project Manager shall arrange to re-inspect the works within two (2) days after the receipt of the notice and the provisions of the above Clause shall reapply on an iterative basis until the Project Manager has determined that Completion of the works has been achieved.

5. Following a determination that Completion has been achieved, the parties then proceed to implement the Handover Plan accepted by the Project Manager.

Where the Project Manager has issued a Minor Defects List, the Contractor shall ensure that the items notified therein are remedied within four (4) weeks of the issue of the Minor Defects List and in accordance with a programme to be agreed between the Client and the Contractor (both acting reasonably) (and where the Contractor has failed to comply with this requirement, the Client may employ others to carry out any necessary work and recover the cost of same from the Contractor as a debt due).

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16 ANCILLARY WORKS

16.1 Value engineering and innovative features

Ref: 321 - Mandatory Requirement

The Contractor shall propose Value Engineering optimisation to the Project Manager, whereby parts or the system or process agreed to be delivered at tender stage maybe replaced by alternative solutions. The Project Manager will be entitled to accept or reject without justification the Contractor's proposal. The Contractor's proposal shall be accompanied with robust technical and economic argument, justifying that technical requirement continued to be met, and the proposal presents a net economical advantage to Irish Rail.

Example of value engineering and innovative feature that could be considered:

- Remote diagnostic of equipment
- Centralised Architecture
- Migration to ETCS Level 2
- Temporary speed Restriction capabilities
- Remote change of configuration data

Ref: 322 - Mandatory Requirement

The Contractor shall organise at least three (3) value engineering workshops with the Project Manager to investigate the value engineering solution. The discussion and conclusion of the workshops shall be document in dedicated report that will be delivered to the Project Manager for acceptance or rejection of the proposed improvements.

16.2 System Changes

Ref: 323 - Mandatory Requirement

The Contractor shall provide a schedule of rates for designing and another schedule of rates for implementing testing and putting into service the following type of changes included in the Maintenance Contract – Schedule 8 Section 1.5 Table 6. The rates will be applied in case of scope or input data change during the design and implementation phase of the project and will be valid until the Maintenance contract is live.

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Change Type	Example Use Case	Impact on Configuration Data
Minor change	- Change of speed sign - Signal relocation - Relocation of point end - Addition of a balise in the linking chain - Addition of 'danger for shunting' information packet without modification of other packets in the message - TSR warning messages	One or several changes in variable values, no major change in telegram structure, e.g. - Change of MA length - Change of static Speed profile (location or value) - Change of Gradient Profile - Change of a signal or track occupation limit location - Change of linking chain - Change of release speed value - Change of section timer value - Addition of one section of gradient - Addition of a packet without modification of other packets in the message - Modification of a text message
Medium change	New balise for performance or safety (e.g., infill, relocation)	Creation/modification of telegram structure.
	Medium number of telegram modifications within the LEU (change of MA, linking, SSP, gradient).	Modification of several configuration parameters of the LEU.
	Introduction of packet 40 in LEU	Major modification of fixed balise telegram.
Major message change	- Additional signal aspect to be addressed by the LEU. - New signal fitment - Additional infill (including complex telegram structure update)	New LEU input table or substantial change to LEU input table.

Table 10 - Schedule of Design Changes Post Commissioning

Ref: 324 - Mandatory Requirement

At the Completion Date of this Contract, the Contractor shall have transferred the capability to the Client to undertake these system changes in full autonomy. Post Completion, any system changes must be

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available to be undertaken by the Client. The Client may choose to contract the Contractor separately as part of a maintenance agreement; however, this is at the Client's sole discretion.

16.3 Capability Transfer for System Changes

Ref: 325 - Mandatory Requirement

The Contractor shall provide all the necessary training procedures and tools (including perpetual license) to enable up to twenty (20) SET department staff to implement changes in the TPS trackside system after it has been handed over, and training to be completed prior to Phase commissioning. That includes:

- ETCS Level 1 Irish Rail Generic Application training
- Functional Design Training
- Infrastructure Database maintenance training.
- Detail Design training.
- Data preparation and validation training
- Safety Assurance training
- Test and commissioning training.
- Asset Register update training
- Train the trainer training to be provided for up to 5nr trainers

16.4 Operational Readiness Support

The Contractor shall review the Client's RU and IM existing procedure and standards affected by the introduction of their trackside ETCS system and update them as required to reach approval and publication. This includes updates to the:

- Irish Rail IM Rule Book, Appendix J, Route information book
- Irish Rail RU Driver Handbook and procedure
- Irish Rail SET Standards for Engineering and Maintenance

16.5 Training of Client Staff in Respect of Maintenance

Ref: 326 - Mandatory Requirement

The Contractor shall design, implement, test and commission a typical ETCS installation at the Thurles training centre, as detailed the Thurles Siding Layout Plan provided in Appendix B (one main signal and one distant signal fitted).

Ref: 327 - Mandatory Requirement

The Contractor shall produce a training plan outlining all training activities and submits a copy of the plan to the Project Manager, for acceptance in accordance with the requirements of the Scope, prior to commencement of training, which includes the following as a minimum:

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- Off-site classroom training.
 - On-site training on all relevant implementation sites on the Irish Rail network.
 - Training facility training at the Client's premises
 - Remote training
 - Refresher training and
 - Visits to relevant sites and/or maintenance depots (where appropriate).

Training is to include operation of the overall system and includes individual Equipment and Materials.

Training refers to the relevant sections of the Maintenance Manuals, Maintenance Instructions, Drawings, Data, Specifications and Standards.

Training to 200nr IE staff to provide a thorough understanding of the purpose and intent of all Equipment and Materials, and systems for the purposes of safe operation, fault finding, maintenance, repair, and overhaul, and include but not be limited to the following:

- Equipment control functions.
- Equipment monitoring and action on alarms.
- Operation of safety systems and interfaces.
- Fault or failure shutdown of system.
- Restarting of the system following shutdown.
- Recovery of fault log files and associated diagnostic information.
- Access to and removal (as necessary) of line replaceable units or individual items of equipment.
- Routine maintenance tasks; and Lifting arrangements.
- Train the trainer training to be provided for up to 5nr trainers.

This training is to be staged in line with system handover to maintenance i.e. training is to be completed in each Phase 14 days prior to handing over the assets to maintenance for Interim Operation.

Without limitation, the Contractor shall design the training to meet the following objectives:

- The Client's technical maintenance & project team should have sufficient knowledge at completion of testing & commissioning to be able to perform 100% of necessary O&M activities autonomously from the Contractor's organisation.
- The Client's technical maintenance & project team should have sufficient knowledge at completion of testing & commissioning phase to undertake remote monitoring of the system to support the maintenance of trackside infrastructure and its condition utilising an onboard ETCS system fitted with remote condition monitoring
- The Client's Electronic Repair Facility staff should have sufficient knowledge at completion of testing & commissioning phase to be able to undertake an assessment of the TPS trackside equipment failure or line-replaceable unit failure and diagnose the fault.

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- The Contractor provides course notes and any visual aid equipment that may be required. The Contractor ensures that only competent trainers deliver sessions – evidence of trainer competence to be provided to the Project Manager prior to commencement of training.

Ref: 328 - Mandatory Requirement

The Contractor shall also provide maintenance, support and obsolescence management included software updates to the installed systems by way of a separate maintenance contract. Refer to the Maintenance Contract in for more detail. Provision of System Training and Work Shadowing During the Course of the Works

Ref: 329 - Mandatory Requirement

Prior to Completion of each section of the works, the Client appoints maintenance staff or a maintenance contractor to undertake the operation and maintenance of the works following handover. The Contractor provides any necessary assistance to the Client's staff during the course of the installation to explain the purpose and function of the works.

The Contractor provides a minimum period of fourteen (14) days prior to Completion, to:

- share any lessons learned in the course of the execution of the works that have relevance and impacted on the overall capital costs of the works.
- Complete the training of the Client's operational staff and maintenance staff the day to day running of the Equipment and Materials, and systems.

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17 OCCUPATIONAL SAFETY

Ref: 330 - Mandatory Requirement

The Client will appoint the Contractor as Project Supervisor for the Construction Stage of the Project (PSCS) in accordance with the Safety Health and Welfare at Work Act (Construction) Regulations 2013. The Contractor shall carry out the role of Project Supervisor for the Construction Stage.

It is noted that the site will return to normal train operations during the day. The Contractor's works will generally take place in rolling or moving geographic areas under specific possessions and generally at night. The Contractors teams will be required to sign in and sign out as PSCS for specific areas of the track where the Contractor's works will take place.

Ref: 331 - Mandatory Requirement

The Client will carry out the role of Project Supervisor Design Process. The Contractor will provide all information as required to permit the Client carry out this role, including all information that shall be fed into the Safety File. The Contractor will be a Designer under the Safety Health and Welfare at Work Act (Construction) Regulations 2013.

Ref: 332 - Mandatory Requirement

In limited situations (where it is not possible to handover a geographic area or site) it may be necessary for the other parties to hold the role of PSCS, in these very rare situations the Contractor shall acts as a contractor under the Safety Health and Welfare at Work Act (Construction) Regulations 2013.

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18 TRACK ACCESS

Ref: 333 - Mandatory Requirement

The Contractor shall comply with IÉ's Rule Book and track access requirements. Please refer to Appendix B for further details.

18.1 General

Ref: 334 - Mandatory Requirement

Generally, works shall be required to be carried out under total possession of the railway at night; unless otherwise arranged and agreed with relevant Control Room.

Ref: 335 - Mandatory Requirement

The sites for the required works are located adjacent to live railway lines which are bounded on either side by either both commercial buildings and/or private residential accommodation and/or fencing. Works are only permitted outside of train operation hours and approved in writing by the Client.

Ref: 336 - Mandatory Requirement

Access points to the railway shall be from the access points shown on the project drawings referred to in Section 7.2.8 and possession plans.

Ref: 337 - Mandatory Requirement

Access via track will require possession of the track infrastructure and will therefore be undertaken under possession. The Contractor shall Manage, Plan & Execute Possessions required for the Works.

Ref: 338 - Mandatory Requirement

Possessions are to be submitted twelve (12) weeks in advance, this does not ensure possessions will be granted. Possessions are reserved via a process of work planning at control room meetings held typically three times each week. Monday at 10 am, Wednesday and Friday at 9 am, these dates will be confirmed with successful tenderer.

Ref: 339 - Mandatory Requirement

The Contractor shall allow for the loss of a minimum of 10% of the number & 25% of the duration of possessions booked due to emergency and/or other unforeseen occurrences. Costs associated with this loss of access is deemed to be included within the Contractor's Fee. Please also refer to Section 7.2.1.

Ref: 340 - Mandatory Requirement

The Client will provide the following protection personnel during possessions, PICOP, HS, ES, PIC and TSC. The Client will make available ES's to the Contractor when undertaking his works under possession.

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18.2 Possession and Isolation Planning - Track Access Request

Where possessions are required to undertake works, the Contractor shall make his representative available to attend with the Project Manager and participate in the control room meetings each week. The purpose of the meetings is to communicate and coordinate work activities of all parties working in or travelling through the particular work site area(s).

The table below outlines the maximum number of worksites that can be occupied for any given phase during the implementation of trackside signalling works, programming and commissioning.

Activity Description (Please also refer to Section 2.2 and Section 7.2.2)	Number of worksites per phase	Group
LOC base and cabinet installation (including the earthworks for LOC base, ducting for LOC cables between new TPS LOC and existing LOC / field cable chamber, and installation of cables)	2	Installation
Cable terminations (at LOC, Disconnection box and balise locations)	2	
Balise installation	1	
Programming and Static testing	1	Testing & Commissioning
Dynamic Testing	1	

Table 11 No. of worksites per Phase - per Implementation and Testing activity

* Note –

- The figures quoted in the table above are susceptible to change. In the event of any changes, the Contractor will be notified well in advance to plan accordingly. Worksite cannot be longer than 400m in track length and consist of no more than 2 work groups.
- While the number of sites for any given activity are outlined in the table above (table 10), at any given point of time during the project, the two group of activities (i.e. Installation and Testing & Commissioning) can be ongoing in 2 phases at a time, i.e., 2 phases for Installation and 2 phases for Testing and Commissioning.

18.3 Eight week lookahead and Detailed Method Statement

The Project Manager, in consultation and coordination with the Contractor, must complete and submit a Track Access Request (TAR) form, along with approved Risk Assessment Method Statement (RAMS) for the proposed works to the IM Track Access Coordinator a minimum of five (5) weeks in advance of the commencement of the eight-week block.

Typical TAR submission dates for 2025 are shown on Table 11 and Table 12 below as an example on the following page: -

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TAR Submission Deadline	2025 Eight-Week Block
17/03/2025	Weeks 17 to 24 inclusive
12/05/2025	Weeks 25 to 32 inclusive
07/07/2025	Weeks 33 to 40 inclusive
01/09/2025	Weeks 41 to 48 inclusive
27/10/2025	Weeks 49 (2025) to 4 (2026) inclusive

Table 12 TAR Submission Date Deadlines (2025)

Periodic calendar 2025													
Period	1	2	3	4	5	6	7	8	9	10	11	12	13
Week	1	5	9	13	17	21	25	29	33	37	41	45	49
Monday	Dec. 30	Jan. 27	Feb. 24	Mar. 24	Apr. 21	May. 19	June. 16	July. 14	Aug. 11	Sep. 8	Oct. 6	Nov. 3	Dec. 1
Tuesday	31	28	25	25	22	20	17	15	12	9	7	4	2
Wednesday	Jan. 1	29	26	26	23	21	18	16	13	10	8	5	3
Thursday	2	30	27	27	24	22	19	17	14	11	9	6	4
Friday	3	31	28	28	25	23	20	18	15	12	10	7	5
Saturday	4	Feb. 1	Mar. 1	29	26	24	21	19	16	13	11	8	6
Sunday	5	2	2	30	27	25	22	20	17	14	12	9	7
Week	2	6	10	14	18	22	26	30	34	38	42	46	50
Monday	Jan. 6	Feb. 3	Mar. 3	Mar. 31	Apr. 28	May. 26	June. 23	July. 21	Aug. 18	Sep. 15	Oct. 13	Nov. 10	Dec. 8
Tuesday	7	4	4	Apr. 1	29	27	24	22	19	16	14	11	9
Wednesday	8	5	5	2	30	28	25	23	20	17	15	12	10
Thursday	9	6	6	3	May. 1	29	26	24	21	18	16	13	11
Friday	10	7	7	4	2	30	27	25	22	19	17	14	12
Saturday	11	8	8	5	3	31	28	26	23	20	18	15	13
Sunday	12	9	9	6	4	June. 1	29	27	24	21	19	16	14
Week	3	7	11	15	19	23	27	31	35	39	43	47	51
Monday	Jan. 13	Feb. 10	Mar. 10	Apr. 7	May. 5	June. 2	June. 30	July. 28	Aug. 25	Sep. 22	Oct. 20	Nov. 17	Dec. 15
Tuesday	14	11	11	8	6	3	July. 1	29	26	23	21	18	16
Wednesday	15	12	12	9	7	4	2	30	27	24	22	19	17
Thursday	16	13	13	10	8	5	3	31	28	25	23	20	18
Friday	17	14	14	11	9	6	4	Aug. 1	29	26	24	21	19
Saturday	18	15	15	12	10	7	5	2	30	27	25	22	20
Sunday	19	16	16	13	11	8	6	3	31	28	26	23	21
Week	4	8	12	16	20	24	28	32	36	40	44	48	52
Monday	Jan. 20	Feb. 17	Mar. 17	Apr. 14	May. 12	June. 9	July. 7	Aug. 4	Sep. 1	Sep. 29	Oct. 27	Nov. 24	Dec. 22
Tuesday	21	18	18	15	13	10	8	5	2	30	28	25	23
Wednesday	22	19	19	16	14	11	9	6	3	Oct. 1	29	26	24
Thursday	23	20	20	17	15	12	10	7	4	2	30	27	25
Friday	24	21	21	18	16	13	11	8	5	3	31	28	26
Saturday	25	22	22	19	17	14	12	9	6	4	Nov. 1	29	27
Sunday	26	23	23	20	18	15	13	10	7	5	2	30	28

Table 13 IE Periodic Calendar 2025

The submitted TAR must include as much detailed information as possible, including inter alia Scope, CCE Division number, access/egress points, site limits (showing mileposts), axle counter resets required, works adjacent to CCTV level crossings OHLE isolations required (if applicable), etc

Early engagement with the Project Manager and in consultation and coordination with the Contractor and with the Client's Track Access Coordinator is required to ensure accuracy of the information contained within the submitted TAR.

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The approved RAMS must have gone through the correct RAMS Review Process as illustrated in the flow chart on Table 14 RAMS Review Process for CI Contractor. The project team must specify in the RAMS, the type of protection and the track protection resources required.

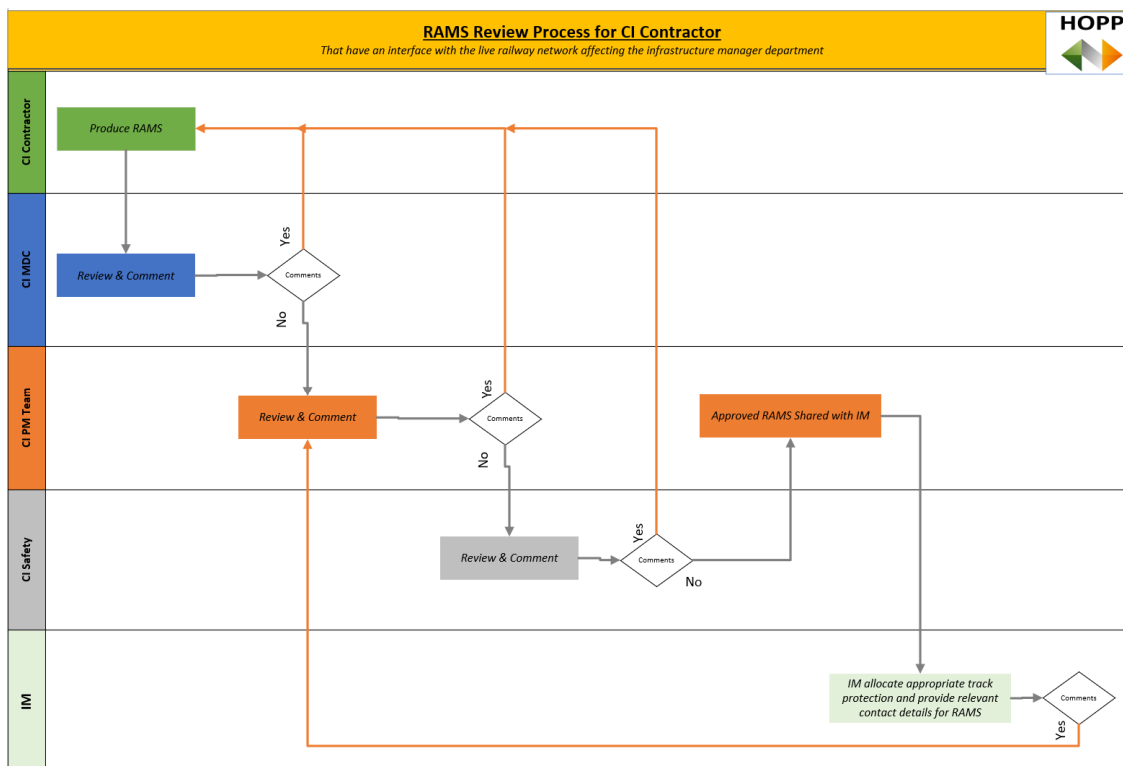


Table 14 RAMS Review Process for CI Contractor

When preparing the RAMS and the TAR, please note that any personnel who go “on or near the line” are required at a minimum to have Personal Track Safety (PTS) certification, Safe Pass certification and Manual Handling certification. All personnel must comply with IM-SMS-026 Personal Track Safety and IM-SMS-026-FAQ-1, which set out the requirements for the safety of persons required to go on or near the railway line and failure to comply will result in track access being refused.

Once submitted by the Project Manager (in consultation and coordination with the Contractor), the Client’s Track Access Coordinator will manage the TAR on behalf of the project, liaise with the relevant CCE and SET Regional Managers as necessary and attend the relevant CCE and SET Control Room Meetings. The Client’s Track Access Coordinator will also advise the Project Manager if there will be any potential clashes or track access issues and advise on any solutions or alternatives. This feedback will be provided at least two weeks in advance of the commencement of the eight-week block. If there are no issues, the required track access will be booked by the Client’s Track Access Coordinator.

18.4 Possession: Handover & Handback

Ref: 341 - Mandatory Requirement

The Contractor shall at all times comply with the Client’s established Handover & Handback PSCS procedures, including completion of the required Handback documentation, refer to Appendix B. Failure to comply with these procedures may result in potential costs incurred by the Client in relation to delayed Handback to the Client. These potential costs will be levied to the Contractor's account. These costs are

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set out below in Section 17.7 below. The contractor must submit the PSCS handover form and a shift report completion post shift. This must be submitted to Project Manager and other Client representatives.

18.5 Nighttime Possessions

The table below outlines the access / night possession times for each phase of the TPS Trackside NRO.

Phase Number	Phase Description	Possession Start	Possession End
1	Cherryville – Waterford	23:30	04:50
2	Maynooth – Sligo	23:40	04:30
3	Portarlinton – Galway	00:10	04:30
4	Athlone - Westport / Ballina	23:40	04:20
5	Wicklow - Rosslare	23:45	04:30
6	Mallow - Tralee	23:40	04:30
7	Limerick - Athenry	23:05	04:30
8	Limerick - Limerick Junction	23:45	04:45
9	Hazelhatch - Mallow	01:00	04:45

Table 15: – Trackside NRO - Access / Night Possession times for each phase

18.6 Special Possessions

Ref: 342 - Mandatory Requirement

The Contractor will be notified in the event that the Client's maintenance team or Others reserve additional possessions outside the aforementioned times referenced in Section 17.5 above. During such times the Contractor may be provided additional access to the relevant worksites.

18.7 Utilisation of Available Possessions

Ref: 343 - Mandatory Requirement

The Contractor shall reserve possessions in accordance with the established procedure for doing so and through the established control room procedure for control of access to rail line. The Contractor is reminded that failure to utilise possessions which have been reserved for its use may have both timing and commercial consequences. If a Contractor is unable to utilise a possession which has been reserved primarily for its works adequate notice must be given to the Client team and appropriate Control Room.

Ref: 344 - Mandatory Requirement

The adequate notice for cancelling planned possession-isolation shall be 48 hours prior to the start of the possession-isolation. This time period is to source and allocate alternate maintenance type works to the possession-isolation. In the event that no other works are undertaken, the cost of the possession-isolation will not be levied to the original scheduled user if cancelled before the 48-hour window.

Cancellation of possessions-isolations after this 48-hour notice period may result in the costs identified, being charged to the original scheduled user as stated. Following cancellation by the Contractor of greater than 15% of available planned possession-isolations, the Client reserves the right to suspend the

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granting of further possession-isolations until a resolution of the issues related to cancellation has been resolved.

Costs associated with any such suspension of possession-isolations will be the responsibility of the Contractor.

Cost of Possession:

- Non-Disruptive - €1,750.00 per hour (No disruption to regular passenger train service);
- Disruptive - €2,500.00 per hour or as advised (Disruption to regular train service).

18.8 Holiday Period/Days in the Republic of Ireland

Ref: 345 - Mandatory Requirement

Holiday period/days may generally not be utilised to work on the rail network and as such will be unavailable for possession works (or any works by the Contractor). The Contractor should allow for these days within their Programme.

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19 REPORTING

Ref: 346 - Mandatory Requirement

The Contractor shall submit a period end report pack every four (4) weeks, by noon on the last Friday of the Client's period (there are 13nr, 4-week periods in the Clients calendar year). The period end report pack shall include, as a minimum, a progress report, the programme submitted for acceptance and narrative, the risk register, cost reports and any other reports required by the Project Manager. The detailed format of all of the reports will be agreed with the Project Manager prior to the submission of the first period end report pack and not later than eight (8) weeks from the contract commencement date.

The progress report shall include the following as a minimum:

- Executive Summary and Key Issues.
- Health and Safety, Environmental.
- Deliverables Tracker
- Programme summary, including a variance table on the important dates which shall be agreed with the Project Manager Contractor's Resources.
- Design and Technical detail.
- Contractor's Procurement activities (if any) in delivering the works.
- Risk Management.
- The Contractor's concerns.
- Quality and Assurance.
- Financial & Commercial summary, high level Red, Amber, Green status.
- Performance dashboard summary, metrics for this to be agreed with the Project Manager.
- Issues Register, Change Request Register.

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20 DELIVERABLES TRACKER

Ref: 347 - Mandatory Requirement

The deliverables tracker is a project management tracker that defines and tracks the review and acceptance of project management documents including plans and systems designs. Within 8 weeks of contract award the Contractor shall provide a deliverables tracker detailing each deliverable to be provided to the Project Manager to deliver the Works. This will include the deliverable name, version number, a description of the deliverable, the owner, forecast or actual delivery date, and status of due diligence reviews.

The deliverables tracker will be maintained throughout the duration of the project and submitted to the Project Manager each period.

Format and content to be agreed with the Project Manager within eight (8) weeks of the contract commencement date.

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21 PROGRAMME

Notwithstanding the requirements from the core clauses of this contract, this section describes the requirements in relation to the Programme for the works.

Ref: 348 - Mandatory Requirement

The Programme shall be managed in Primavera P6.

Ref: 349 - Mandatory Requirement

The use of hard constraints shall be avoided in the Programme. The standard calendar should be five-day working and include non-working days for all Irish public holidays and two (2) weeks non-working for Christmas. Additional calendars can be used for different working times but must be highlighted in the narrative. The use of negative lags shall be avoided, and any positive lags shall be identified in the programme narrative and kept to a minimum.

21.1 Programme Standards

Ref: 350 - Mandatory Requirement

The Programme shall as a minimum:

- Include the entire Scope.
- Show all activities with a predecessor and successor, apart from the first and last task.
- Have a critical path with no negative float,
- Have no durations greater than thirty (30) days, except in procurement & manufacturing.
- Keep constrained dates to a minimum and document the reasons for them in the notes for the task.
- Record a version and period number. Version number format as follows:

Period YYYY-XX YYYYMMDD.

Show the "Data Date" titled "progress to xxxx" and the period number in the layout for the PDF.
Full PDF layout to be agreed with the Project Manager.

21.2 The First Programme

Ref: 351 - Mandatory Requirement

The first programme issued for acceptance shall include the order and timing of the following:

- When the Contractor plans to submit particulars of the design required by the Scope.
- When the Contractor plans to submit particulars of the design of any items of Equipment and Materials required by the Scope.
- The work of the Client and Others either as stated in the scope of works or as later agreed with them by the Contractor.
- All milestones as described in the Scope.
- The dates when, to provide the works in accordance with his programme, the Contractor will need information.

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- Information provided by the Project Manager, the Client or Others.
 - Equipment and Materials and other things to be provided by the Client.
 - Procurement of any Equipment or Materials provided by the Contractor.
 - Details of third-party approval submissions and related approvals, allowing sufficient time for each stage of the process with allowance for resubmission. Any Client review periods should be 20 working days in duration unless otherwise agreed with the Project Manager.
 - The Contractor is to clearly identify activities related to any extended possessions (as agreed with the client). The extended possession period shall be clearly identified.
 - In advance of any works that require taking trains out of service, to provide an additional 'hour-by-hour' and resource loaded programme for acceptance by the Project Manager in a timescale agreed with the Project Manager.
 - Critical interfaces/dependencies.
 - Clearly identify milestones for Client inputs for information or decisions required by the Contractor (e.g., approvals, consents).
 - Clearly identify milestones for Contractor outputs to the Client for all information or physical delivery by the Contractor taking into account agreed response timescales.
 - **This Programme is to be submitted in the timescale identified within the Contract.**

21.2.1 Programme Structure

Ref: 352 - Mandatory Requirement

The proposed work breakdown structure of the programme and its components are detailed below. The programme shall include these items as a minimum:

1. POAP - Plan on a page
2. Key Milestones (including Sectional Completion Dates, and Key Dates)
3. Generic design.
4. Phase
 - a. Client Inputs.
 - b. Procurement.
 - c. Manufacturing and Delivery.
 - d. Survey and site correlation.
 - e. Specific design and data production.
 - f. Installation.
 - g. Testing & Commissioning.
 - h. Safety Assurance.

The Contractor shall submit the Work Breakdown Structure to the Project Manager for acceptance.

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21.2.2 First Programme Narrative

Ref: 353 - Mandatory Requirement

The first programme issued for acceptance shall be supported by a written narrative including but not limited to the following headings:

- Executive summary including phase, contract and planned start and finish dates.
- A description of the sequence of works including the critical path.
- Key milestones (including Commencement Dates, Access Dates, Key Dates and Sectional Completion Dates) and dependencies.
- Key risks.
- Assumptions including resource assumptions, productivity rates and number of shifts for durations for each set of activities.
- Calendars and Coding Structures used.

21.3 Programme Cost Loading

Ref: 354 - Mandatory Requirement

The Activity Schedule and the Programme shall be aligned. The Programme shall be cost loaded to produce the cost profile each period. The financial period calendar within Primavera P6 shall be aligned to the Client's 13 periods.

21.4 Programme Change

Ref: 355 - Mandatory Requirement

Any change to the Programme contained within a programme issued for acceptance, such as addition/deletion of activities, changes to logic, changes to dependencies or durations will be recorded and reported in the supporting programme narrative.

21.5 Programme updates

Ref: 356 - Mandatory Requirement

The revised programme issued for acceptance is to be submitted on a four-weekly basis to align with the Client's 13 period reporting cycle. This revision will be submitted to the Project Manager by noon on the Friday before the Client period end. The outputs from this update will be issued as part of the period report pack, which will include a copy of the native .xer file, a PDF of the programme and the programme narrative. The data date to be used for the period update will be the last Sunday of the period.

Activities that are in progress are updated to show the physical % complete and remaining duration, where the remaining duration is the best estimate of the time it will take to complete the activity.

There will also be no activities that have actual dates shown in the future.

21.5.1 Programme update Narrative

Ref: 357 - Mandatory Requirement

The revised programme issued for acceptance shall also be supported by a written programme narrative including but not limited to the following headings:

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-
- Executive summary including any variance to the contract finish date and critical issues.
 - Key milestones and dependencies along with any variance.
 - Key risks.
 - Progress summary.
 - Mitigations.
 - Critical path analysis.
 - Assumptions.
 - Reports –
 - ◆ A programme change log,
 - ◆ a variance report, including CPI and SPI
 - ◆ a two and four-week lookahead,
 - ◆ a key activities report – detailing key activities that were planned to be achieved and haven't not been, key activities that have been achieved and key activities that are planned to be achieved in the next period.
 - ◆ Other reports - may be requested by the Project Manager as required. Formats of all reports are to be agreed with the Project Manager.

Layout to be agreed with the Project Manager.

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22 CONTRACTOR'S TEAM

In the spirit as intended under the Contract, collaborative working throughout delivery (so as to inform the outcome), is required across the various technical, operational, and commercial departments of Iarnród Éireann.

Ref: 358 - Mandatory Requirement

Within four (4) weeks of commencement date the Contractor shall provide to the Project Manager for acceptance an organisational chart showing the Contractor's project team deemed necessary to deliver the requirements, as set out in this Scope. Any change to the organisation chart must be notified in advance of the change to the Project Manager for acceptance. The proposed replacement will be of an equivalent level of qualifications and experience.

Ref: 359 - Mandatory Requirement

The Contractor shall submit CV's together with qualifications, competencies and number of year experience for key management and engineering team members. These are as follows and will be classed as Key Persons identified in the Contract Data:

- Contractor's Project Manager
- Safety Assurance Manager
- Scheduler
- Technical Design Lead
- Engineering Manager
- Safety Coordinator

The Contractor shall also provide the same details for the proposed Deputies for the roles above.

Ref: 360 - Mandatory Requirement

The Contractor shall include in the period progress report, a resource report showing 'planned headcount' v 'actual headcount' and in such detail as to enable the Project Manager to fully review.

Ref: 361 - Mandatory Requirement

The Contractor's project management staff will be required to attend face to face meetings in Ireland, the frequency is to be determined but a minimum allowance required is once per period (note client operates a 13 period calendar) and this should include the Contractor's Project Manager and Planner. Attendance by the Contractor's other project team members will be required from time to time.

Ref: 362 - Mandatory Requirement

The Contractor shall comply with all the relevant competency standards required to carry out the works.

Ref: 363 - Mandatory Requirement

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The Contractor shall submit a training plan detailing any training required by their team to support the project delivery and the proposed delivery method and timing. This will include, but not limited to, PTS, Safe Pass, manual handling. The training plan shall be submitted to the Project Manager for acceptance within four (4) weeks of the contract commencement date.

Ref: 364 - Mandatory Requirement

The Contractor shall maintain a team competency register for all project staff, detailing number of year experience, qualifications and competencies, together with competency renewal dates and proposed training. The competency register shall be maintained for the duration of the project and available for review by the Project Manager. See list of standards including required competencies in Appendix B.

22.1 Responsible, Accountable, Consulted and Informed (RACI) matrix

Ref: 365 - Mandatory Requirement

The Contractor and the Project Manager shall develop a Responsible, Accountable, Consulted and Informed (RACI) matrix within four (4) weeks of the contract commencement date. Thereafter, the RACI matrix shall be updated by the Contractor and provided to Project Manager every twenty-four (24) weeks throughout the contract duration.

Ref: 366 - Mandatory Requirement

The (RACI) matrix shall describe the Contractor's internal responsibilities clearly describing the interfaces existing between the various offices or teams involved in the design and implementation.

Ref: 367 - Mandatory Requirement

The (RACI) matrix shall clearly describe the expected inputs from the Project Manager and Client and their responsibilities. The Project Manager has the right to reject the proposal from the Contractor if they are not deemed aligned with the principle of this Scope.

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23 PROJECT EXECUTION PLAN

Ref: 368 - Mandatory Requirement

The Contractor shall submit a project execution plan within four (4) weeks of the contract commencement date for acceptance.

The project execution plan shall describe the works to be delivered, the delivery approach, duties and responsibilities of the sub delivery teams including their location, competency and relationship to the Contractor. The project execution plan shall describe the project controls process and tools to be deployed to manage the works. The project execution plan shall describe the sequence and timing of the works. The project execution plan shall be informed by Irish Rail Project Management Standards and procedures, IE PMP 01 to IE PMP 10 (See Appendix B).

The project execution plan shall include but not be limited to the following.

- Describe the Scope
- Describe the Execution Strategy
- Project structure and team organisation
- Description of Contractor resource management process, obtaining, retaining and managing Contractor's resource, team development and competency management, resource management and resource location.
- Details of how the Contractor is going to coordinate regional teams (if appropriate)
- Stakeholder Management and communications
- Describe the design management process, including but not limited to, product approval, designer competency approval, sub-contractor designer approvals.
- Describe the Health and Safety and Environmental Management process and tools.
- Describe the implementation management process, including daily work planning, possession planning, access planning, site reporting,
- Describe the test and commissioning process
- Describe the safety assurance process including their party approvals at all stages of delivery.
- Describe the project control procedures that will be employed and any tools.
- Describe the procurement, contract and cost management approach and tools.
- Describe the risk management approach and risk management deliverables.
- Describe the Document management process & tools, together with the key document structure.
- Describe the deliverables management process and tools.
- Describe the approach to data management, use of Building Information Model (BIM)
- Describe the Performance management process including, KPIs.

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24 COST MANAGEMENT

Ref: 369 - Mandatory Requirement

The contractor shall submit a cost management plan within 4 weeks of the contract start date. The plan shall describe the process, tools and team proposed to manage the cost throughout the duration of the contract. The contractor shall maintain adequate records of all costs and expenditures and make them available to the client for review and audit on request.

Ref: 370 - Mandatory Requirement

Each period the Contractor shall provide labour, materials, subcontractors detailed records at transactional level, together with a summary report at the agreed cost structure level, including cost to date, cost per period, periodic forecast to completion. The subcontractor shall comply with the Clients requirements, including the Project Management requirements as if it were the contractor.

Ref: 371 - Mandatory Requirement

The Contractor shall produce a variance report detailing cost deviations from the agreed forecast, together with commentary and mitigating actions. Periodic cost report formats and content to be agreed with the Project Manager.

Ref: 372 - Mandatory Requirement

The payment cycle will follow the Client 4 weekly period reporting cycle (there are 13nr, 4-week periods in the Clients calendar year). The contractor will issue a tax invoice for the assessment amount, including VAT where applicable and payment shall be within four weeks of the date of receipt of the Contractor's tax invoice, where relevant.

Ref: 373 - Mandatory Requirement

All payments are deemed to be interim on accounts payments until such time as the Final Assessment has been issued .Any errors in an interim payment on account may be adjusted for in subsequent interim payments on account.

Ref: 374 - Mandatory Requirement

The contractor will list all group and associated companies engaged in the delivery of the contract, identifying the legal jurisdiction of each entity. The relationship of each company to the contractor and the parent company should also be identified.

Ref: 375 - Mandatory Requirement

The contractor will identify all transactions which are between associated or group companies.

Ref: 376 - Mandatory Requirement

The contractor will identify intercompany markups and transfer pricing adjustments

Ref: 377 - Mandatory Requirement

For each group company or jurisdiction, a sample calculation of the costs of employment shall be provided at tender stage prior to award of contract.

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Ref: 378 - Mandatory Requirement

The contractor shall identify any restrictions or limitations, legal or otherwise, relating to the provision of detailed payroll and cost information relating to its employees or subcontractors.

Ref: 379 - Mandatory Requirement

The contractor shall identify any restrictions or limitations relating to General Data Protection Regulation (GDPR) or General Data Protection principles, relating to the provision of detailed payroll and cost information relating to its employees or subcontractors.

Ref: 380 - Mandatory Requirement

All staff will submit timesheets on a weekly basis, detailing start and finish times, allocation of hours worked on each project, each activity and location of work. Any non-productive time to be recorded on the timesheets (holidays, training, etc). Time booked to other contracts and projects to be recorded and available for inspection.

Ref: 381 - Mandatory Requirement

All timesheets to be approved by contractor's line management on a weekly basis

Ref: 382 - Mandatory Requirement

The contractor will facilitate access to any attendance or time recording recording systems used.

Ref: 383 - Mandatory Requirement

The contractor will facilitate access to any attendance or time recording recording systems used.

Ref: 384 - Mandatory Requirement

The contractor will facilitate access to all documents to support Staff costs (and the payment of these costs) – including but not limited to; timesheets, payslips, invoices, bank statements, expense claim forms, journal vouchers, petty cash voucher, employment contracts, subsistence claims, pension returns

Ref: 385 - Mandatory Requirement

The contractor provide a list of all staff working on the contract with their roles and rates each period .

Ref: 386 - Mandatory Requirement

The contractor will provide an updated organisation charge each period.

Ref: 387 - Mandatory Requirement

The contractor will provide details of payment terms for all categories of cost components in every Jurisdiction.

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Ref: 388 - Mandatory Requirement

The contractor will provide evidence of actual payments for all costs including but not limited to, bank statements remittance advices and payslips.

Ref: 389 - Mandatory Requirement

Any costs in relation to the provision of Information Technology (IT) services or equipment to the contractors staff is deemed to be in the fee.

Ref: 390 - Mandatory Requirement

Any costs in relation to the provision of mobile phones to the contractors staff is deemed to be included in the fee.

Ref: 391 - Mandatory Requirement

Any costs in relation to indirect or head office staff is deemed to be included in the fee.

Ref: 392 - Mandatory Requirement

Any costs in relation to the provision of Office accommodation other than accommodation for site based staff is deemed to be in the fee. Any cost for utilities, including but not limited to electricity, gas and water supply in relation to office accommodation is deemed to be included in the fee.

Ref: 393 - Mandatory Requirement

The contractor will use an activity based costing system, using WBS (Work Breakdown structure) or CBS (Cost breakdown structure) in its accounting and time recording systems to identify spend to date and cost to completion for each category or phase of expenditure.

Ref: 394 - Mandatory Requirement

The contractor will calculate productivity rates for those activities with measurable outputs.

Ref: 395 - Mandatory Requirement

The contractor will confirm which employees are entitled to overtime payments and which are not.

Ref: 396 - Mandatory Requirement

Agency and subcontracted staff are required to produce timesheets as if they were employees.

Ref: 397 - Mandatory Requirement

Accounting transactions relating to agency and subcontracted staff are required to show units (hours/days), rate, total, purchase order number, and supplier name.

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25 PROCUREMENT AND CONTRACT MANAGEMENT

Ref: 398 - Mandatory Requirement

The Contractor shall submit a procurement plan to the Project Manager for acceptance within four (4) weeks of the contract commencement date detailing the major element of equipment, material, plant, subcontracts or services to be awarded and how they will be procured.

Ref: 399 - Mandatory Requirement

The Contractor shall provide supporting evidence of competencies and submit to the Project Manager for acceptance prior to appoint the subcontractor.

Ref: 400 - Mandatory Requirement

The Contractor shall submit a material delivery schedule to the Project Manager for acceptance.

Ref: 401 - Mandatory Requirement

The Contractor shall submit a procurement report each period. The report will include as a minimum subcontractor tender status, contract change registers , including agreed and proposed changes material delivery schedules, long lead item procurement and delivery.

Ref: 402 - Mandatory Requirement

The Contractor and the Project Manager shall use an electronic hosted web server-based contract administration management system provided by the Client. The Client will setup and administer the system and provide access and training in the system to the Contractor.

The Contractor shall demonstrate their experience in the use of contract management systems.

Ref: 403 - Mandatory Requirement

The contractor will present subcontractors control registers to the client each period in a format to be agreed with the client.

Ref: 404 - Mandatory Requirement

The contractor will present any proposed changes to the subcontracts to the client prior to formal issue and agreement.

Ref: 405 - Mandatory Requirement

The contractor will provide commitment reports to the client each period in a format to be agreed with the project manager.

Ref: 406 - Mandatory Requirement

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The contractor will provide a list of rebates and discounts received from suppliers, either at project , local or corporate level.

Ref: 407 - Mandatory Requirement

The contractor will maintain a register of on hire equipment and vehicles

Ref: 408 - Mandatory Requirement

The contractor will maintain a register of small tools.

Ref: 409 - Mandatory Requirement

The contractor will maintain a register of all materials used and in stock , in a format to be agreed with the project manager.

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26 DOCUMENT CONTROL

Ref: 410 - Mandatory Requirement

The Contractor shall produce a Document Management plan to be submitted to the Project Manager for acceptance within four (4) weeks of the contract commencement date. The plan shall detail how the Contractor will manage, control and track all documentation and communications in relation to the works. The Contractor shall track all incoming and outgoing correspondence, which shall be kept up to date and be accessible to the Project Manager on request.

Ref: 411 - Mandatory Requirement

The Contractor and the Project Manager shall use an electronic hosted web server-based document management system provided by the Client. The Client will setup and administer the system and provide access and training in the system to the Contractor.

Ref: 412 - Mandatory Requirement

The Document Management system shall contain all project documents, including but not limited to, drawings, specifications, site queries, photographs, inspection and testing reports shall be accessible to the Project Manager.

Ref: 413 - Mandatory Requirement

The Contractor shall follow the Client document review process as set out in Appendix B (SET DRN process).

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27 ACCESS AND TRAINING FOR SYSTEMS

The Client will setup and administer the Contract Administration Management and the Document Management system.

The Client will provide the Contractor with access to systems operating on hosted web servers. The Client will be responsible for support and maintenance, including database administration.

The Project Manager will approve the system user accounts.

Ref: 414 - Mandatory Requirement

The Contractor shall nominate users for these systems and notify the Project Manager immediately of any changes necessary to user accounts to avoid misuse of the systems. The Contractor will be held liable for any misuse of these systems by its nominated users.

Ref: 415 - Mandatory Requirement

The Client will provide training in the use of these systems to the Contractor's nominated representatives on a 'train the trainer' basis. It will be the responsibility of the Contractor to train any further users.

Ref: 416 - Mandatory Requirement

The Contractor is to nominate an interface point within the Contractor's team to execute the functions required (including agreeing appropriate distribution lists for notices and communications).

Ref: 417 - Mandatory Requirement

The Contractor shall comply with the Client's document controls procedures PMP-007_Document Management Control & Procedure unless deviations are agreed with the Project Manager in advance.

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28 RISK MANAGEMENT

Ref: 418 - Mandatory Requirement

The Contractor shall develop a Risk & Opportunities Management plan and provide to the Project Manager within eight (8) weeks of the contract commencement date. This shall detail how the process, responsibilities and output from the risk management process will be applied by the Contractor to manage risks arising from the Contractor's activities (including the management of its suppliers and sub-contractors). The plan shall also address how the Contract early warning process (as set out in the core clauses) will be applied including the escalation of risks outside of the Contractors control.

All Risk and Opportunities Management shall be in compliance with PMP-002 Risk & Contingency Management Procedure.

Ref: 419 - Mandatory Requirement

The Contractor shall maintain a Risk Register which includes at a minimum the risk description, risk owner, risk proximity date, risk assessment information, mitigation actions. The risk register shall be submitted each period in MS excel format for the Project Manager's review.

Ref: 420 - Mandatory Requirement

The Contractor shall participate in joint risk reviews with the Project Manager on a four (4) weekly basis, to be held throughout the duration of the project.

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29 QUALITY MANAGEMENT

The Contractor shall implement and document comprehensive quality planning processes, in a project specific Quality management Plan and submit it to the Project Manager for acceptance within eight (8) weeks of contract commencement date.

Ref: 421 - Mandatory Requirement

The Contractor shall ensure the quality of all activities involved in the design, delivery, integration, and commissioning of the ETCS trackside system, and shall comply with ISO 9001 in his works. The Contractor's quality processes shall incorporate the principle of plan-do-check-act as defined in ISO 9001:2015 into each of the engineering management plan.

Ref: 422 - Mandatory Requirement

The Contractor shall comply with the processes defined in the Client's IM Standards, particularly the Quality Manual (I-SET-1013) issued by the Client's IM Signalling, Electrical and Telecom (SET) Department.

Ref: 423 - Mandatory Requirement

The Contractor shall appoint a Quality Assurance Manager responsible for ensuring implementation of the Quality Management System.

Ref: 424 - Mandatory Requirement

The Contractor shall evaluate the competence of key personnel to ensure necessary resources, experience, systems, and technical capabilities are available to meet project requirements. The Contractor shall maintain an internal competence register for all roles in their database.

Ref: 425 - Mandatory Requirement

The Contractor shall maintain an internal competence management procedure compliant with ISO 9001.

Ref: 426 - Mandatory Requirement

The Contractor shall provide their staff with all product and project specific trainings required for the delivery of this Scope and retain evidence of competence for all project staff, including the completion of the identified training needs.

Ref: 427 - Mandatory Requirement

The Contractor shall assess and implement the independence of roles requirements as defined in EN 50126 and EN 50716 standards.

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Ref: 428 - Mandatory Requirement

The Contractor shall implement a workload management process to ensure quality of the deliverables and installations. Such measures may include Work in Progress (WIP) limits for concurrent tasks per key roles, monitoring of regular work outside core hours, and a delay attribution process to identify low resourcing levels.

Ref: 429 - Mandatory Requirement

The Contractor shall define Key Performance Indicators (KPIs) for each process and monitor implementation of process quality.

Ref: 430 - Mandatory Requirement

The Contractor's Quality Assurance Manager shall conduct regular and ad-hoc project internal quality audits and report results to the Project Manager. These audits shall cover site inspections and audits of the processes covered by the Quality Management Plan.

Ref: 431 - Mandatory Requirement

The Contractor shall perform quality gate reviews and quality certifications at each baseline release. The Contractor shall demonstrate clear links and traceability between outputs at each project stage and the quality assurance activities completed during the stage to create a fully auditable trail.

Ref: 432 - Mandatory Requirement

The Contractor shall document results of quality issues investigations in the gate review report.

Ref: 433 - Mandatory Requirement

The Contractor shall implement principles specified by ISO processes for requirements engineering and the INCOSE guide for requirements quality.

Ref: 434 - Mandatory Requirement

The Contractor shall implement and monitor a quality assurance process for geodetic survey and data processing methods.

Ref: 435 - Mandatory Requirement

The Contractor shall provide KPI quality reporting for design processes, including design pass rate, open RFIs, and open non-conformities.

Ref: 436 - Mandatory Requirement

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The Contractor shall release declarations of compliance for all identified equipment and purchased products according to relevant standards and the ISO/IEC 17050-1 conformity assessment process.

Ref: 437 - Mandatory Requirement

The Contractor shall develop and complete inspection checklist reports in accordance with product-specific instructions.

Ref: 438 - Mandatory Requirement

The Contractor shall conduct all external laboratory tests in a CEI ISO17025 accredited laboratory.

Ref: 439 - Mandatory Requirement

The Contractor shall conclude each installation stage with a quality certificate referencing all inspection and testing activities performed on site.

Ref: 440 - Mandatory Requirement

The Contractor shall ensure testing staff have documented proof of competence in accordance with SET-SMS-004.

Ref: 441 - Mandatory Requirement

The contractor shall implement the QMS Risk Assessment Matrix to take advantage of opportunities and mitigate project risks.

Ref: 442 - Mandatory Requirement

The Contractor shall circulate a consolidated report that includes both progress and quality reporting.

Ref: 443 - Mandatory Requirement

The Contractor shall present quality information in monthly reports in both narrative and quantitative form, including text, charts, graphs, and photographs.

Ref: 444 - Mandatory Requirement

The Contractor shall include a summary of quality inspections and activities conducted on site during the reporting period.

Ref: 445 - Mandatory Requirement

The Contractor shall report the status of Quality Audit findings when applicable.

Ref: 446 - Mandatory Requirement

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The Contractor shall produce bi-weekly quality reporting during site installation stages of the project.

Ref: 447 - Mandatory Requirement

The Contractor shall identify, investigate, control, and rectify non-conforming material and unintended incidents.

Ref: 448 - Mandatory Requirement

The Contractor shall immediately escalate Critical non-compliances through the steering committee.

The Contractor shall submit investigation reports and recovery measures within twenty (20) days of any critical or major incident. During this period, the Contractor shall provide daily reporting of quality for critical and major non-compliances.

Ref: 449 - Mandatory Requirement

The Contractor shall record moderate and minor incidents in the issues register with identified owners for corrective actions.

Ref: 450 - Mandatory Requirement

The Contractor shall update status of moderate incidents in weekly or monthly reports based on the delivery stage requirements.

Ref: 451 - Mandatory Requirement

The Contractor's quality assurance process shall support the tracking of non-compliances and the implementation of the Fault Reporting, Analysis and Corrective Action (FRACAS) Process.

Ref: 452 - Mandatory Requirement

The Contractor shall include updates of quality inspections, failure reporting, and open non-conformities in bi-weekly reporting.

Ref: 453 - Mandatory Requirement

The Contractor shall conduct project reviews and Lessons Learned sessions, in coordination with the Project Manager.

Ref: 454 - Mandatory Requirement

The Contractor shall perform stakeholder satisfaction consultations as part of quality monitoring.

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30 LESSONS LEARNED

Ref: 455 - Mandatory Requirement

The Contractor shall participate in a joint Client led lesson learned workshops with the Project Manager. The scope of these workshops will include but not limited to:

- A review of historic lessons learned from other projects of similar scope.
- A review of historic lessons learned from the design, installation works, testing and safety assurance.
- Lessons learned workshop following the completion of the design stage, hardware design phases and site activities (including implementation and programming and all testing), assurance activities for all delivery phases as set out in Section 3 of this document.
- Lessons learned workshop for procurement activities and subcontractor performance.

Ref: 456 - Mandatory Requirement

The Contractor shall maintain their own lesson learned register. The lesson learned register will be maintained for the duration of the project and available for review by the Project Manager.

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31 MEETING MANAGEMENT

Effective meeting management is essential for efficient management of the works.

Ref: 457 - Mandatory Requirement

The Contractor shall ensure meetings are well managed, including a clear purpose and agenda together with a record of actions and decisions. The Contractor shall attend or manage the following meetings to be agreed with the Project Manager and ensure a suitable representative attend.

- 2 weekly progress review meetings
- 2 weekly Contractor review meetings
- 1 Weekly Technical review meetings. Additional meetings as required
- Design meetings, HAZOPS as required
- Assurance management meetings as required
- Early warning meetings as required (in accordance with the core clauses)
- Risk Management meetings
- Periodic Contract management meetings
- Periodic Cost management meetings
- Periodic Steering group meetings

Ref: 458 - Mandatory Requirement

The Contractor shall notify the Project Manager of meetings between themselves and their subcontractors. The Project Manager or their nominated representative will have the right to attend, and/or receive the outputs from these meetings. The Contractor's procurement process shall include the requirement for subcontractors to attend meetings as required by the Project Manager.

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32 BUILDING INFORMATION MODEL

Ref: 459 - Mandatory Requirement

The Contractor shall develop a Data and BIM Execution Plan (BEP) detailing processes for information management, data exchange, and file-sharing protocols. The Contractor shall deliver the Building Information Model (BIM) in accordance with ISO 19650: Part 1 & 2, ISO 16739, and ISO 12006-3.

Ref: 460 - Mandatory Requirement

The Contractor shall develop 3D model (blocks) in format IFC4 of the generic components that will be installed by them or the civil contractor:

- Balises with all type of mounting system
- All types of location cases
- Junction boxes and termination boxes

Ref: 461 - Mandatory Requirement

The 3D model (blocks) developed by the Contractor shall be allocated to each applicable entry of the Asset Register defined in section 10.4:

Ref: 462 - Mandatory Requirement

The attribute of each individual 3D model product (blocks) shall be directly fed from the Asset Register as defined in section 10.4.

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33 HANDOVER DOCUMENTATION

Ref: 463 - Mandatory Requirement

The Contractor shall be responsible for the setup, coordination, production and support of the projects Digital Construction Handover Information (DCHI). This shall be a comprehensive information source and guide for the Client and provide a complete understanding of the systems, their intended performance and thus, enabling it to be operated and maintained efficiently and safely. The Contractor shall nominate a specific Digital Construction Handover Information manager to compile the safety file (this position should be clearly identified in the Contractor's Organisation chart).

Ref: 464 - Mandatory Requirement

The Contractor shall complete a comprehensive Handover Manual (incorporating the Safety File) and as-built drawings on completion of the Work for the acceptance by the Project Manager. The contents and requirements are to be as per the Client's performance specification for digital construction handover document. The digital handover shall include but not be limited to:

- ETCS Supplier Documents
 - Element Estimation
 - Asset Register
 - Release notes
 - Products specifications
 - Generic Application & Requirements Documents
 - Generic Safety Case
 - Manuals
 - Training Documents
 - Materials Information (Technical File)
 - Specific Application Safety Case
 - Specific Application Functional Design
 - Specific application Detailed Design
 - Balise Positioning Forms
 - Maintenance Data
 - Testing Specifications
 - V&V & Testing Reports
 - EMC Documents
 - RAM Documents
 - Cyber Security Documents
 - Safety File
 - LEU and balises configuration files
 - Infrastructure database

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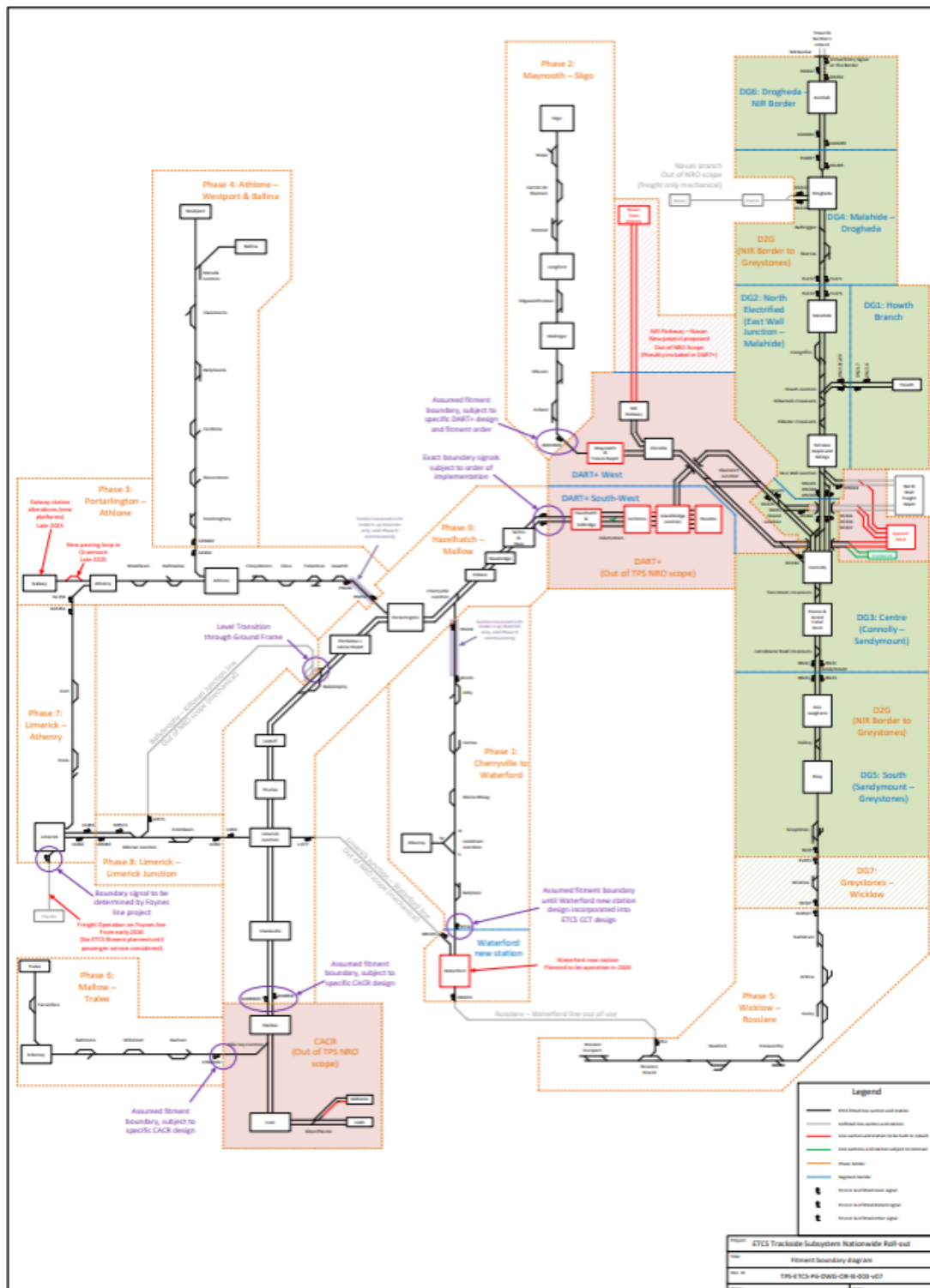
-
- Location case documents:
 - Location Case Fabrication Drawings
 - Certificates of conformity
 - Location Case Design Drawings (site specific)
 - Equipment Specifications
 - Testing Documentation and Records (See Testing Requirements)
 - Data Sheet Per Location Case:
 - Version & Serial Numbers for Equipment Installed where appropriate

Documents to be summarised on an excel spreadsheet containing the following headings or similar: Document, Code Number, Reference Number, Version, Submission Date.

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APPENDIX A WORKS SECTIONS

For readability, refer to TPS NRO - Fitment boundary diagram TPS-ETCS-P6-DWG-OR-IE-003 in Appendix B.



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The table below provides indicative number of signals, locations cases, balises and km of balise cables required for each phase

Phase	No. signals fitted	No. Locations cased	No. Fixed Balises	No. controlled Balises	Balise cable (Km)
1	92	64	118	146	52km
2	113	76	165	175	67km
3	107	78	126	200	66km
4	157	105	187	220	79km
5	62	38	130	101	37km
6	98	64	116	151	62km
7	76	46	94	140	55km
8	23	19	33	29	5km
9	206	157	361	370	101km

APPENDIX B APPLICABLE REGULATIONS, STANDARDS, GUIDELINES, EXAMPLES & POLICY DOCUMENTS

The below provides a non-exhaustive list of documents for information which are referenced or included in the relevant appendices and shall be adhered to as part of the works. The Contractor shall apply and fulfil all relevant legislation and industry standards and any other EU Directives or Commission Regulations with applicability to the works not listed below which are described in this Scope. Applicable Client's Standards & Guidelines for 2023 are also listed below.

Document Number	Document Description	Sharpoint Location	Use / Purpose
			(Warranted / for Information only)
Directives & Applicable Regulations – applicable to the Works			
Directive (EU) 2016/797	Interoperability of the Rail System within the European Union	Copy not provided	
Regulation (EU) 2016/919 and 2023/1695	Technical Specification for Interoperability relating to the ‘control-command and signalling’ (CCS TSI) subsystems.	Copy not provided	
Regulation (EU) 2014/1302 and 2023/1694	Technical Specification for Interoperability relating to the ‘rolling stock — locomotives and passenger rolling stock’ (LOC&PAS TSI) subsystem.	Copy not provided	
Directive (EU) 2016/798	Railway Safety: defines common principles for the management, regulation and supervision of railway safety.	Copy not provided	
IRS-203	EMC Co-Ordination	Copy not provided	
IRS-304	Requirements for Class A ETCS CCO Systems in the Republic of Ireland	Copy not provided	
IRS-305	Requirements for Class A ETCS CCT Systems in the Republic of Ireland	Copy not provided	
CRR-G-009	Guidance on Application for Authorisation and Application for Acceptance for Heavy Rail Fixed Installations and Vehicles	Copy not provided	
EN Standards Referenced - applicable to the Works			
EN 50125-1:2014	Railway applications - Environmental conditions for equipment – Part 1: Rolling stock and CCO equipment.	Copy not provided	
EN 50121-3-2:2017	Railway applications - Electromagnetic compatibility Part 2: Emission of the whole railway system to the outside world. Part 3-2: Rolling stock- Apparatus	Copy not provided	

Document Number	Document Description	Sharpoint Location	Use / Purpose
			(Warranted / for Information only)
EN 50121-3-1:2017	Railway applications - Electromagnetic compatibility Part 3-1: Rolling stock — Train and complete vehicle	Copy not provided	
EN 50126-1-2	Railway Applications - The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS)	Copy not provided	
EN 50716	Railway Applications - Requirements for software development	Copy not provided	
EN 60529:1992+A2:2013	Degrees of protection provided by enclosure (IP)	Copy not provided	
EN 60947-3:2021	Control circuit devices having a rated voltage not exceeding AC 1 000 V (at a frequency not exceeding 1 000 Hz) or DC 600 V.	Copy not provided	
EN 62262:2002	Classification of the degrees of protection provided by enclosures against external mechanical impacts	Copy not provided	
EN 62625-1	Electronic railway equipment	Copy not provided	
ERA ERTMS Subsets Referenced – applicable to the Works			
Subset 026	System Requirements Specification	Copy not provided	
Subset 036	FFFIS for Eurobalise	Copy not provided	
Subset 040	Dimensioning and Engineering Rules	Copy not provided	
Subset 041	Performance Requirements for Interoperability.	Copy not provided	
Subset 091	Safety Requirements for the Technical Interoperability of ETCS in Levels 1 & 2	Copy not provided	
Subset 104	ETCS System Version Management	Copy not provided	
TD/011REC1028	ESC/RSC technical document	Copy not provided	
	ERTMS/ATO Glossary	Copy not provided	
ISO Standards Referenced applicable to the Works			
ISO 26800:2011	Ergonomics. General approach, principles and concepts	Copy not provided	
ISO 6385:2016	Ergonomic principles in the design of work systems	Copy not provided	

Document Number	Document Description	Sharpoint Location	Use / Purpose
			(Warranted / for Information only)
ISO 9241	Ergonomics of human-system interaction, particularly	Copy not provided	
Irish Rail - Project Scope document			
TPS-ETCS-P6-DWG-OR-IE-003	TPS NRO - Fitment boundary diagram	D1-A-07	
TPS-ETCS-P6-PRO-DE-IE-001 v06	TPS NRO – ETCS Migration Strategy	D1-A-02	
TPS-NRO-P3-P8-P9-ENG-REP-0001 v01	TPS NRO – ETCS Correlation Report Phase 09/08 and 03	D1-A-03	
IM-CI Strategic Planner v.6b	Irish Rail – Strategic Project Planner	D1-A-04	
NRO Trackside POAP, Programme Update (Data Date - 26-Jan-25)	TPS NRO – Trackside POAP	D1-A-04	
NRO Trackside Key Project Milestones Tracker – 26-Jan-25	TPS NRO – Project Milestones Tracker	D1-A-04	
TPS-NRO-TS-P1-PLA-SA-IE-002 v01	TPS NRO – Quality Management Plan		
Fleet Types Diagram v01	Typical Train Types to be considered for Infill purposes		
Irish Rail -Technical Requirements – copy included in Appendix			
TPS-ETCS-P6-REQ-SL-IE-001 v05	Irish Rail – TPS Common Application Requirements	D1-B-01	
TPS-ETCS-P6-REQ-SL-IE-003-v02	Irish Rail – TPS Balise Product Requirements	D1-B-02	
TPS-ETCS-P6-REQ-SL-IE-004-v03	Irish Rail – TPS Encoder Product Requirements	D1-B-01	
TPS-D2G-P4-PLA-SI-IE-001 v03	Irish Rail – TPS Balise Layout Methodology	D1-B-01	
TPS-NRO_TS-P5-REQ-SL-IE-001-v01.	TPS Lamp Parameters	D1-B-01	
Irish Rail -Reference Design			
581 BLP Waterford line Rev1.6	Irish Rail – Balise Layout plans – Waterford line	D1-A-01	
523 BLP Longford Sligo Rev1.4	Irish Rail – Balise Layout plans – Longford Sligo	D1-A-01	
522 BLP Maynooth Longford Rev1.4	Irish Rail – Balise Layout plans – Maynooth Longford	D1-A-01	
531 BLP Portarlinton Athlone Rev1.6-Model	Irish Rail – Balise Layout plans – Portarlinton Athlone	D1-A-01	
533 BLP Galway line Rev2.2	Irish Rail – Balise Layout plans – Galway line	D1-A-01	

Document Number	Document Description	Sharpoint Location	Use / Purpose
			(Warranted / for Information only)
532 BLP Mayo line Rev1.2	Irish Rail – Balise Layout plans – Mayo line	D1-A-01	
516 BLP Greystones Rosslare Rev1.4	Irish Rail – Balise Layout plans – Greystones Rosslare	D1-A-01	
561 BLP Tralee line Rev1.2	Irish Rail – Balise Layout plans – Tralee line	D1-A-01	
541 BLP Limerick Athenry Rev1.2	Irish Rail – Balise Layout plans – Limerick Athenry	D1-A-01	
542 BLP Limerick Station Rev2.3	Irish Rail – Balise Layout plans – Limerick Station	D1-A-01	
503 BLP Hazelhatch Cherryville Rev1.4	Irish Rail – Balise Layout plans – Hazelhatch Cherryville	D1-A-01	
504 BLP Portarlinton Rev1.5	Irish Rail – Balise Layout plans – Portarlinton	D1-A-01	
505 BLP Portlaoise Lisduff Rev1.4	Irish Rail – Balise Layout plans – Portlaoise Lisduff	D1-A-01	
506B BLP Thurles Limerick Junction Rev2.3	Irish Rail – Balise Layout plans – Thurles Limerick Junction	D1-A-01	
507 BLP Charleville Mallow Rev1.4	Irish Rail – Balise Layout plans – Charleville Mallow	D1-A-01	
N/A	Irish Rail - SET Technical Development Centre, Thurles - Siding Layout Plan	D1-A-01	
LJ 08-73 22.08.2023	Irish Rail – Distant Signal and Existing LoCs with Signal Circuits Example	D1-A-05	
XW048 REB 25-01-2024	Irish Rail – REB Example	D1-A-05	
LC1	Irish Rail – Location Case 1 Example	D1-A-05	
LC3	Irish Rail – Location Case 3 Example	D1-A-05	
Lisduff Relay Room Shts A-E_13.06.24	Irish Rail – Relay Rooms Drawing Example Lisduff	D1-A-05	
Lisduff Relay Room Shts F-H_13.06.24	Irish Rail – Relay Rooms Drawing Example Lisduff	D1-A-05	
N/A	Irish Rail – Existing Signalling Details with Options 1-4	D1-A-05	
HB09-48-TPS_1.0	Irish Rail – TPS Location Case Template Drawings	D1-A-05	
SN11-05-TPS_1.0	Irish Rail – TPS Location Case Template Drawings	D1-A-05	
SN11-46-TPS_1.0	Irish Rail – TPS Location Case Template Drawings	D1-A-05	

Document Number	Document Description	Sharpoint Location	Use / Purpose
			(Warranted / for Information only)
N/A	Irish Rail – TPS Full width (5-8 LEUs)/ Half width (3-4 LEUs)/Pedestal (1-2 LEUs)	D1-A-05	
LJ 10-85 23.08.2023	Irish Rail – TPS Track Circuit LoCs for Power	D1-A-05	
TPS – Example/template of engineering documents from TPS D2G project			
IE-TPS-GS-SIA-001	Generic Specification for Survey of Infrastructure Assets	D1-A-06	
CTA05B-IR-REP-MPM-000009 06	Engineering Rules	D1-A-06	
CTA05B-IR-REP-MPM-000007 06	ETCS Signalling and Application Principles	D1-A-06	
CTA05B-IR-SPE-ESG-000004 05	Encoder Standard configuration document	D1-A-06	
CTA05B-IR-SPE-ESG-000005 03	Gradient segmentation Methodology	D1-A-06	
CTA05B-IR-DRG-ESG-000004 07	DG-3 ETCS Signalling Layout	D1-A-06	
CTA05B-IR-SPE-ESG-000023 07	DG-3 Signal Data Table	D1-A-06	
CTA05B-IR-SPE-ESG-000013 06	DG-3 LEU Control Table	D1-A-06	
AEG-000081	DG3 Gradient Profile Table	D1-A-06	
	Dynamic testing derogation and method statement		
	Test sequences and box plan;		
	Static test box plans protection etc.	Tomas/ Monika	
DIV 3_Heuston to Cork	Geodetic survey example - Lidar	Mark N	
	Balise plate drawing	D1-A-06	
	Balise mounting drawings/specs used in D2G	D1-A-06	
S.511	Newry Malahide (Ver.2.5) Master Signalling Plan with Main TPS Layer.	ID to share	Being drafted
	Junction and termination boxes drawings	D1-A-06	
E1.1 AASR 2024.03.07 200529	TPS Appropriate Assessment Screening Report Cherryville Junction to Waterford		
E2.1 PEA 2024.03.07 200529	TPS Ecological Impact Assessment Cherryville Junction to Waterford		

Document Number	Document Description	Sharpoint Location	Use / Purpose
			(Warranted / for Information only)
	TPS Track Patrolling Feature Forms Phase 1B		
	TPS Track Patrolling Feature Forms Phase 1A		
TPS – Example/template of engineering documents from other projects			
MN S 41604	Sydney Trains – Alstom ETCS Maintenance Manual	D1-A-06	
Project Management Standards for Iarnród Éireann Capital Investment Division - copy included in Appendix			
PMP-003	Project Schedule Procedure	D1-B-04	
PMP-001	Project Management Procedure	D1-B-04	
PMP-002	Risk and Contingency Management Procedure	D1-B-04	
PMP-004	Project Governance & Assurance Procedure	D1-B-04	
PMP-005	Change Control Procedure	D1-B-04	
PMP-006	Stakeholder & Communication Management Procedure	D1-B-04	
PMP-007	Document Management & Control Procedure	D1-B-04	
PMP-008	Cost Management Procedure	D1-B-04	
PMP-009	Design Management Procedure	D1-B-04	
PMP-010	Project Reporting Procedure	D1-B-04	
Interdepartmental Safety Standard - copy included in Appendix			
RU-SMS-014 Part A	Safety Approval of Changes in Plant, Equipment, Infrastructure and Operations (PEIO)	D1-B-03	
RIMU-SMS-014 v5.0	Safety Approval of Changes in Plant, Equipment, Infrastructure and Operations (PEIO)	D1-B-03	
IM-SMS-023	IM-SMS-023 Implementation of the Construction Regulations		
CCE Departmental Technical Standards - copy included in Appendix – need to be completed			
CME Departmental Technical Standards - copy included in Appendix – need to be completed			

Document Number	Document Description	Sharpoint Location	Use / Purpose
			(Warranted / for Information only)
SET Departmental Technical Standards - copy included in Appendix			
021 SET-SMS-002	SET Safety Statement	D1-B-05	
011 SET-SMS-001	Safety Management System	D1-B-05	
033 SET-SMS-003-003	Site Safety Briefing and Risk Assessment for Yourself	D1-B-05	
040 SET-SMS-004	Competency Assessments and Training	D1-B-05	
0401 SET-SMS-004-001	Competency Requirement for Signalling Designers Specialists Testers	D1-B-05	
060 SET-SMS-006	Hazards and Risk Assessments	D1-B-05	
070 SET-SMS-007	Reporting and Investigation of Accidents and Incidents	D1-B-05	
	European Railway Agency - Guidelines for the practical arrangements for the vehicle authorisation process	copy not provided	
I-SET-1012	SET Quality Principles	D1-B-05	
I-SET-1014	SET Internal Audits	D1-B-05	
I-SET-1101	SET Operational Technology Cyber Security	D1-B-05	
I-SET-1102	SET Operational Technology Cyber Security Procurement Guidance	D1-B-05	
I-SET-5051	Product Acceptance Procedure	D1-B-05	
I-SET-5101	SET Geospatial Asset Survey Standard	D1-B-05	
I-SET-5201	Quarantine of Equipment	D1-B-05	
I-SET-5202	Issuing SET 221 Locks and Keys	D1-B-05	
I-SET-5203	Calibration of Test and Measurement Equipment	D1-B-05	
I-SET-5204	Preservation of Evidence Following an Incident	D1-B-05	
I-SET-5205	Control of SET Keys	D1-B-05	
I-SET-5301	NMC Fault Management Process	D1-B-05	
I-SET-5303	SET Permit to Access for Authorised Contractors	D1-B-05	
I-SET-5401	SET PWI Control Room Areas	D1-B-05	

Document Number	Document Description	Sharpoint Location	Use / Purpose
			(Warranted / for Information only)
I-SET-6000	Specification for Signalling Electrification and Telecoms Cable	D1-B-05	
I-SET-6002	Specification for Multi-Pair ETCS EuroBalise Cable	D1-B-05	
I-SET-6004	Concrete Cable Troughing	D1-B-05	
I-SET-6005	Elevated Cable Troughing	D1-B-05	
I-SET-6200	ETCS Trackside Equipment Location Case Specification	D1-B-05	
I-SET-9010	Engineering Change Request	D1-B-05	
I-SET-9100	Engineering Management for Projects	D1-B-05	
I-SET-9102	Management and Review of Engineering Deliverables	D1-B-05	
I-SET-9305	Management of Overlapping and Parallel Design	D1-B-05	
I-SET-9308	Design Media Techniques and Drawing Control	D1-B-05	
I-SET-9309	Design Change Presentation and Convention	D1-B-05	
I-SET-9601	Standard for Location Case Installation	D1-B-05	
I-SET-9603	Cable Route and Cable Installation	D1-B-05	
I-SET-9604	Site Preparation for Signals and Location Cases	D1-B-05	
I-SET-9610	Construction of Under Road Crossings (URX)	D1-B-05	
I-SIG-2001	Management of Signalling and Related Equipment	D1-B-05	
I-SIG-2002	Signalling Interface Management	D1-B-05	
I-SIG-2005	Asset Management in Signalling for Safety and Performance	D1-B-05	
I-SIG-2007	Management Requirements for SET Design Production	D1-B-05	
I-SIG-2008	Requirements for Safe Design	D1-B-05	
I-SIG-2010	Validation and Verification Requirements	D1-B-05	
I-SIG-2101	Signalling Scheme Plans	D1-B-05	
I-SIG-2102	Signalling Control Tables	D1-B-05	
I-SIG-2103	Signalling Symbols for Scheme Plans	D1-B-05	

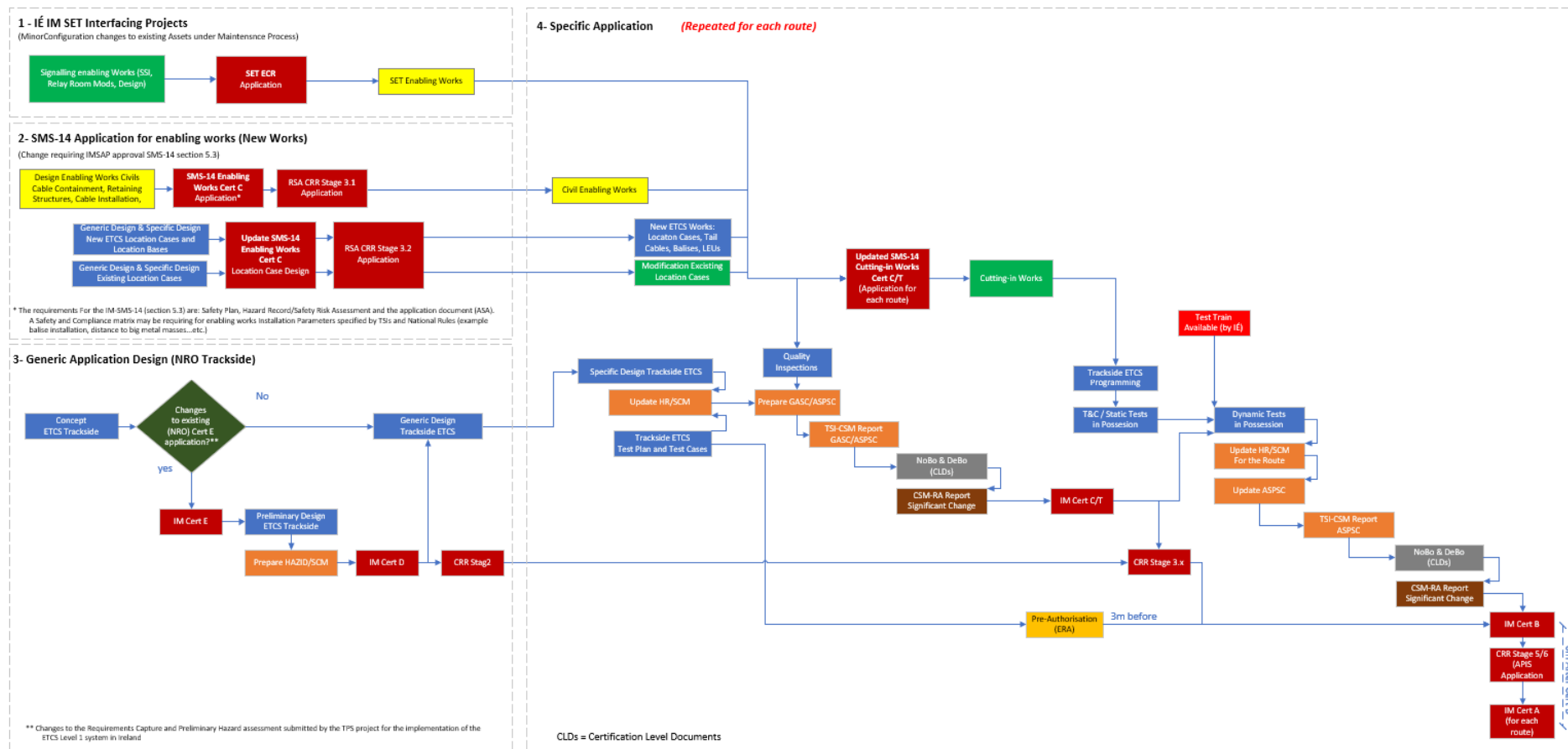
Document Number	Document Description	Sharpoint Location	Use / Purpose
			(Warranted / for Information only)
I-SIG-2104	Symbols for Signalling Circuit Diagrams	D1-B-05	
I-SIG-2175	Allocation of Values to ETCS Variables	D1-B-05	
I-SIG-2180	Signalling Permit to Access for 3rd Party Contractors	D1-B-05	
I-SIG-2190	Signalling Compliance Verification	D1-B-05	
I-SIG-2201	Signalling Installation Handbook	D1-B-05	
I-SIG-2210	Hand Tracing of Wiring	D1-B-05	
I-SIG-2222	Procedure for Equipment Room Installation	D1-B-05	
I-SIG-2241	Procedure for Colour Light Signal Installation	D1-B-05	
I-SIG-2301	Wrong-Side Failure and Incident Testing	D1-B-05	
I-SIG-2329	Test Procedure for Lineside Apparatus Cases	D1-B-05	
SET-SMS-004-011	3rd Party Signalling Competency Requirements	D1-B-05	
I-SIG-2172	Management of ETCS Trackside Configuration Data	D1-B-05	
NA	SET Operational Technology Security Policy V1.0	D1-B-05	
NA	Social Media Policy	D1-B-05	
NA	SET Fire Safety Policy V1.1	D1-B-05	
Departmental Safety Management Standards			
IÉ Railway Safety Requirements			
Drugs and Alcohol Policy Booklet	IE Drugs & Alcohol Policy	D1-C-02	
Contractor PTS Application Form & Information			
Contractor Personal Track Safety Medical Form & Guidelines			
Iarnród Éireann - Code of Conduct for Suppliers & Business Partners			
Applicable Policies Guidelines and Standards related to the Civils Implementation works of the ETCS Contractor			

Document Number	Document Description	Sharpoint Location	Use / Purpose
			(Warranted / for Information only)
NR-I3-CIV-039	Protective Treatment Specification	D1-C-02	
NR-L3-CIV-040	Protective Treatment Use	D1-C-02	
I-FBD-8210	Installation of Electrical Low Voltage (LV) Ducts and Access Chambers	D1-C-02	
I-PLM-5001	Certification Requirements for RoadRail Vehicles v2.3	D1-C-02	
I-PLM-5001-OPS-001	Operation of RRV High Heavy Haulage System v3.1	D1-C-02	
I-PLM-5003	Certification Process for Road Rail Vehicles and Associated Equipment v2.2	D1-C-02	
NW-QMS-4-8-2	Ver 0.9 Draft	D1-C-02	
I-PLM-5110	Management of Plant on Site	D1-C-02	
I-PLM-5110	Management of Road Rail Vehicles	D1-C-02	
CCE-TEB-2013-004	Working around Track Support Zone	D1-C-02	
SIG-TAN-0016_	Cable_Damage_during_Installation	D1-C-02	
I-PWY-1101	Requirements for Track and Structures Clearances	D1-C-02	
PRJ-SSW-001	Use of Ander Troughing Lifter	D1-C-02	
SET-SSOW-0002	Safe Use of SEB CD40 Cable Drum Trailer	D1-C-02	
	Possession Plans	D1-C-02	
	Contractor Performance Assessment Form Blank	D1-C-02	
	ETCS Train Protection System Project Installation Guidance-Balise	D1-C-02	
	Balise mark-up Siting Form-Example	D1-C-02	
CCE-IMS-008-002	Noise Management	D1-C-02	
	Cable Testing Record Sheet-Blank	D1-C-02	
	Certificate of Handover for Signalling Works Testing	D1-C-02	

Document Number	Document Description	Sharpoint Location	Use / Purpose
			(Warranted / for Information only)
	Special Class Lineman Course Cable Sections-Cable Identification-2-59	D1-C-02	
CCE-IMS-008-001	Ecological Assessment Railway Maintenance & Building Maintenance Activities	D1-C-02	
CCE-STR-PSD-005	Technical Approval for Civil Engineering Structures V1.0	D1-C-02	
TPS	TPS NRO Works SET Prelim SH Plan V1.0	D1-C-02	
	IE Drugs & Alcohol Policy	D1-C-02	

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APPENDIX C SAFETY ASSURANCE PROCESS



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APPENDIX D DRAFT PROJECT PROGRAMME

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APPENDIX E SCOPE DELIVERABLES TIMEFRAMES

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Indicative Dates: Only to be included in Pre-tender scope issue - Not part of ITT Documents
(Contract Dates will be included as part of contract documents)

Phase	Section Deliverables	Sectional Completion	Indicative Forecast Date	Key Task / Deliverables	Key Dates	Indicative Forecast Date	Work Package	Completion Criteria
All	Generic Design	SC 1	Jan-27	Generic Design	KD 01	Jan-27	A	IE acceptance (Without site testing – included in SC - 4,5,6,8)
Phase 1 / Phase 2 / Phase 3 / Phase 4	Specific Design	SC 2	Nov-27	Specific Design Phase 1	KD 02	Jul-27	B	IE acceptance of Specific design
				Specific Design Phase 2	KD 03	Jul-27	B	
				Specific Design Phase 3	KD 04	Nov-27	B	
				Specific Design Phase 4	KD 05	Nov-27	B	
Phase 5 / Phase 6 / Phase 7 / Phase 8	Specific Design	SC 3	Nov-28	Specific Design Phase 5	KD 06	Mar-28	B	IE acceptance of Specific design
				Specific Design Phase 6	KD 07	Mar-28	B	
				Specific Design Phase 7	KD 08	Jul-28	B	
				Specific Design Phase 8	KD 09	Nov-28	B	
Phase 1 / Phase 2	Installation, T&C and APIS 5 approval	SC 4	Jul-29	Supply and Install Phase 1	KD 10	Mar-28	C	Handover certificate from Alstom to IE
				GA - T & C Phase 1	KD 11	Oct-28	C	Master test certificate issued to IE
				SA - T & C Phase 1	KD 12	Oct-28	C	Master test certificate issued to IE
				Assurance Phase 1	KD 13	May-29	C	APIS 5 Approval
				Supply and Install Phase 2	KD 14	Mar-28	C	Handover certificate from Alstom to IE
				GA - T & C Phase 2	KD15	Jan-29	C	Master test certificate issued to IE
				SA - T & C Phase 2	KD 16	Jan-29	C	Master test certificate issued to IE
				Assurance Phase 2	KD 17	Jul-29	C	APIS 5 Approval
Phase 3 / Phase 4 / Phase 5	Installation, T&C and APIS 5 approval	SC 5	Jun-30	Supply and Install Phase 3	KD 18	Sep-28	C	Handover certificate from Alstom to IE
				GA - T & C Phase 3	KD 19	Jun-29	C	Master test certificate issued to IE
				SA - T & C Phase 3	KD 20	Jun-29	C	Master test certificate issued to IE
				Assurance Phase 3	KD 21	Jan-30	C	APIS 5 Approval
				Supply and Install Phase 4	KD 22	Nov-28	C	Handover certificate from Alstom to IE
				GA - T & C Phase 4	KD 23	Nov-29	C	Master test certificate issued to IE
				SA - T & C Phase 4	KD 24	Nov-29	C	Master test certificate issued to IE
				Assurance Phase 4	KD 25	Jun-30	C	APIS 5 Approval
				Supply and Install Phase 5	KD 26	Jan-29	C	Handover certificate from Alstom to IE
				GA - T & C Phase 5	KD 27	Aug-29	C	Master test certificate issued to IE
				SA - T & C Phase 5	KD 28	Aug-29	C	Master test certificate issued to IE

**Indicative Dates: Only to be included in Pre-tender scope issue - Not part of ITT Documents
(Contract Dates will be included as part of contract documents)**

Phase	Section Deliverables	Sectional Completion	Indicative Forecast Date	Key Task / Deliverables	Key Dates	Indicative Forecast Date	Work Package	Completion Criteria
				Assurance Phase 5	KD 29	Mar-30	C	APIS 5 Approval
Phase 6 / Phase 7 / Phase 8	Installation, T&C and APIS 5 approval	SC 6	Feb-31	Supply and Install Phase 6	KD 30	May-29	C	Handover certificate from Alstom to IE
				GA - T & C Phase 6	KD 31	Mar-30	C	Master test certificate issued to IE
				SA - T & C Phase 6	KD 32	Mar-30	C	Master test certificate issued to IE
				Assurance Phase 6	KD 33	Oct-30	C	APIS 5 Approval
				Supply and Install Phase 7	KD 34	Sep-29	C	Handover certificate from Alstom to IE
				GA - T & C Phase 7	KD 35	Jun-30	C	Master test certificate issued to IE
				SA - T & C Phase 7	KD 36	Jun-30	C	Master test certificate issued to IE
				Assurance Phase 7	KD 37	Jan-31	C	APIS 5 Approval
				Supply and Install Phase 8	KD 38	Nov-29	C	Handover certificate from Alstom to IE
				GA - T & C Phase 8	KD 39	Jul-30	C	Master test certificate issued to IE
				SA - T & C Phase 8	KD 40	Jul-30	C	Master test certificate issued to IE
				Assurance Phase 8	KD 41	Feb-31	C	APIS 5 Approval
Phase 9	Specific Design	SC 7	Jul-28	Specific Design Phase 9	KD 42	Jul-28	C	IE acceptance of Specific design
Phase 9	Installation, T&C and APIS 5 approval	SC 8	Jan-32	Supply and Install Phase 9	KD 43	Feb-30	C	Handover certificate from Alstom to IE
				GA - T & C Phase 9	KD 44	Jun-31	C	Master test certificate issued to IE
				SA - T & C Phase 9	KD 45	Jun-31	C	Master test certificate issued to IE
				Assurance Phase 9	KD 46	Jan-32	C	APIS 5 Approval